MOCCA 2025
Fundamentals of Character Animation and Motion Control

Lecture 08

Actuating Simulated Characters

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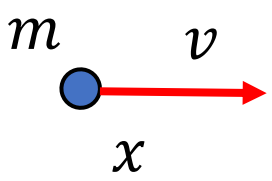
Outline

- Simulating & Actuating Characters
  - Joint torques
- PD (Proportional-Derivative) control
  - Stable PD control
- Feedforward Motion Control
  - Residual force
- Feedback Motion Control
  - Static balance

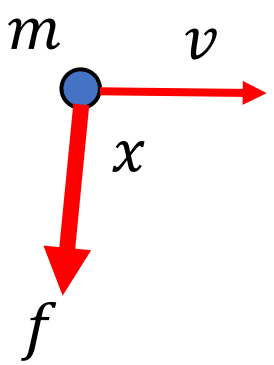
Recap: Dynamics of a Point Mass



x, v

$p = mv$

Recap: Dynamics of a Point Mass

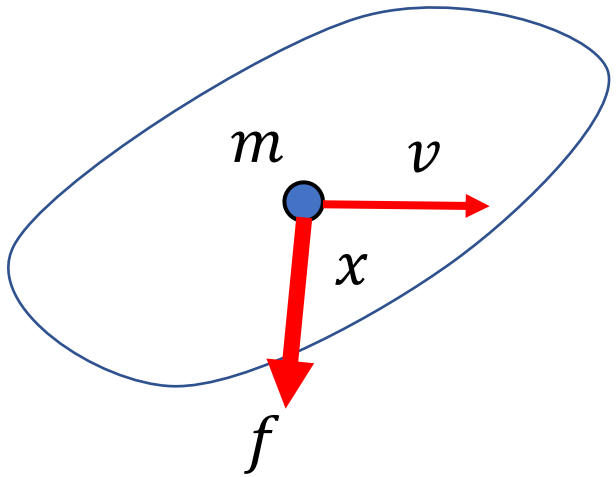


x, v

$p = mv$

Newton's Second Law: $\frac{dp}{dt} = f \Rightarrow m\dot{v} = f$

Recap: Rigid Body Dynamics



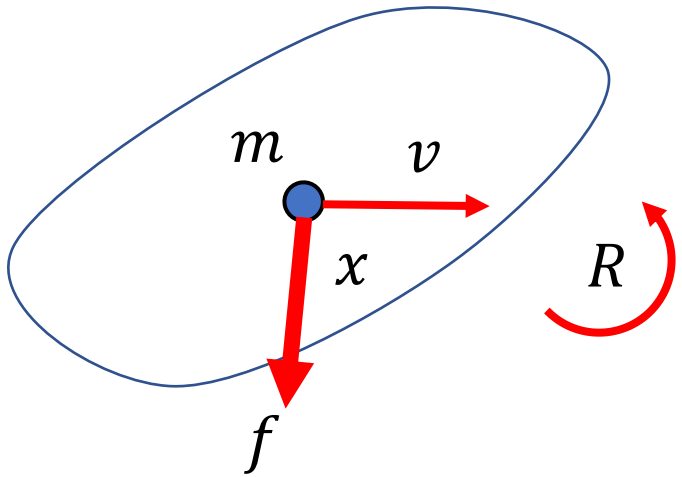
Linear

x, v

$p = mv$

Newton's Second Law: $\frac{dp}{dt} = f$ ➡ $m\dot{v} = f$

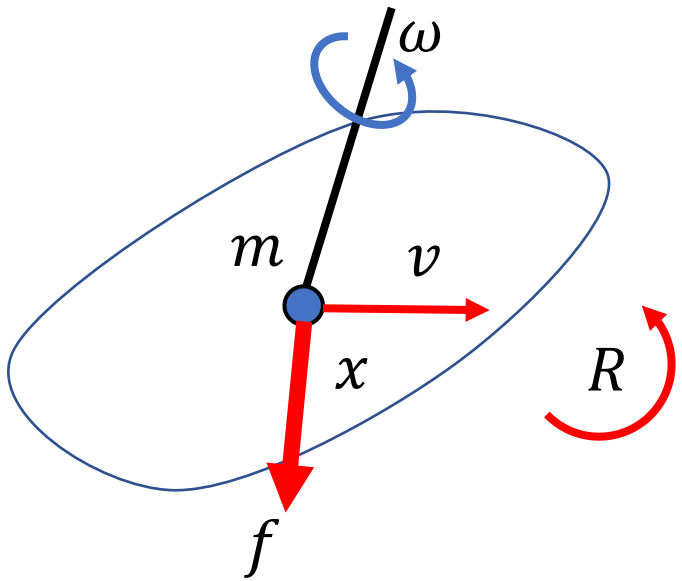
Recap: Rigid Body Dynamics



Linear	Angular
x, v	R
$p = mv$	

Newton's Second Law: $\frac{dp}{dt} = f \Rightarrow m\dot{v} = f$

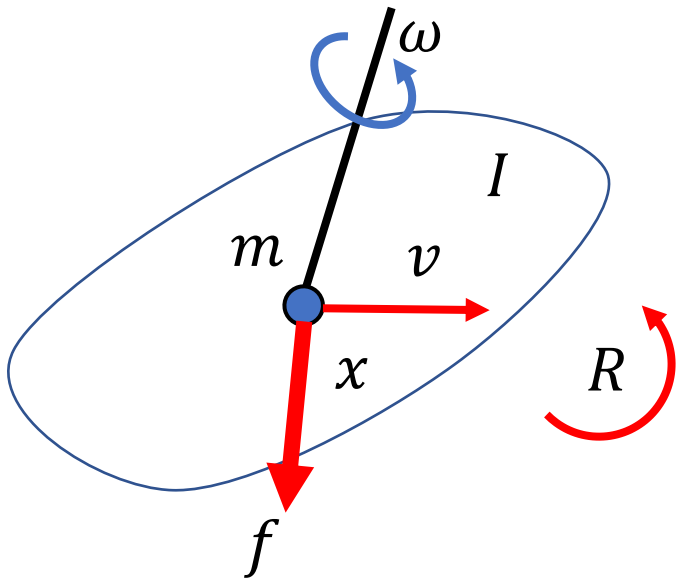
Recap: Rigid Body Dynamics



Linear	Angular
x, v	R, ω
$p = mv$	

Newton's Second Law: $\frac{dp}{dt} = f \Rightarrow m\dot{v} = f$

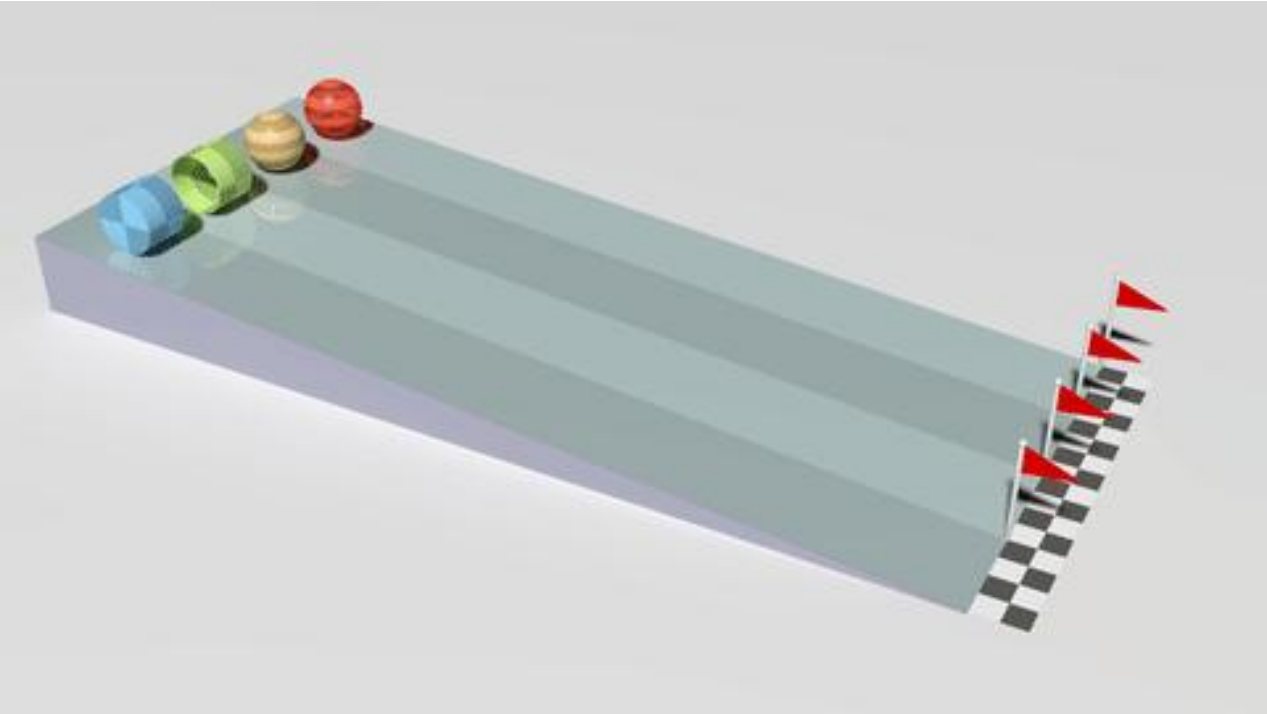
Recap: Rigid Body Dynamics



Linear	Angular
x, v	R, ω
$p = mv$	$L = I\omega$

Newton's Second Law: $\frac{dp}{dt} = f \Rightarrow m\dot{v} = f$

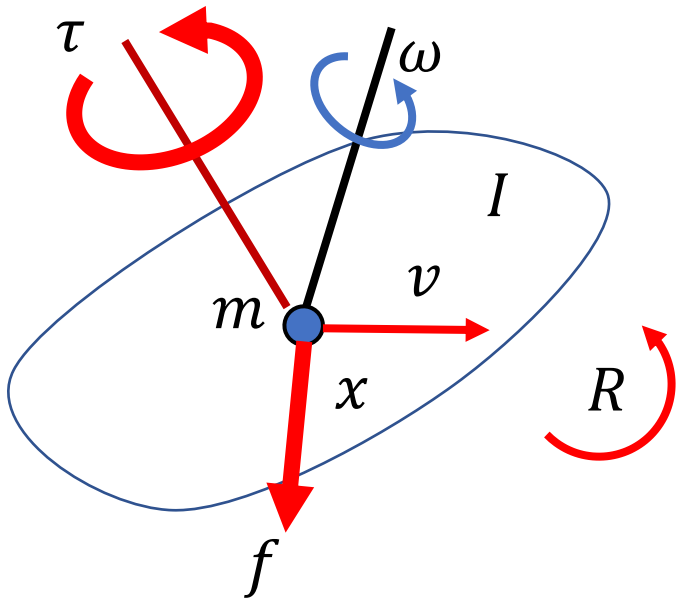
Recap: Moment of Inertia



Same mass, different shapes

Different Moments of Inertia

Recap: Rigid Body Dynamics

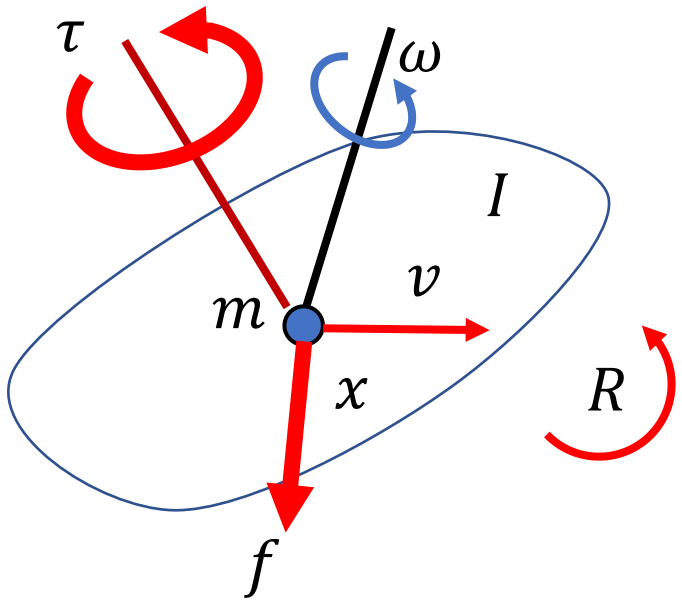


Linear	Angular
x, v	R, ω
$p = mv$	$L = I\omega$

Newton's Second Law: $\frac{dp}{dt} = f$ ➡ $m\dot{v} = f$

Euler's laws of motion: $\frac{dL}{dt} = \tau$ ➡ $I\dot{\omega} + \omega \times I\omega = \tau$

Recap: Rigid Body Dynamics



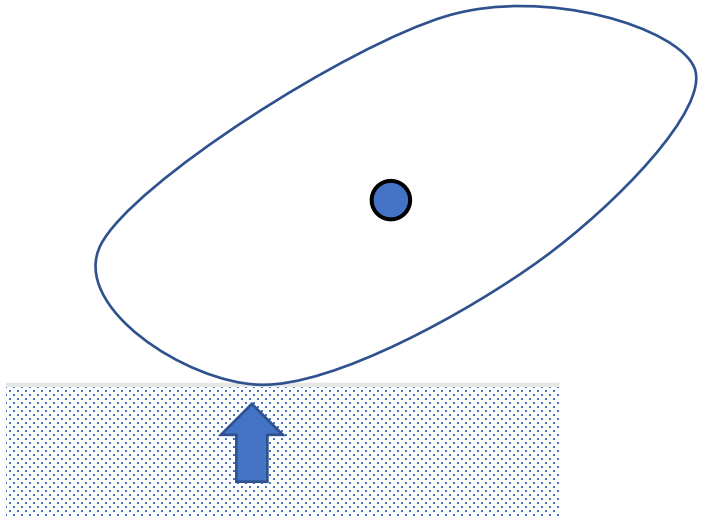
Linear	Angular
x, v	R, ω
$p = mv$	$L = I\omega$

Newton's Second Law:

Euler's laws of motion:

$$\begin{bmatrix} m\mathrm{I}_3 & 0 \\ 0 & I \end{bmatrix} \begin{bmatrix} \dot{v} \\ \dot{\omega} \end{bmatrix} + \begin{bmatrix} 0 \\ \omega \times I\omega \end{bmatrix} = \begin{bmatrix} f \\ \tau \end{bmatrix}$$

Defining a Rigid Body



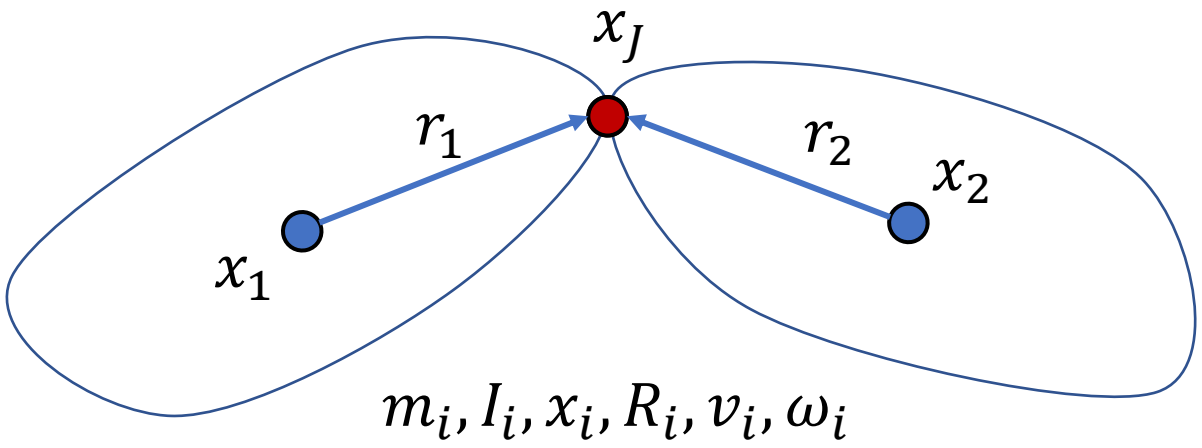
Masses: m, I

Kinematics: $\boldsymbol{x}, \boldsymbol{v}, R, \boldsymbol{\omega}$

Geometry:

- Box, Sphere, Capsule, Mesh, …
- Collision detection
- Compute m, I

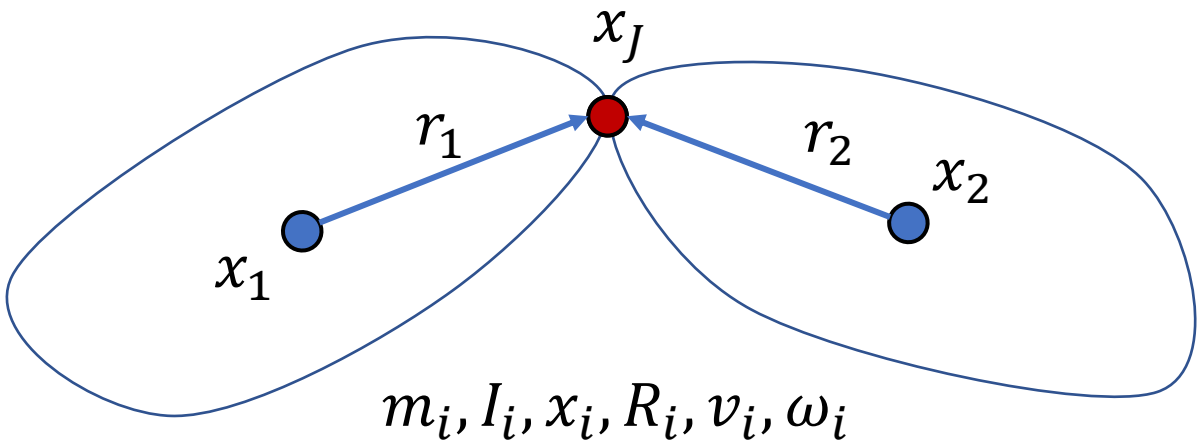
Recap: Dynamics of Articulated Rigid Bodies



$$\boxed{Jv = 0}$$

$$\begin{bmatrix} I_3 & -[r_1]_\times & -I_3 & [r_2]_\times \end{bmatrix} \begin{bmatrix} v_1 \\ w_1 \\ v_2 \\ w_2 \end{bmatrix} = 0$$

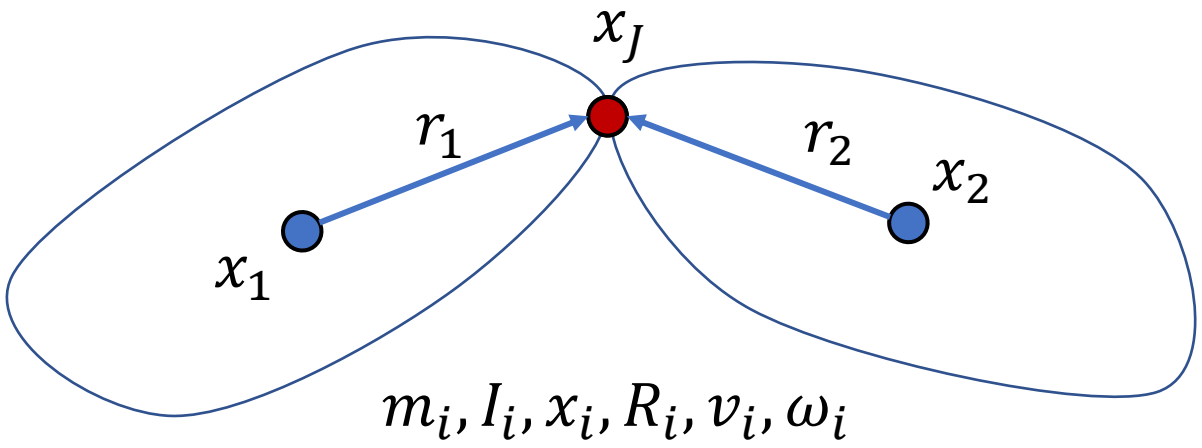
Recap: Dynamics of Articulated Rigid Bodies



$$\begin{bmatrix} m_1 \mathrm{I}_3 & & & \\ & I_1 & & \\ & & m_2 \mathrm{I}_3 & \\ & & & I_2 \end{bmatrix} \begin{bmatrix} \dot{v}_1 \\ \dot{\omega}_1 \\ \dot{v}_2 \\ \dot{\omega}_2 \end{bmatrix} + \begin{bmatrix} 0 \\ \omega_1 \times I_1 \omega_1 \\ 0 \\ \omega_2 \times I_2 \omega_2 \end{bmatrix} = \begin{bmatrix} f_1 \\ \tau_1 \\ f_2 \\ \tau_2 \end{bmatrix} + \begin{bmatrix} I_3 \\ [r_1]_\times \\ -I_3 \\ -[r_2]_\times \end{bmatrix} \lambda$$

$$Jv = 0$$

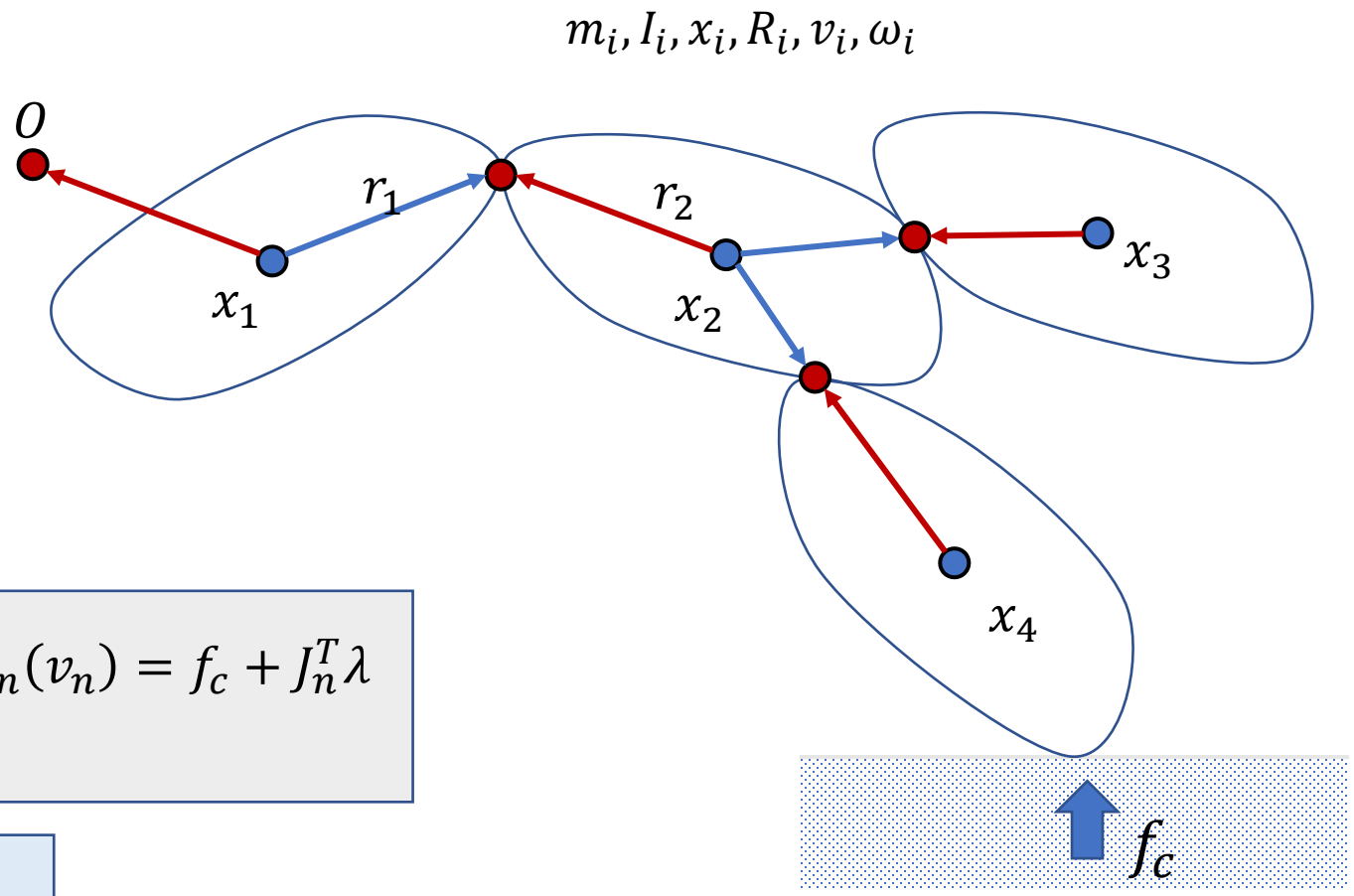
Recap: Dynamics of Articulated Rigid Bodies



$$M\dot{\boldsymbol{v}} + C(\boldsymbol{x}, \boldsymbol{v}) = \boldsymbol{f} + \boldsymbol{J}^{\boldsymbol{T}}\boldsymbol{\lambda}$$

$$Jv = 0$$

Recap: Simulation of a Rigid Body System



$$I_n = R_n I_0 R_n^T$$

$$M_n(v_{n+1} - v_n)/_h + C_n(v_n) = f_c + J_n^T \lambda$$

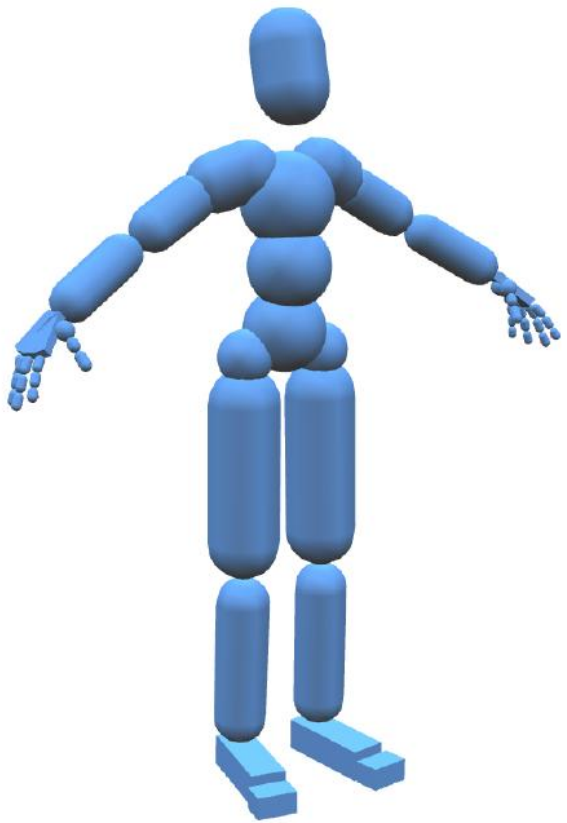
$$J_n v_{n+1} = c_n$$

$$x_{n+1} = x_n + h v_{n+1}$$

$$q_{n+1} = q_n + \frac{h}{2} \bar{\omega}_{n+1} q$$

Defining a Simulated Character

[Liu et al 2018]



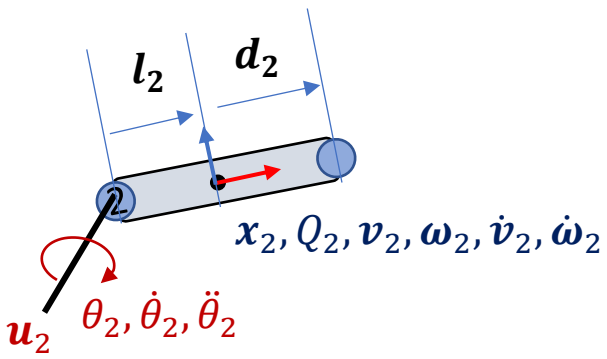
Rigid bodies:

- $m_i, I_i, \boldsymbol{x}_i, R_i$
- Geometries

Joints:

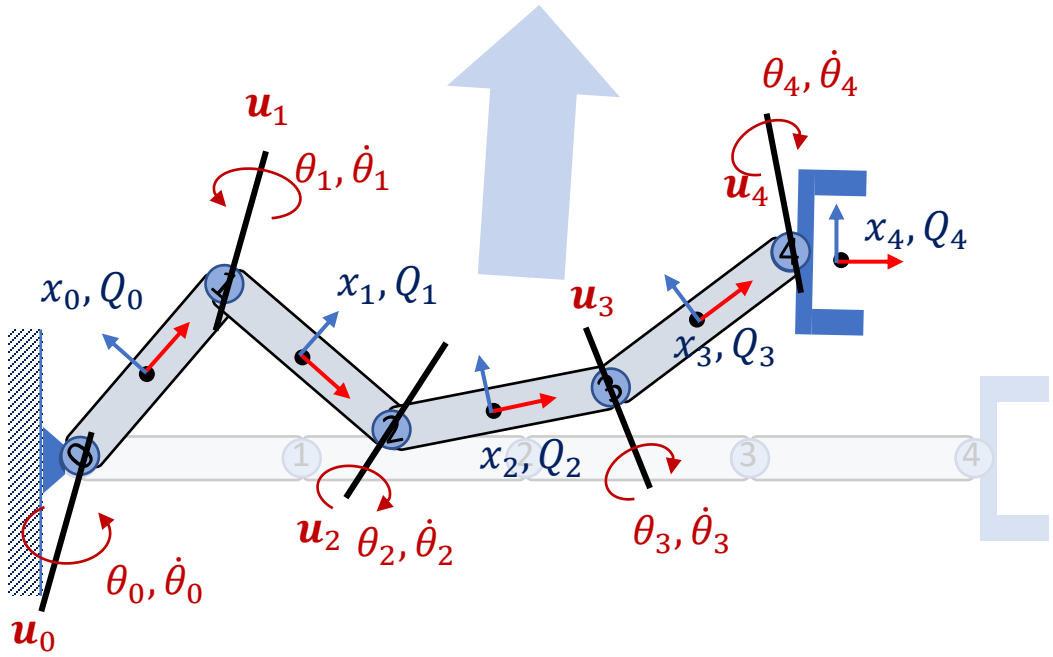
- Position
- Type
- Bodies

Maximized Coordinates & Generalized Coordinates



Maximized Coordinates:

$$\boldsymbol{x}_i, Q_i \qquad \boldsymbol{v}_i, \boldsymbol{\omega}_i \qquad \dot{\boldsymbol{v}}_i, \dot{\boldsymbol{\omega}}_i$$

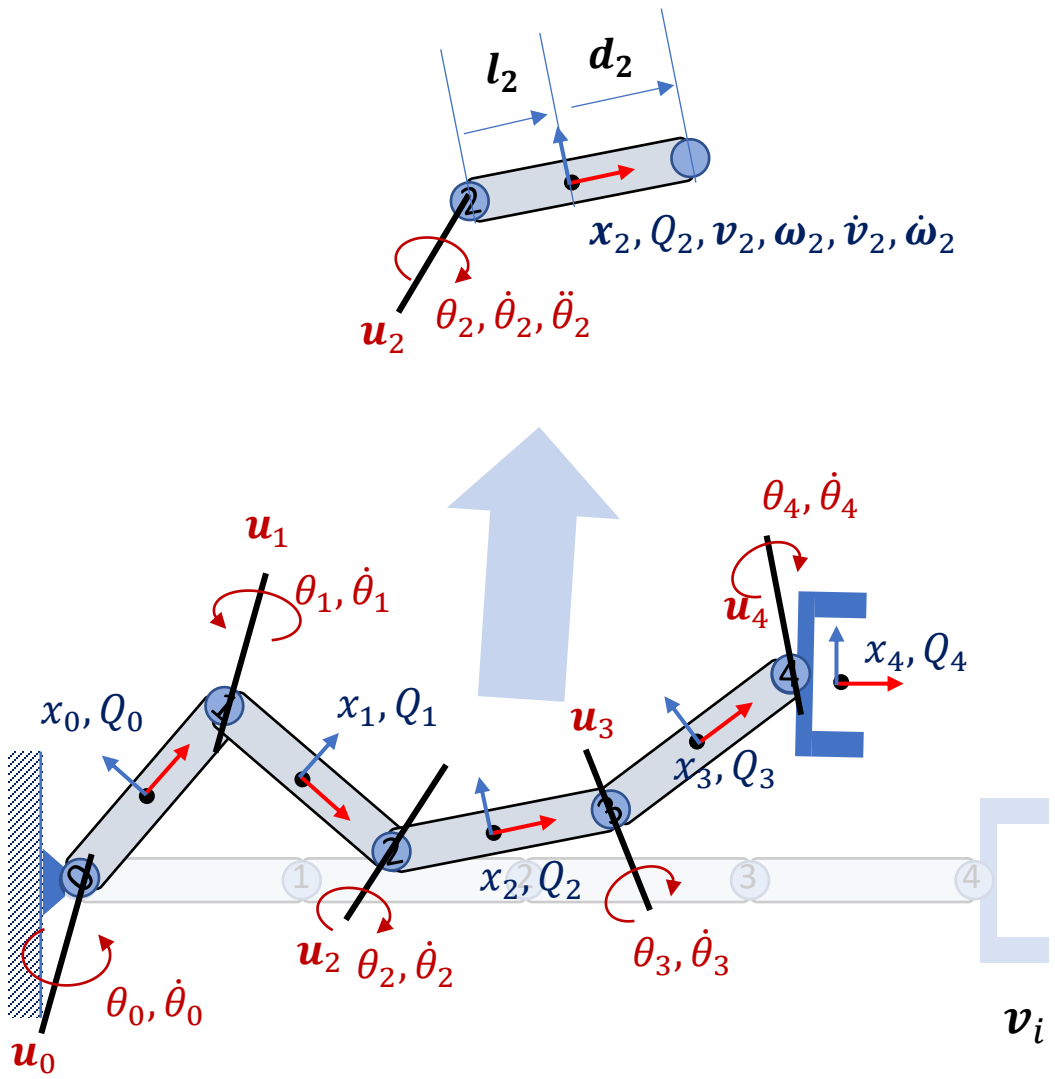


Generalized Coordinates:

$$\boldsymbol{\theta} = (\theta_0, \theta_1, \dots)$$

$$\boldsymbol{\theta}, \dot{\boldsymbol{\theta}}, \ddot{\boldsymbol{\theta}}$$

Maximized Coordinates & Generalized Coordinates



Generalized Coordinates

$$\boldsymbol{\theta}, \dot{\boldsymbol{\theta}}, \ddot{\boldsymbol{\theta}}$$

Maximized Coordinates:

$$\boldsymbol{x}_i, Q_i \quad \boldsymbol{v}_i, \boldsymbol{\omega}_i \quad \dot{\boldsymbol{v}}_i, \dot{\boldsymbol{\omega}}_i$$

Forward kinematics:

$$Q_i = \prod_{j=0}^{i} R(\theta_j \boldsymbol{u}_j)$$

$$Q_0 = R(\theta_0 \boldsymbol{u}_0)$$

$$Q_1 = R(\theta_0 \boldsymbol{u}_0) R(\theta_1 \boldsymbol{u}_1)$$

$$\boldsymbol{x}_i = \sum_{j=0}^{i-1} Q_j(\boldsymbol{l}_j + \boldsymbol{d}_j) + Q_i \boldsymbol{l}_i$$

$$\boldsymbol{x}_0 = Q_0 \boldsymbol{l}_0$$

$$\boldsymbol{x}_1 = Q_0(\boldsymbol{l}_0 + \boldsymbol{d}_0) + Q_1 \boldsymbol{l}_1$$

$$\boldsymbol{\omega}_i = \sum_{j=0}^{i} Q_{j-1} \dot{\theta}_j \boldsymbol{u}_j$$

$$\boldsymbol{\omega}_0 = \dot{\theta}_0 \boldsymbol{u}_0$$

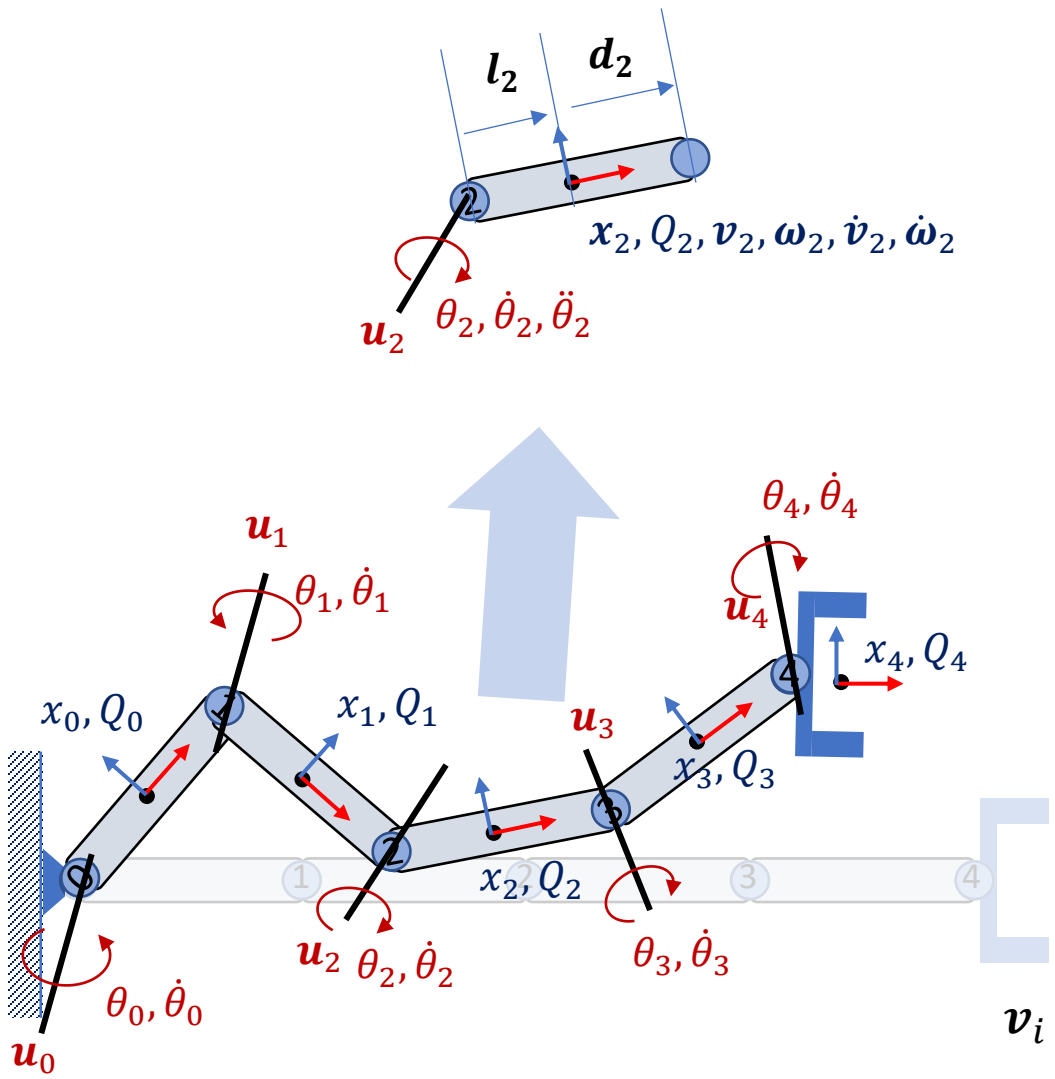
$$\boldsymbol{\omega}_1 = \dot{\theta}_0 \boldsymbol{u}_0 + Q_0 \dot{\theta}_1 \boldsymbol{u}_1$$

$$\boldsymbol{v}_i = \sum_{j=0}^{i-1} \boldsymbol{\omega}_j \times \left(Q_j(\boldsymbol{l}_j + \boldsymbol{d}_j)\right) + \boldsymbol{\omega}_i \times \boldsymbol{l}_i$$

$$\boldsymbol{v}_0 = \boldsymbol{\omega}_0 \times \boldsymbol{l}_0$$

$$\boldsymbol{v}_1 = \boldsymbol{\omega}_0 \times \left(Q_0(\boldsymbol{l}_0 + \boldsymbol{d}_0)\right) + \boldsymbol{\omega}_1 \times Q_1 \boldsymbol{l}_1$$

Maximized Coordinates & Generalized Coordinates



Generalized Coordinates

$$\boldsymbol{\theta}, \dot{\boldsymbol{\theta}}, \ddot{\boldsymbol{\theta}}$$

Maximized Coordinates:

$$\boldsymbol{x}_i, Q_i \quad \boldsymbol{v}_i, \boldsymbol{\omega}_i \quad \dot{\boldsymbol{v}}_i, \dot{\boldsymbol{\omega}}_i$$

Forward kinematics:

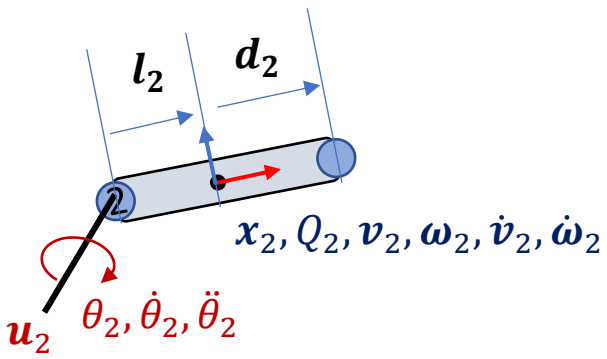
$$Q_i = \prod_{j=0}^{i} R(\theta_j \boldsymbol{u}_j) \qquad Q_i = Q_{i-1} R(\theta_i \boldsymbol{u}_i)$$

$$\boldsymbol{x}_i = \sum_{j=0}^{i-1} Q_j(\boldsymbol{l}_j + \boldsymbol{d}_j) + Q_i \boldsymbol{l}_i \qquad \boldsymbol{x}_i = \boldsymbol{x}_{i-1} + Q_{i-1}\boldsymbol{d}_{i-1} + Q_i \boldsymbol{l}_i$$

$$\boldsymbol{\omega}_i = \sum_{j=0}^{i} Q_{j-1} \dot{\theta}_j \boldsymbol{u}_j \qquad \boldsymbol{\omega}_i = \boldsymbol{\omega}_{i-1} + Q_{i-1} \dot{\theta}_i \boldsymbol{u}_i$$

$$\boldsymbol{v}_i = \sum_{j=0}^{i-1} \boldsymbol{\omega}_j \times \left(Q_j(\boldsymbol{l}_j + \boldsymbol{d}_j)\right) + \boldsymbol{\omega}_i \times \boldsymbol{l}_i \qquad \boldsymbol{v}_i = \boldsymbol{v}_{i-1} + \boldsymbol{\omega}_{i-1} \times (Q_{i-1}\boldsymbol{d}_{i-1}) + \boldsymbol{\omega}_i \times (Q_i \boldsymbol{l}_i)$$

Maximized Coordinates & Generalized Coordinates



Generalized Coordinates

$$\boldsymbol{\theta}, \dot{\boldsymbol{\theta}}, \ddot{\boldsymbol{\theta}}$$

Maximized Coordinates:

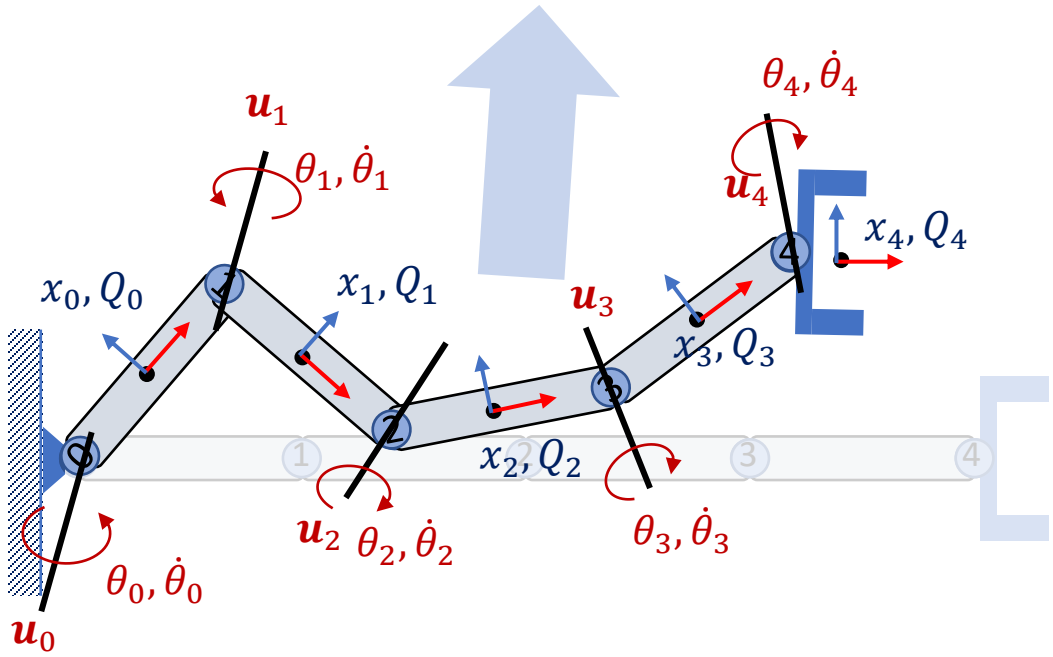
$$\boldsymbol{x}_i, Q_i \quad \boldsymbol{v}_i, \boldsymbol{\omega}_i \quad \dot{\boldsymbol{v}}_i, \dot{\boldsymbol{\omega}}_i$$

Forward kinematics:

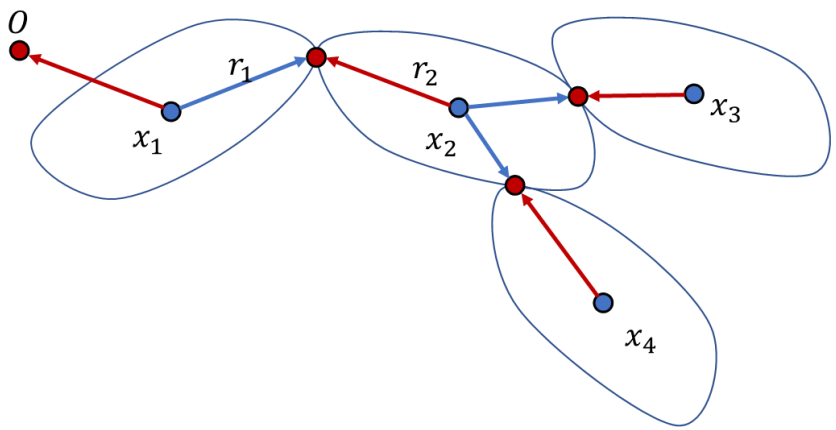
$$\dot{\boldsymbol{\omega}}_i = \dot{\boldsymbol{\omega}}_{i-1} + \boldsymbol{\omega}_i \times \left(Q_i \dot{\theta}_i \boldsymbol{u}_i\right) + Q_i \ddot{\theta}_i \boldsymbol{u}_i$$

$$\dot{\boldsymbol{v}}_i = \dot{\boldsymbol{v}}_{i-1} + \dot{\boldsymbol{\omega}}_{i-1} \times (Q_{i-1}\boldsymbol{d}_{i-1}) + \boldsymbol{\omega}_{i-1} \times \left(\boldsymbol{\omega}_{i-1} \times (Q_{i-1}\boldsymbol{d}_{i-1})\right)$$

$$+\dot{\boldsymbol{\omega}}_i \times (Q_i \boldsymbol{l}_i) + \boldsymbol{\omega}_i \times \left(\boldsymbol{\omega}_i \times (Q_i \boldsymbol{l}_i)\right)$$



Maximized Coordinates & Generalized Coordinates



Maximized Coordinates:

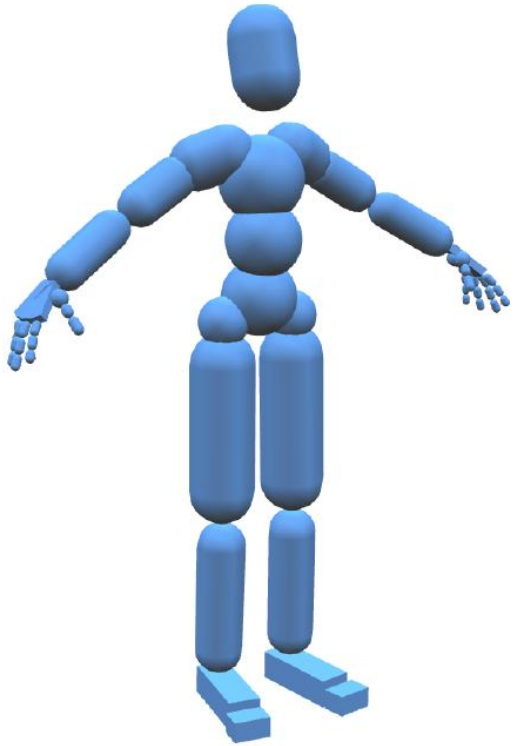
$$\boldsymbol{x}_i, R_i \qquad \boldsymbol{v}_i, \boldsymbol{\omega}_i \qquad \dot{\boldsymbol{v}}_i, \dot{\boldsymbol{\omega}}_i$$

Generalized Coordinates:

$$\boldsymbol{\theta} = (\boldsymbol{t}_0, \boldsymbol{\theta}_0, \boldsymbol{\theta}_1, \boldsymbol{\theta}_2, \dots\dots)$$

$$\boldsymbol{\theta}, \dot{\boldsymbol{\theta}}, \ddot{\boldsymbol{\theta}}$$

Character Kinematics



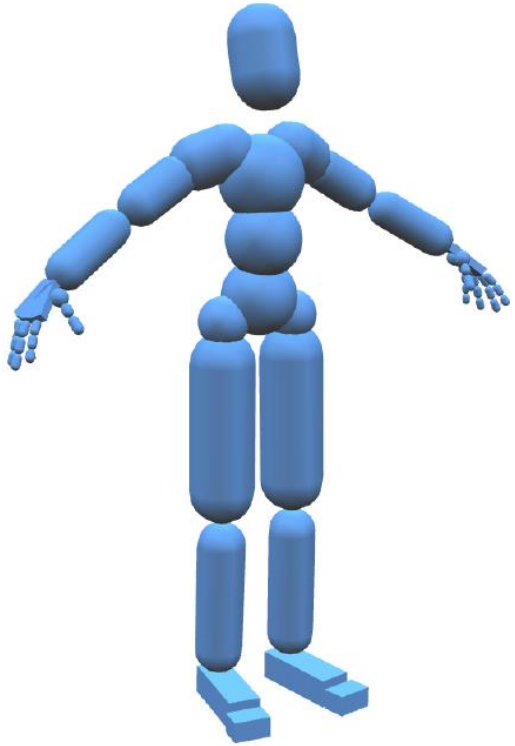
Forward Kinematics

$$\boldsymbol{\theta} = (\boldsymbol{t}_0, \boldsymbol{\theta}_0, \boldsymbol{\theta}_1, \boldsymbol{\theta}_2, \dots\dots) \longrightarrow \boldsymbol{x}_i, R_i$$

$$\dot{\boldsymbol{\theta}} = (v_0, \omega_0, \omega_1, \omega_2, \dots\dots) \longrightarrow \boldsymbol{v}_i, \boldsymbol{\omega}_i$$

$$\ddot{\boldsymbol{\theta}} = (\dot{v}_0, \dot{\omega}_0, \dot{\omega}_1, \dot{\omega}_2, \dots\dots) \longrightarrow \dot{\boldsymbol{v}}_i, \dot{\boldsymbol{\omega}}_i$$

Simulating a Character

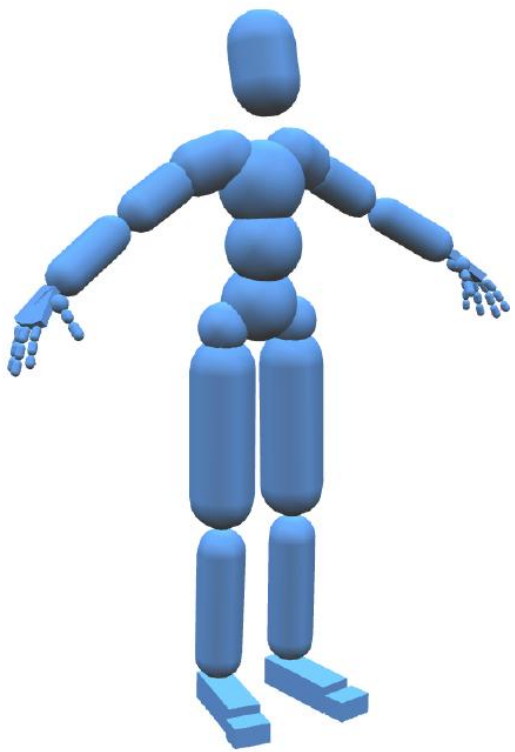


Current state:
$\boldsymbol{x}_t, \boldsymbol{v}_t, R_t, \boldsymbol{\omega}_t$

Dynamic Simulator

Next state:
$\boldsymbol{x}_{t+1}, \boldsymbol{v}_{t+1}, R_{t+1}, \boldsymbol{\omega}_{t+1}$

Simulating a Character

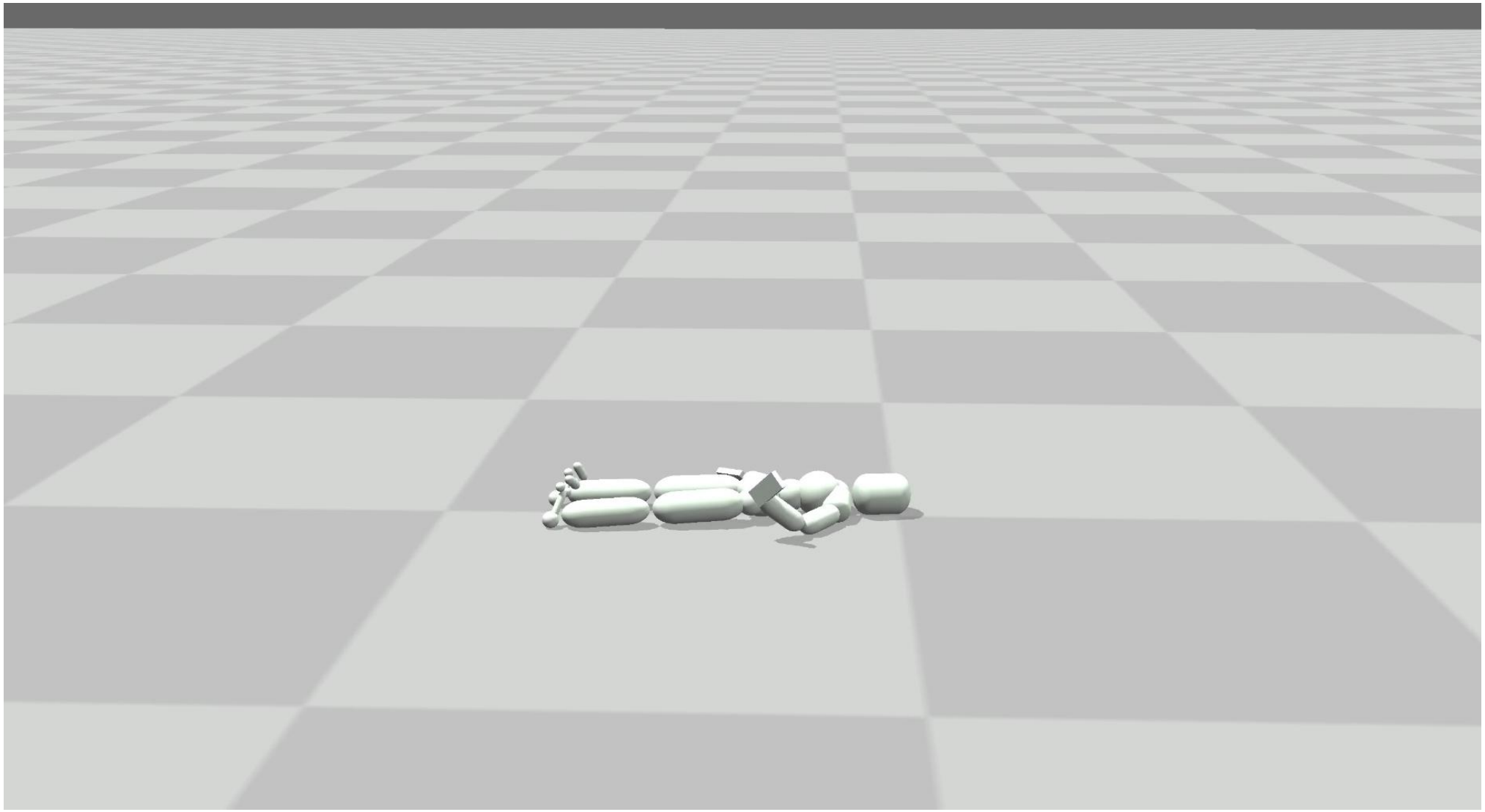


Current state:
$\boldsymbol{x}_t, \boldsymbol{v}_t, R_t, \boldsymbol{\omega}_t$

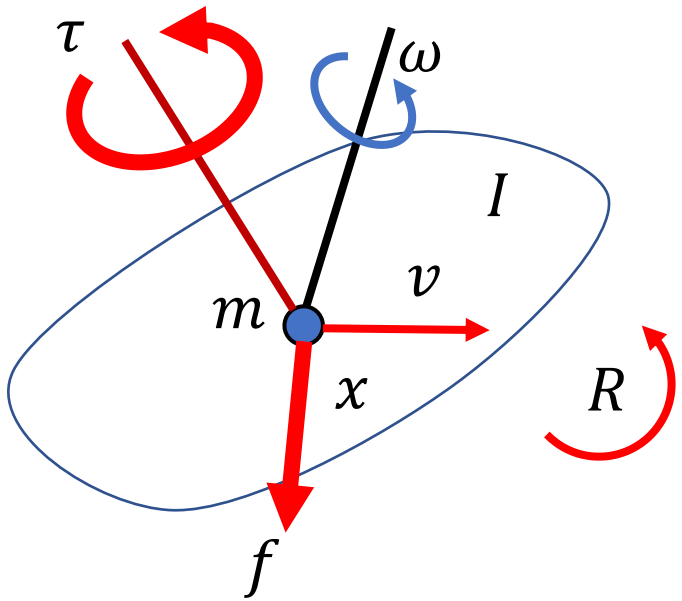
- Collision Detection
- Contact Handling
- Constraint Handling
- Construct Equations of Motion
- Solve Equations of Motion
- Update Velocity $\boldsymbol{v}_{t+1}, \boldsymbol{\omega}_{t+1}$
- Update Position $\boldsymbol{x}_{t+1}, \boldsymbol{R}_{t+1}$

Next state:
$\boldsymbol{x}_{t+1}, \boldsymbol{v}_{t+1}, R_{t+1}, \boldsymbol{\omega}_{t+1}$

Ragdoll Simulation



Actuating a Rigid Body



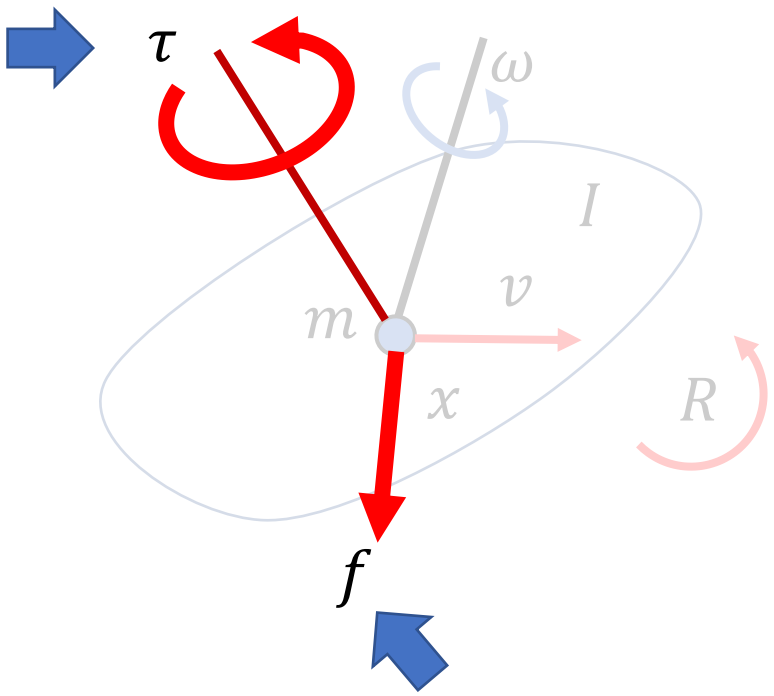
Linear	Angular
x, v	R, ω
$p = mv$	$L = I\omega$

Newton's Second Law:

Euler's laws of motion:

$$\begin{bmatrix} m\mathrm{I}_3 & 0 \\ 0 & I \end{bmatrix} \begin{bmatrix} \dot{v} \\ \dot{\omega} \end{bmatrix} + \begin{bmatrix} 0 \\ \omega \times I\omega \end{bmatrix} = \begin{bmatrix} \boldsymbol{f} \\ \boldsymbol{\tau} \end{bmatrix}$$

Actuating a Rigid Body



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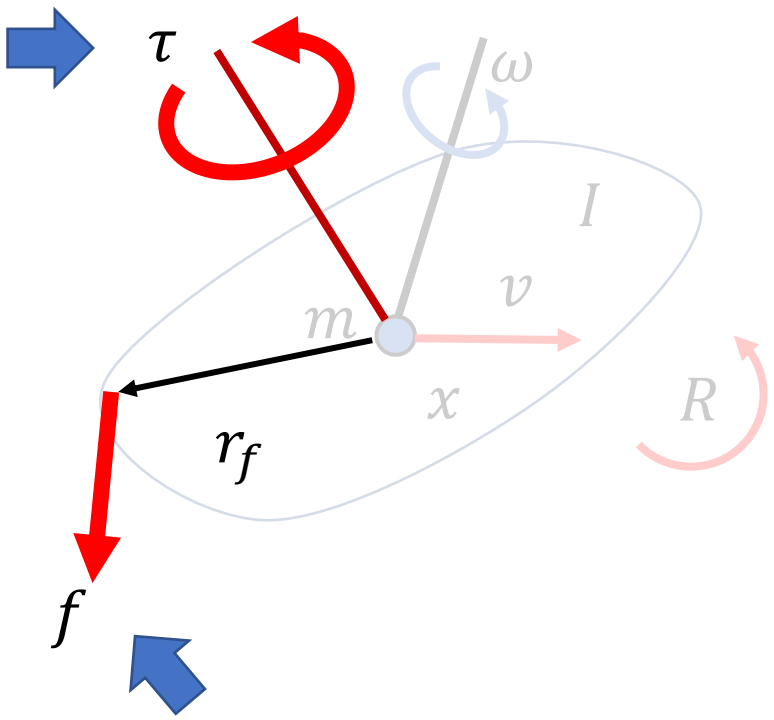
Actuating a Rigid Body



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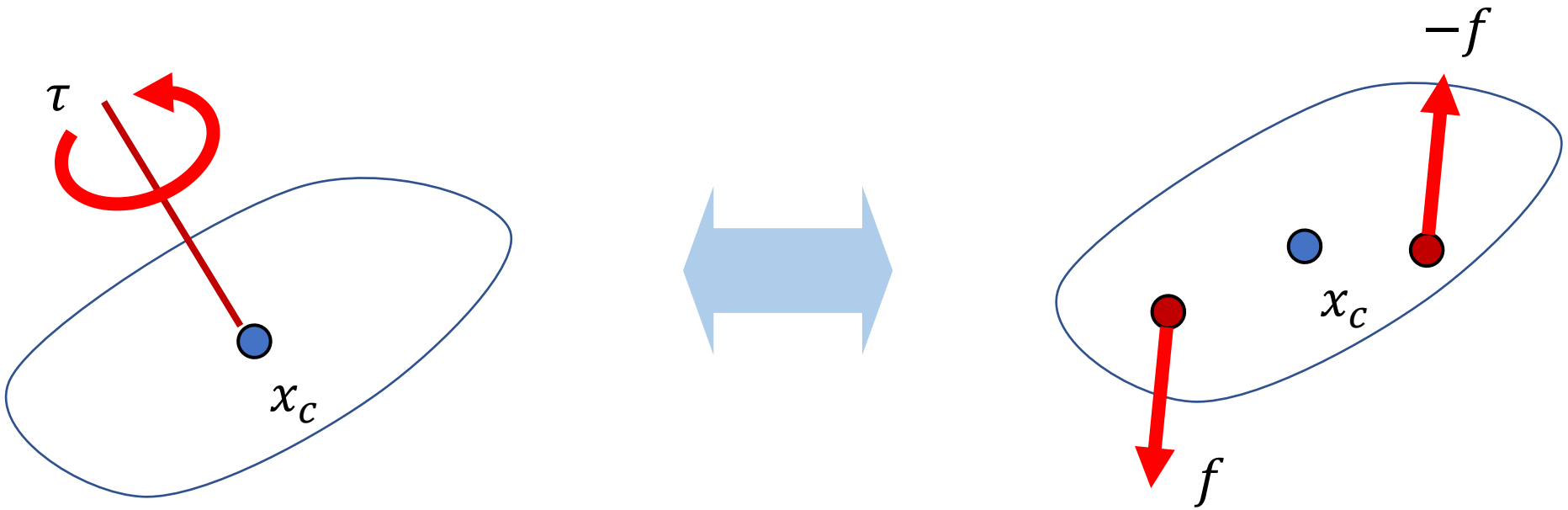
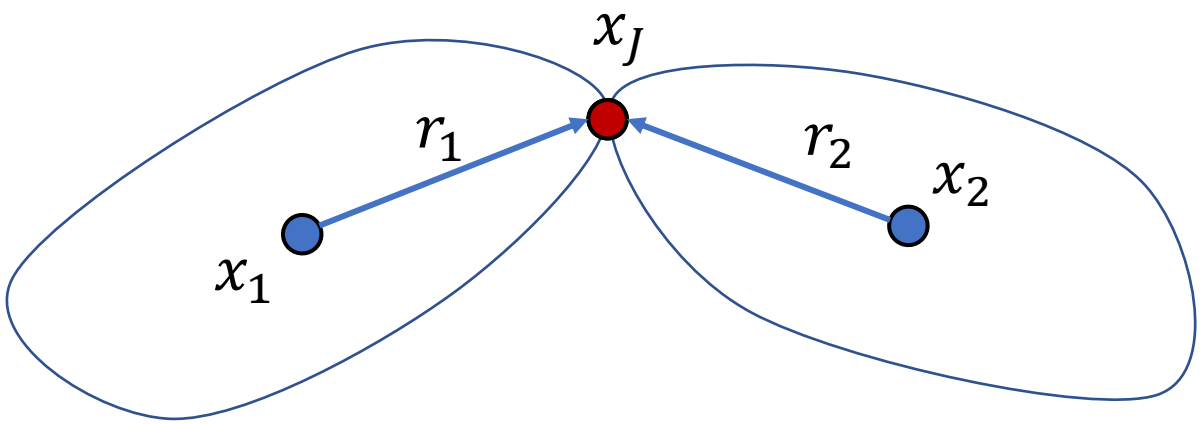
Actuating a Rigid Body



Actuating Articulated Rigid Bodies

$m_i, I_i, x_i, R_i, v_i, \omega_i$



$$M\dot{\boldsymbol{v}} + C(\boldsymbol{x}, \boldsymbol{v}) = \boldsymbol{f} + \boldsymbol{J}^{\boldsymbol{T}}\boldsymbol{\lambda}$$

$$Jv = 0$$

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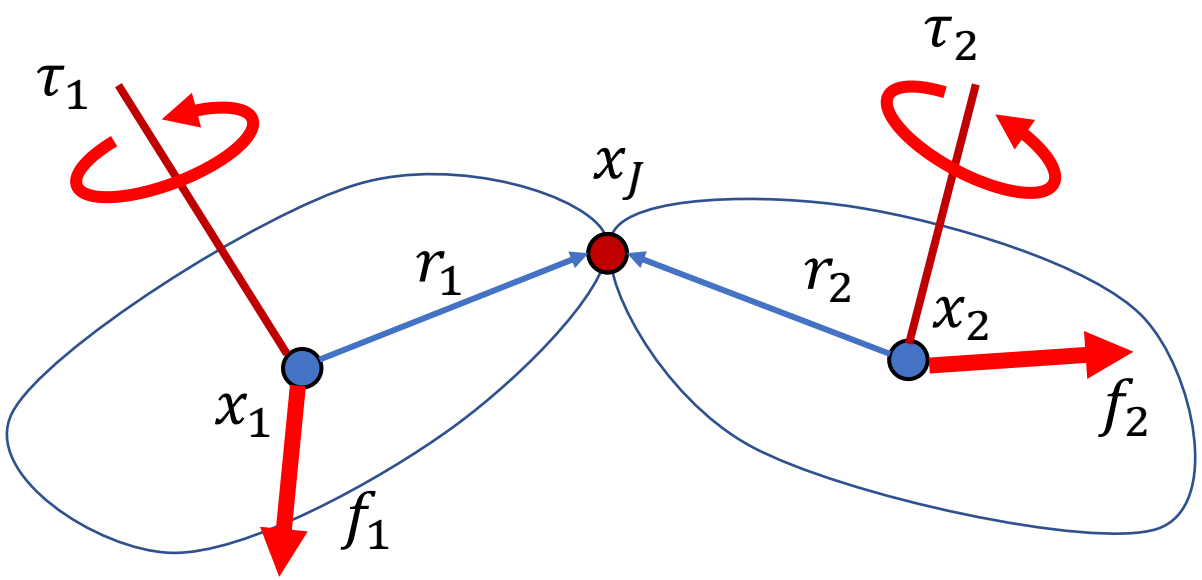

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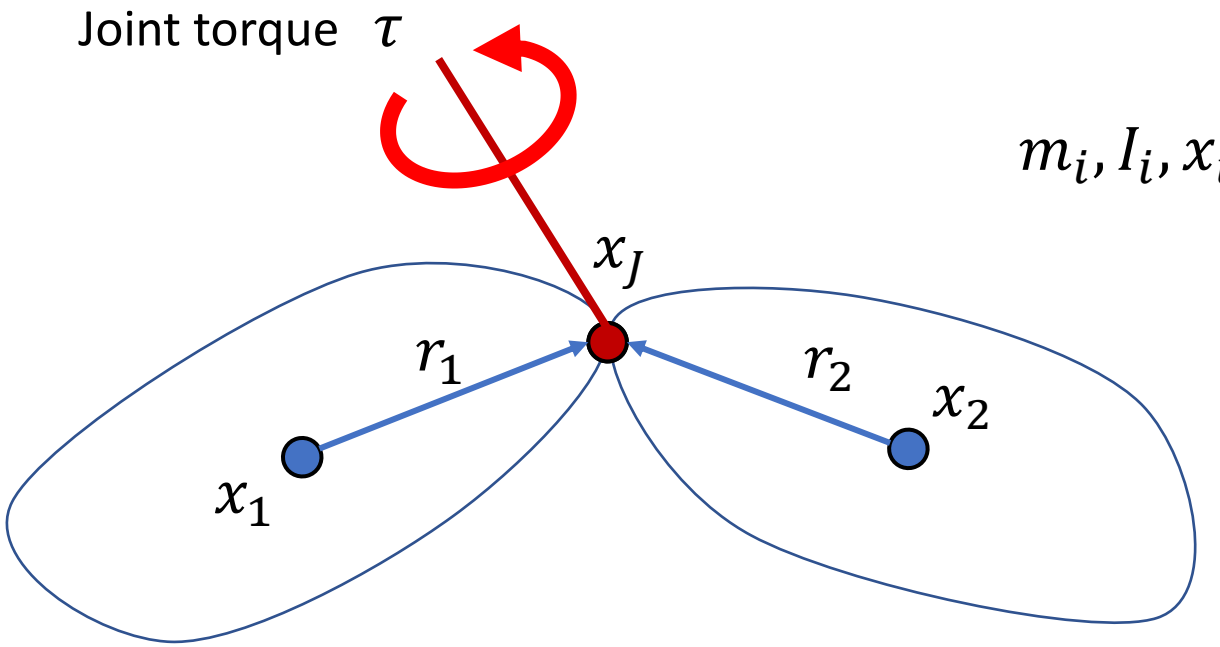

$$m_i, I_i, x_i, R_i, v_i, \omega_i$$

$$M\dot{\boldsymbol{v}} + C(\boldsymbol{x}, \boldsymbol{v}) = \boldsymbol{f} + \boldsymbol{J}^{\boldsymbol{T}}\boldsymbol{\lambda}$$

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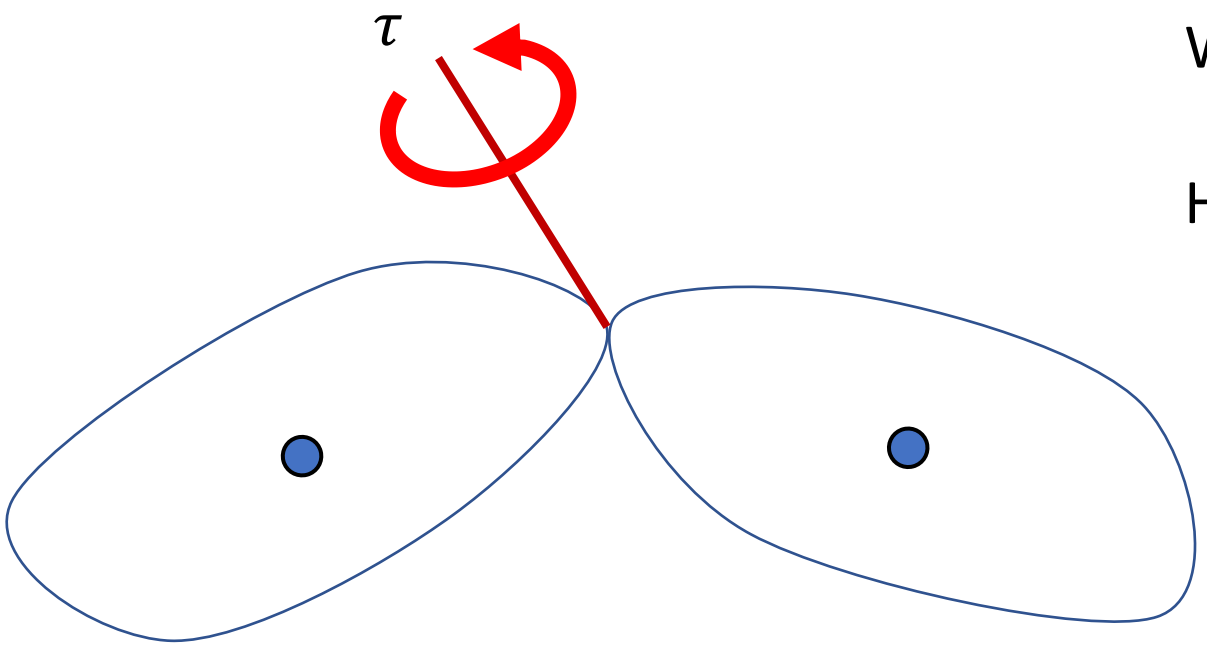
Joint Torques



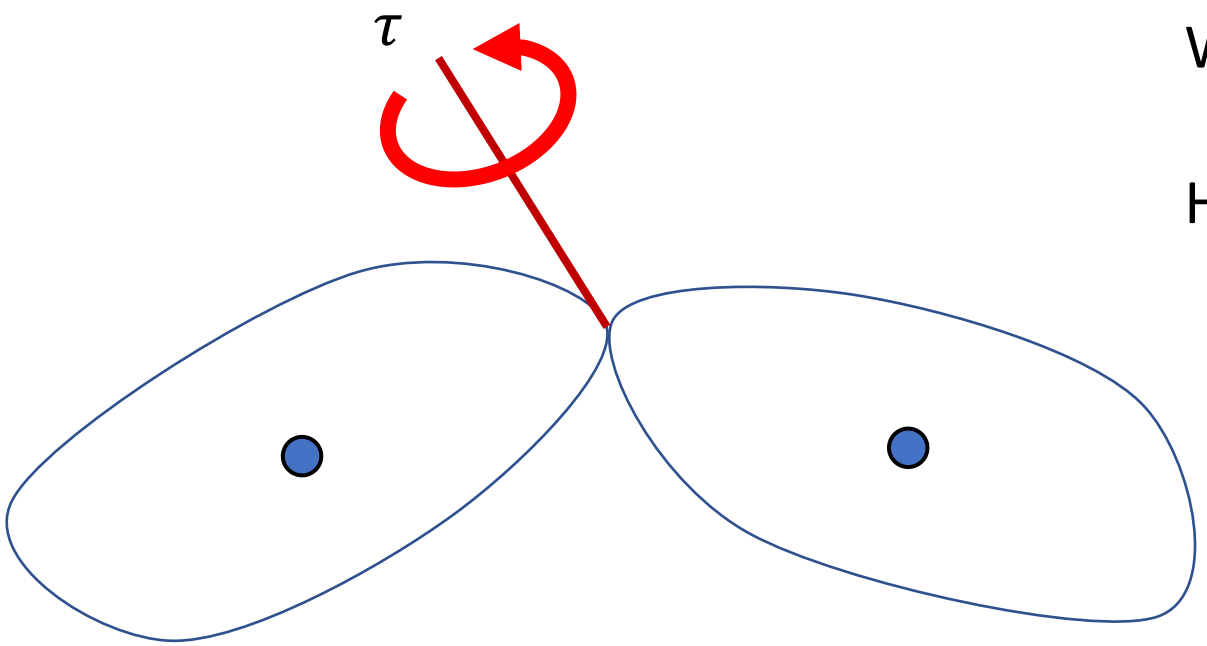
What is a joint torque?

How to apply a joint torque?

$$M\dot{\boldsymbol{v}} + C(\boldsymbol{x}, \boldsymbol{v}) = \boldsymbol{f} + \boldsymbol{J}^{\boldsymbol{T}}\boldsymbol{\lambda}$$

$$Jv = 0$$

Joint Torques



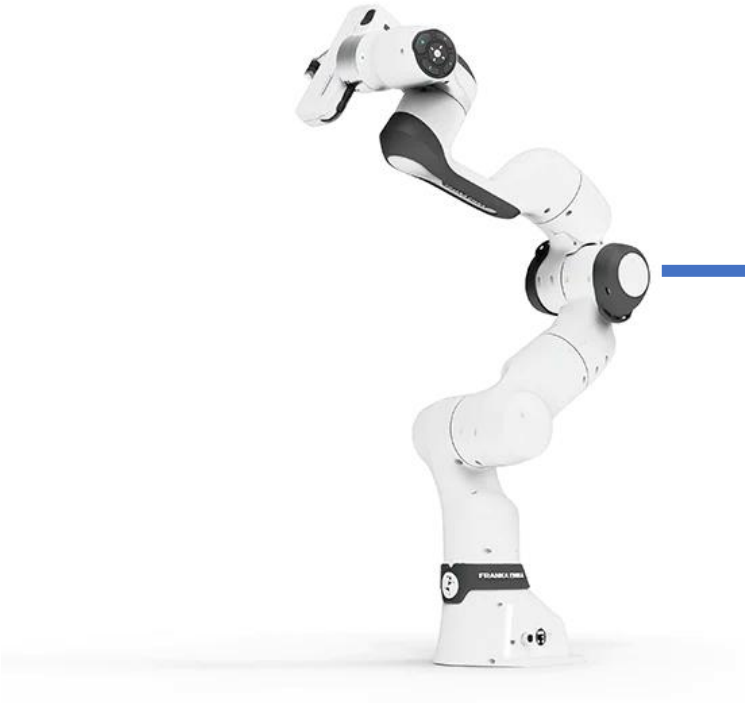
What is a joint torque?

How to apply a joint torque?

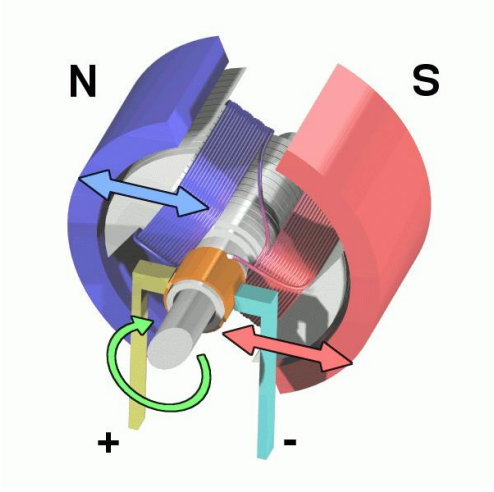
$$\begin{bmatrix} m_1 \mathrm{I}_3 & & & \\ & I_1 & & \\ & & m_2 \mathrm{I}_3 & \\ & & & I_2 \end{bmatrix} \begin{bmatrix} \dot{v}_1 \\ \dot{\omega}_1 \\ \dot{v}_2 \\ \dot{\omega}_2 \end{bmatrix} + \begin{bmatrix} 0 \\ \omega_1 \times I_1 \omega_1 \\ 0 \\ \omega_2 \times I_2 \omega_2 \end{bmatrix} = \begin{bmatrix} f_1 \\ \tau_1 \\ f_2 \\ \tau_2 \end{bmatrix} + J^T \lambda$$

$$Jv = 0$$

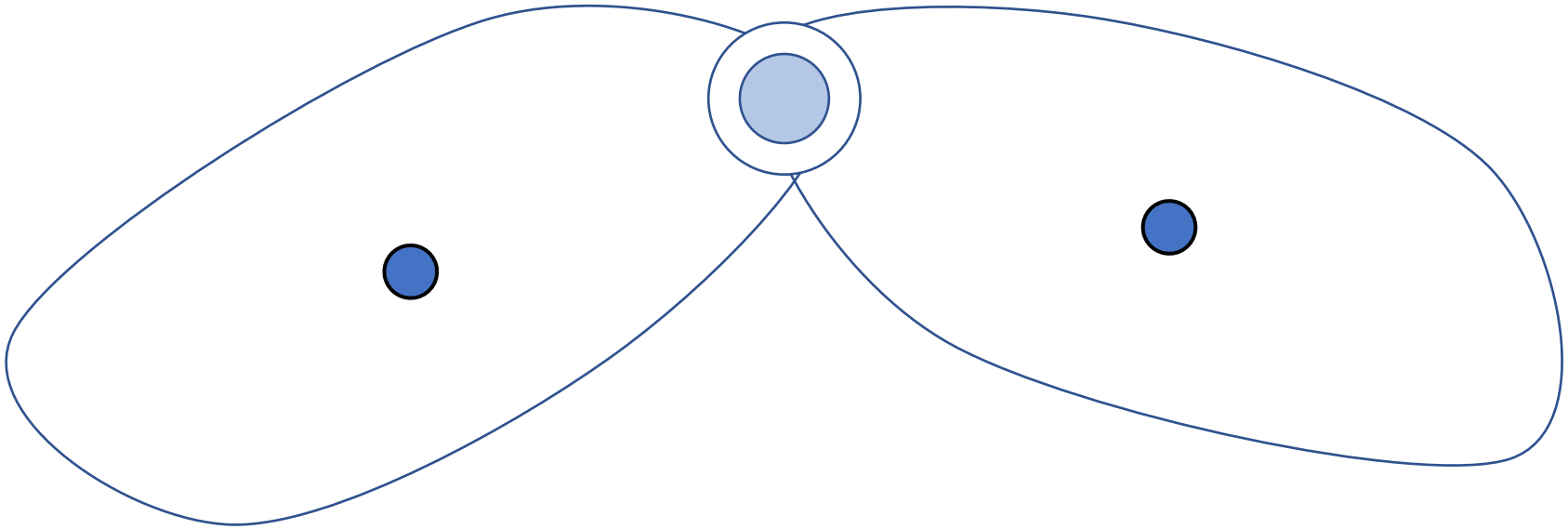
Joint Torques



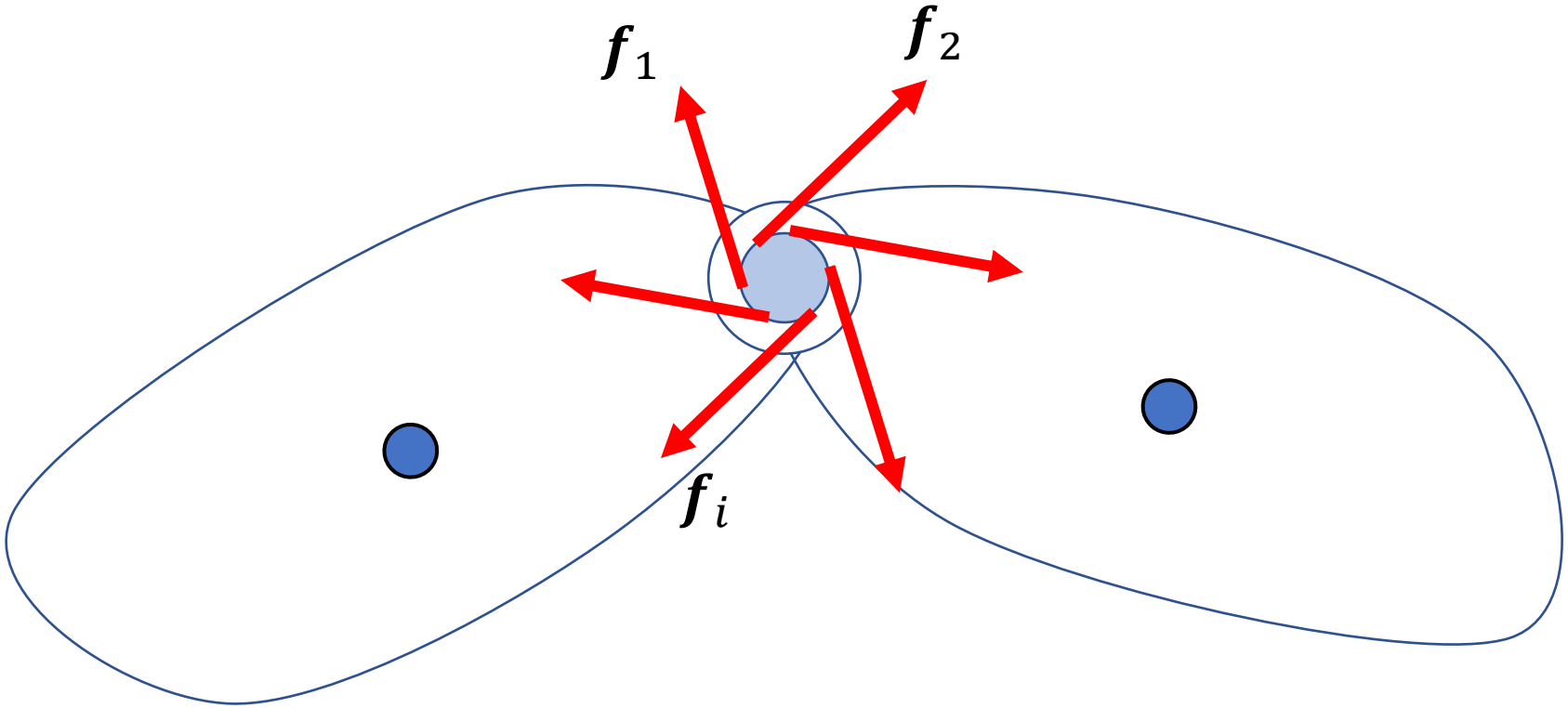
Franka



Joint Torques



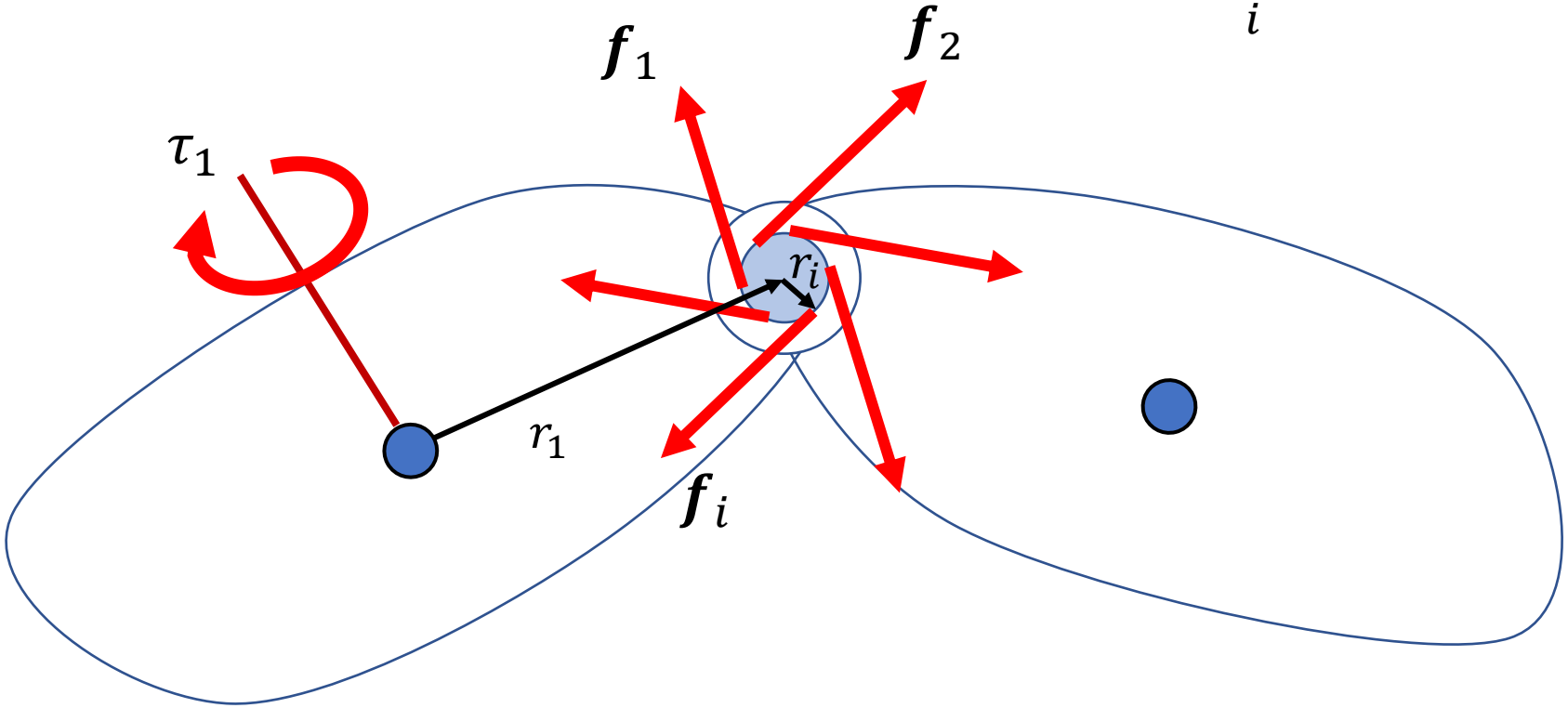
Joint Torques



$$\sum_i f_i = 0$$

Joint Torques

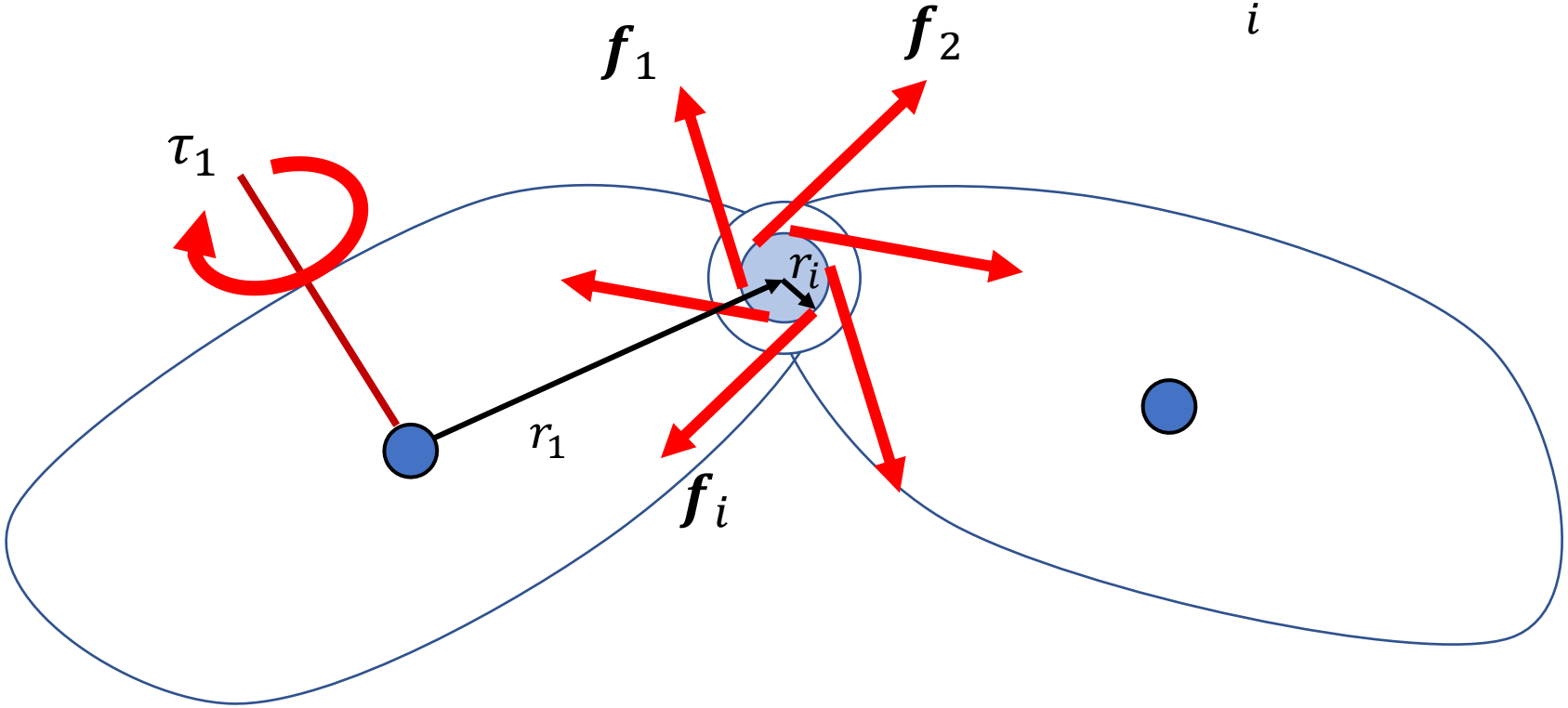
$$\sum_i f_i = 0$$



$$\tau_1 = \sum_i (r_1 + r_i) \times f_i = r_1 \times \sum_i f_i + \sum_i r_i \times f_i$$

Joint Torques

$$\sum_i f_i = 0$$



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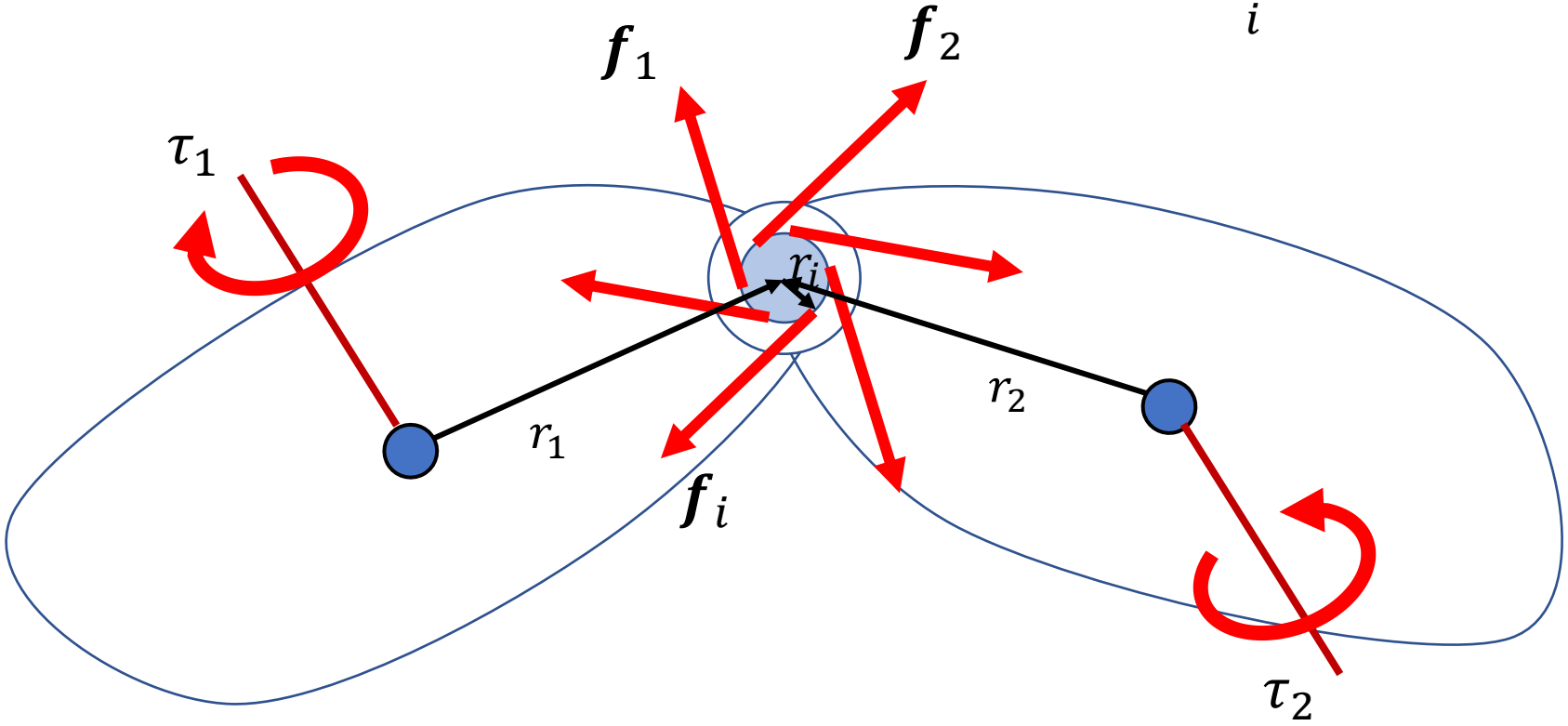
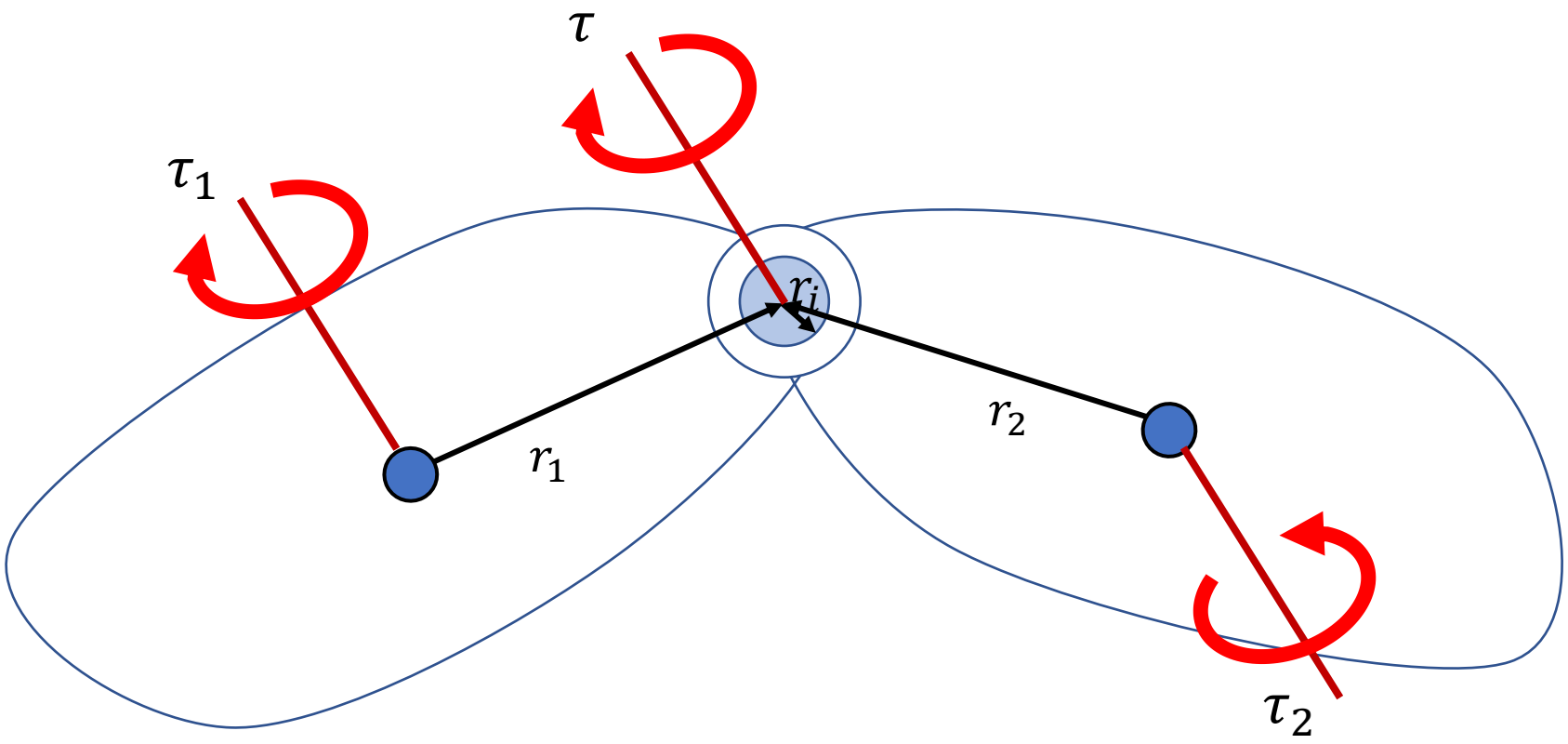


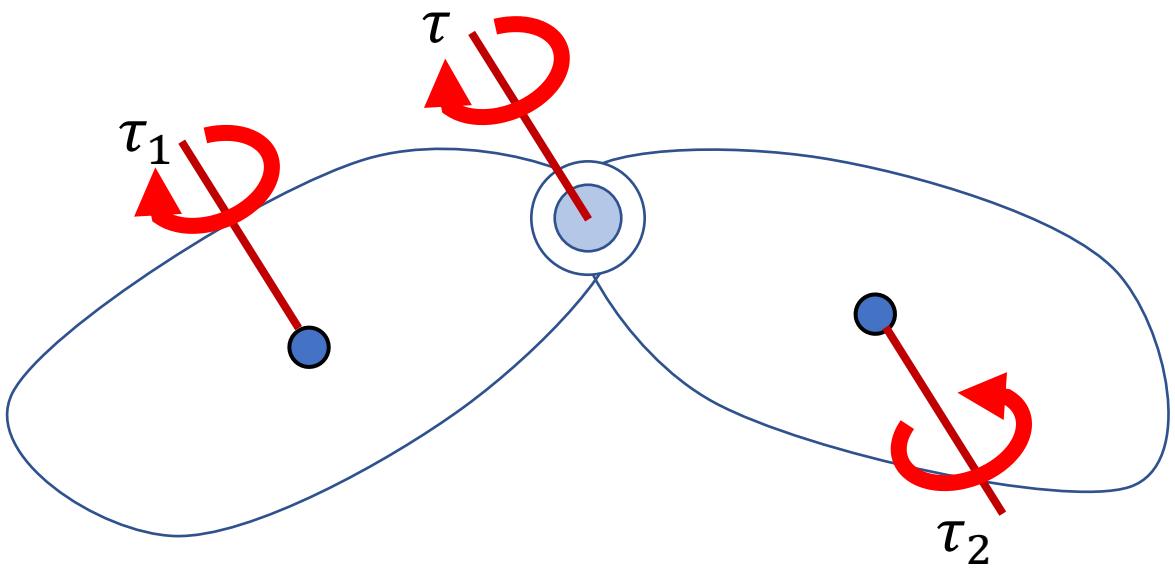
$$\tau_1 = \sum_i r_i \times f_i \qquad \tau_2 = -\sum_i r_i \times f_i$$

Joint Torques



$\tau_1 = \tau$ $\tau_2 = -\tau$

Joint Torques



$\tau_1 = \tau$

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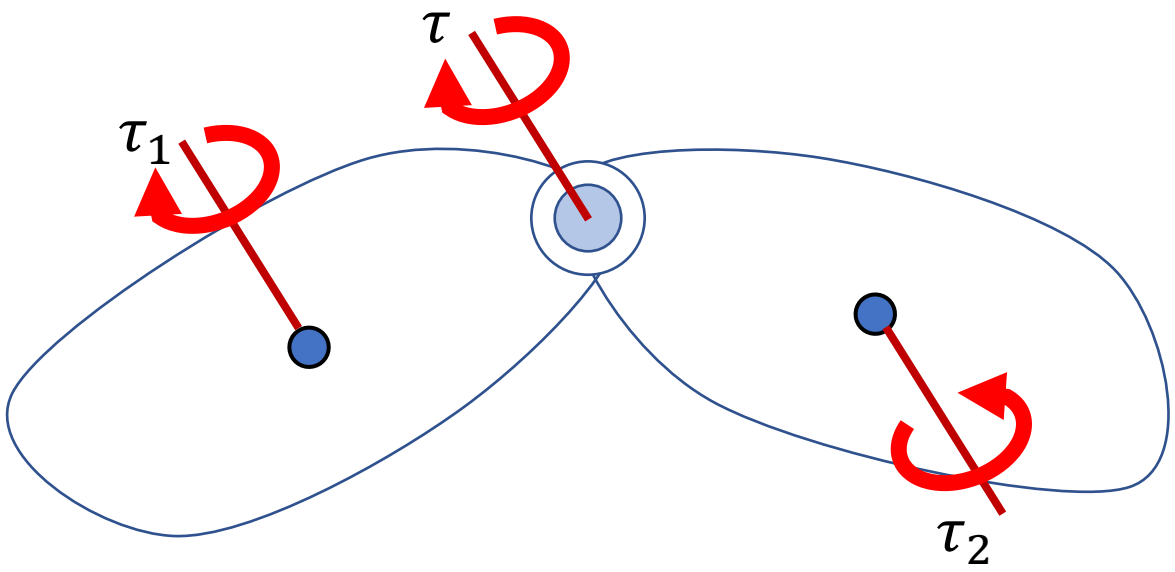
Applying a joint torque τ:

- Add τ to one attached body
- Add $-\tau$ to the other attached body

Joint Torques



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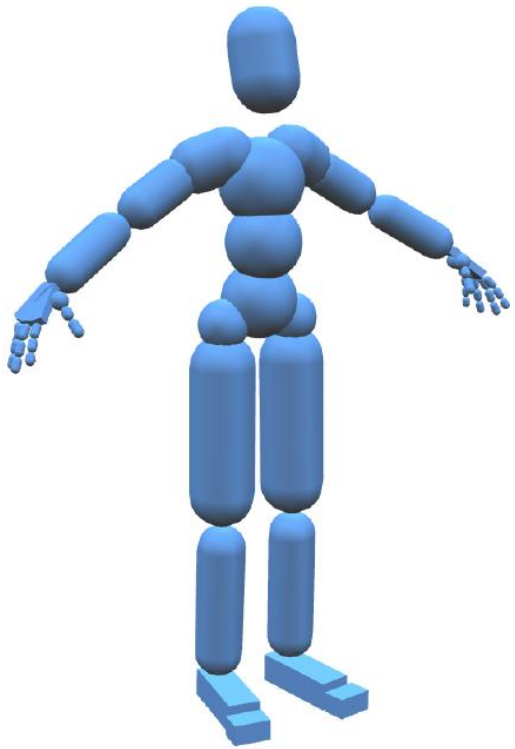
- Add τ to one attached body
- Add $-\tau$ to the other attached body

$$M \begin{bmatrix} \dot{v}_1 \\ \dot{\omega}_1 \\ \dot{v}_2 \\ \dot{\omega}_2 \end{bmatrix} + \begin{bmatrix} 0 \\ \omega_1 \times I_1 \omega_1 \\ 0 \\ \omega_2 \times I_2 \omega_2 \end{bmatrix} = \begin{bmatrix} 0 \\ \tau \\ 0 \\ -\tau \end{bmatrix} + J^T \lambda$$

$$Jv = 0$$

Simulating a Character

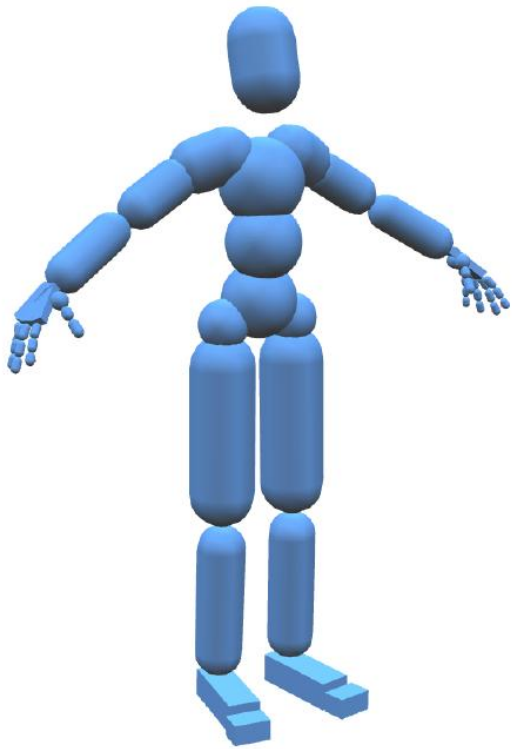


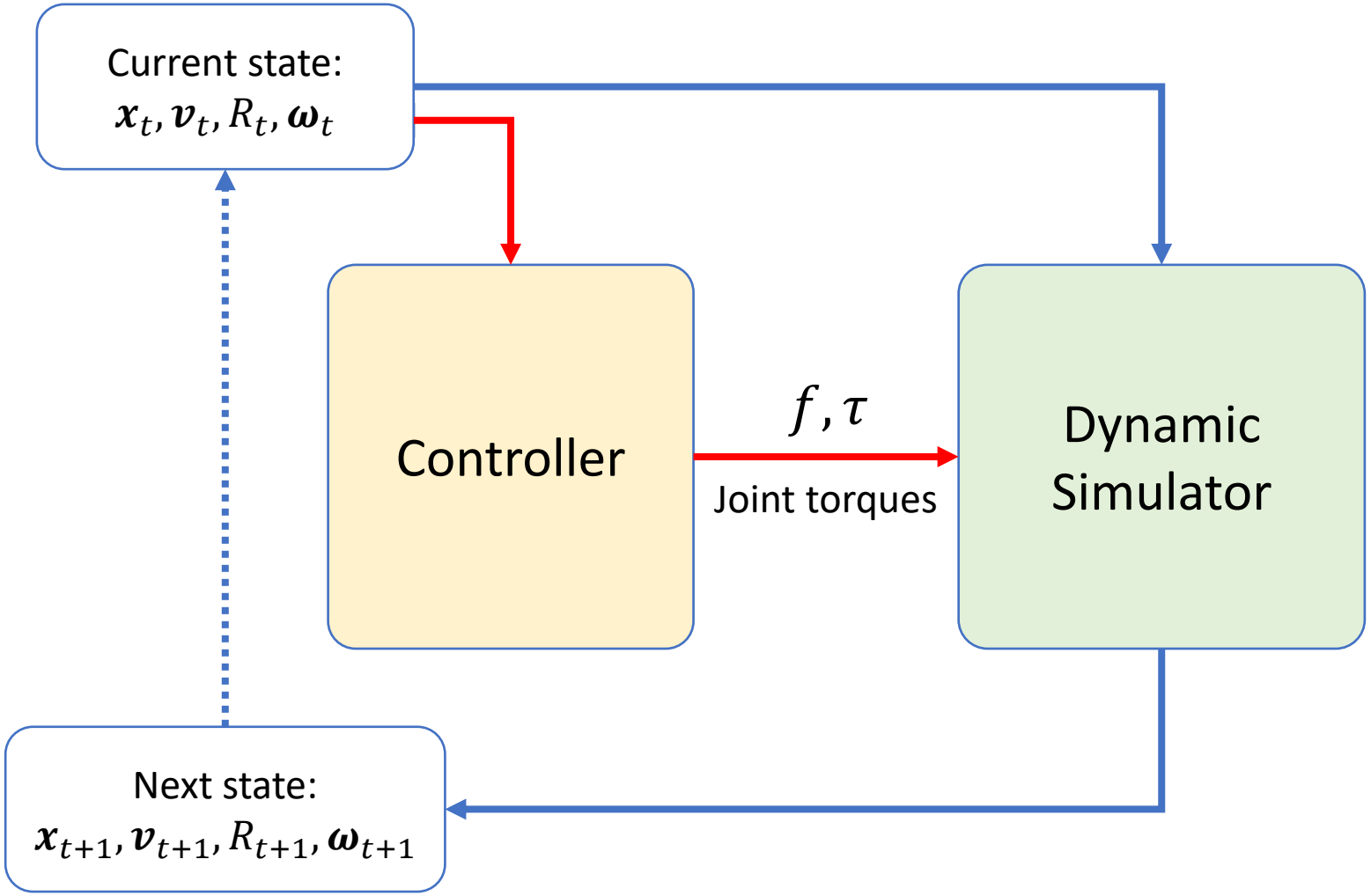
Current state:
$\boldsymbol{x}_t, \boldsymbol{v}_t, R_t, \boldsymbol{\omega}_t$

Dynamic Simulator

Next state:
$\boldsymbol{x}_{t+1}, \boldsymbol{v}_{t+1}, R_{t+1}, \boldsymbol{\omega}_{t+1}$

Simulating + Controlling a Character





Forward Dynamics vs. Inverse Dynamics

Equations of motion of the system:

$$M\dot{\boldsymbol{v}} + C(\boldsymbol{x}, \boldsymbol{v}) = \boldsymbol{f} + J^T\lambda \qquad Jv = 0$$

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Forward dynamics: $[\boldsymbol{x}, \boldsymbol{v}, \boldsymbol{f}] \mapsto \dot{\boldsymbol{v}}$

given a set of force/torques, compute the motion

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Equations of motion of the system:

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given a set of force/torques, compute the motion

Inverse dynamics: $[\boldsymbol{x}, \boldsymbol{v}, \dot{\boldsymbol{v}}] \mapsto \boldsymbol{f}$

given a motion, compute the forces/torques that give rise to it

Forward Dynamics vs. Inverse Dynamics

Equations of motion of the system:

$$M\dot{\boldsymbol{v}} + C(\boldsymbol{x}, \boldsymbol{v}) = \boldsymbol{f} + J^T\lambda \qquad Jv = 0$$

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dynamic simulator

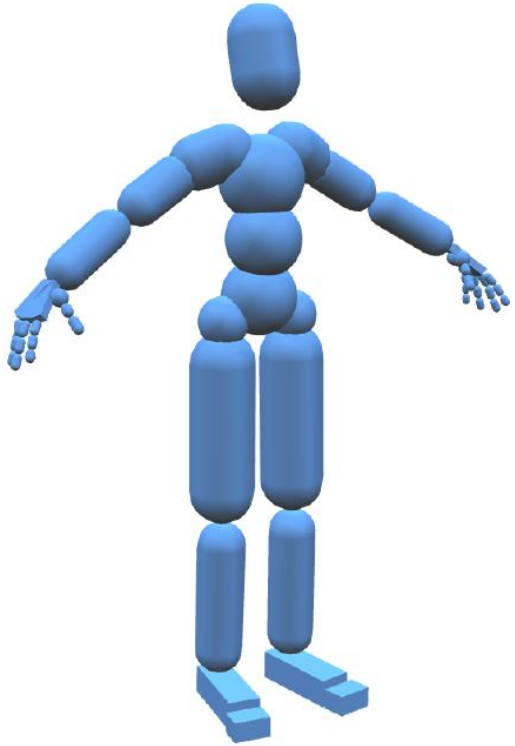
given a set of force/torques, compute the motion

Inverse dynamics: $[\boldsymbol{x}, \boldsymbol{v}, \dot{\boldsymbol{v}}] \mapsto \boldsymbol{f}$

controller

given a motion, compute the forces/torques that give rise to it

Simulating + Control



If #actuators < #dofs,
the system is **underactuated**

Current state:
$\boldsymbol{x}_t, \boldsymbol{v}_t, R_t, \boldsymbol{\omega}_t$

Controller

f, τ

Joint torques

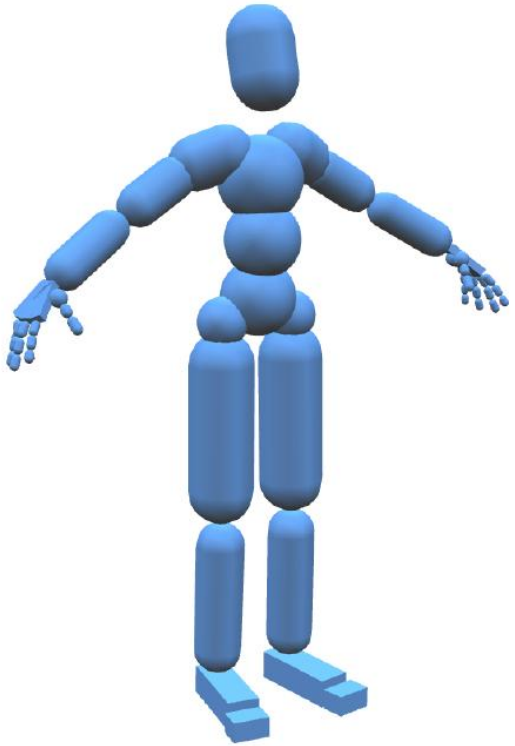
Dynamic Simulator

Next state:
$\boldsymbol{x}_{t+1}, \boldsymbol{v}_{t+1}, R_{t+1}, \boldsymbol{\omega}_{t+1}$

Fully-Actuated vs. Underactuated

Fully-Actuated

Underactuated



欠驱动系统

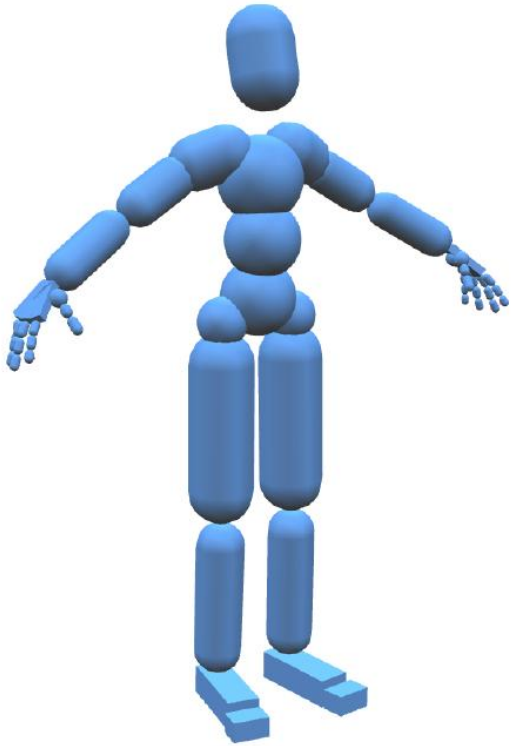
If #actuators ≥ #dofs, the system is **fully-actuated**

If #actuators < #dofs, the system is **underactuated**

Fully-Actuated vs. Underactuated

Fully-Actuated

Underactuated

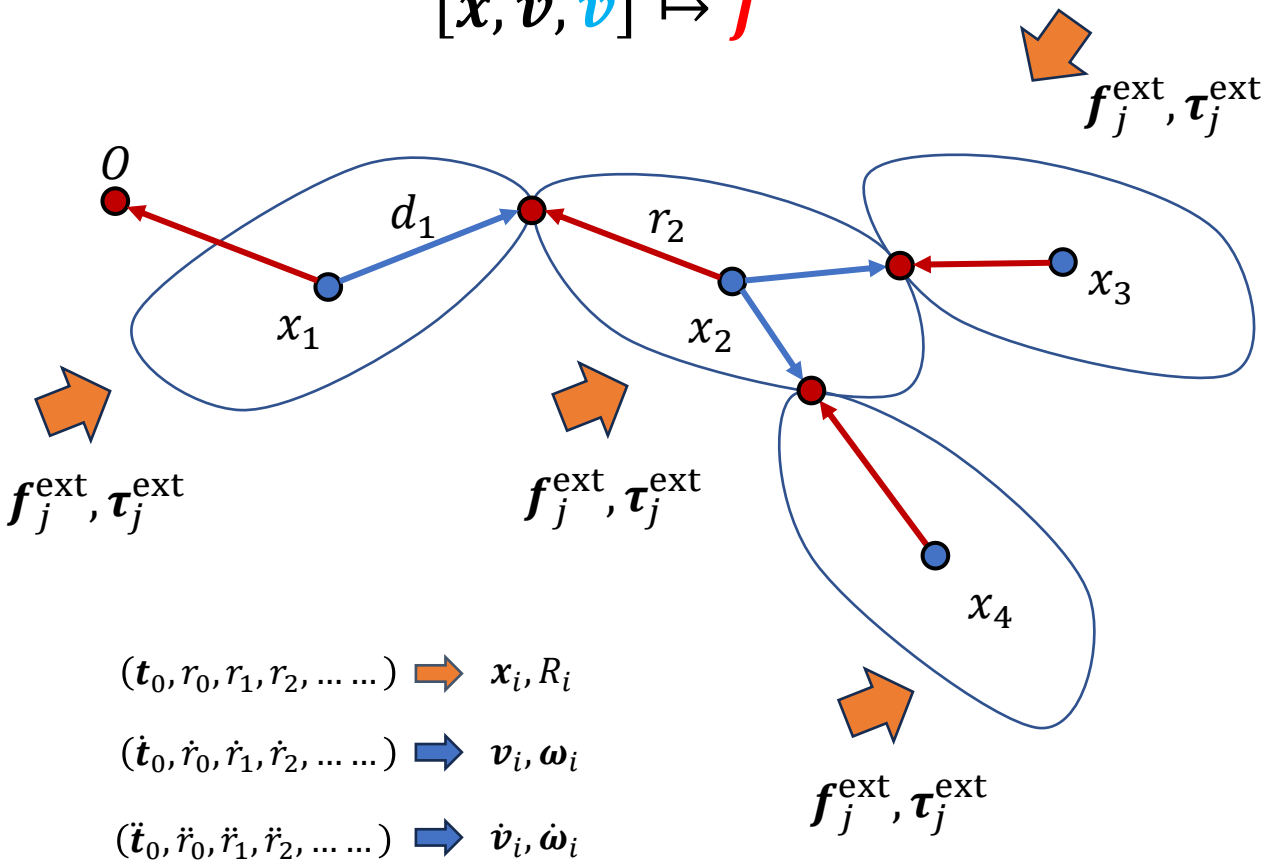


For any $[\boldsymbol{x}, \boldsymbol{v}, \dot{\boldsymbol{v}}]$, there exists an $\boldsymbol{f}$ that produces the motion

For many $[\boldsymbol{x}, \boldsymbol{v}, \dot{\boldsymbol{v}}]$, there is no such $\boldsymbol{f}$ that produces the motion

Inverse Dynamics

$$[\boldsymbol{x}, \boldsymbol{v}, \dot{\boldsymbol{v}}] \mapsto \boldsymbol{f}$$



$(\boldsymbol{t}_0, r_0, r_1, r_2, \dots\dots) \Rightarrow \boldsymbol{x}_i, R_i$

$(\dot{\boldsymbol{t}}_0, \dot{r}_0, \dot{r}_1, \dot{r}_2, \dots\dots) \Rightarrow \boldsymbol{v}_i, \boldsymbol{\omega}_i$

$(\ddot{\boldsymbol{t}}_0, \ddot{r}_0, \ddot{r}_1, \ddot{r}_2, \dots\dots) \Rightarrow \dot{\boldsymbol{v}}_i, \dot{\boldsymbol{\omega}}_i$

Given: $m_i, I_i, \boldsymbol{x}_i, R_i, \boldsymbol{v}_i, \boldsymbol{\omega}_i$

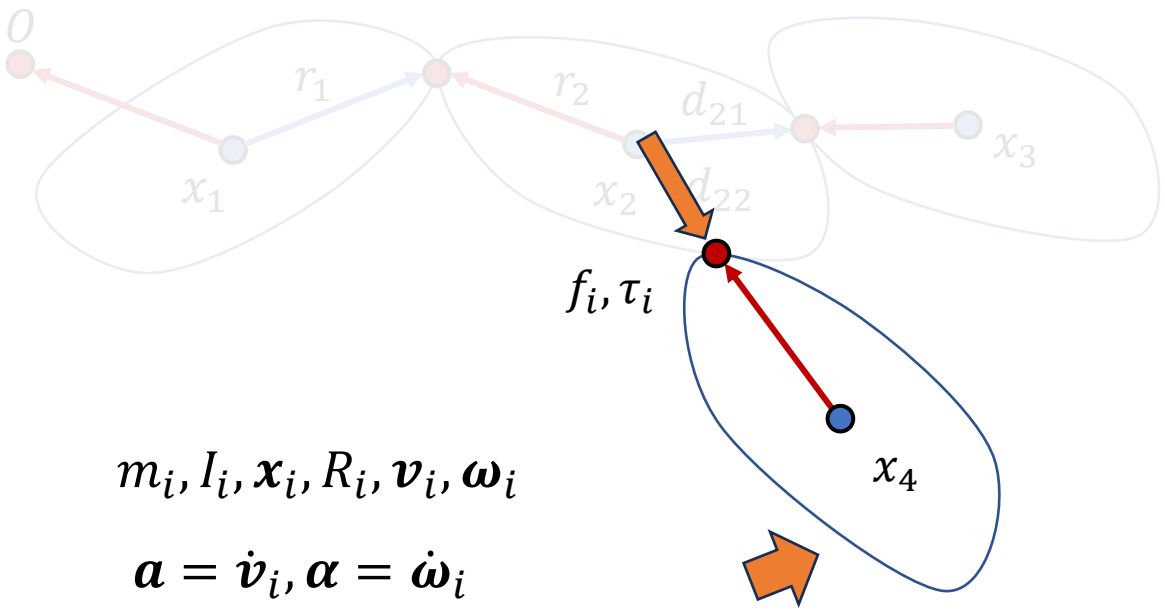
$\boldsymbol{a} = \dot{\boldsymbol{v}}_i, \boldsymbol{\alpha} = \dot{\boldsymbol{\omega}}_i$

$\boldsymbol{f}_j^{\text{ext}}, \boldsymbol{\tau}_j^{\text{ext}}$

Compute: $\boldsymbol{f}_k, \boldsymbol{\tau}_k$

Inverse Dynamics

$$[\boldsymbol{x}, \boldsymbol{v}, \dot{\boldsymbol{v}}] \mapsto \boldsymbol{f}$$



$m_i, I_i, \boldsymbol{x}_i, R_i, \boldsymbol{v}_i, \boldsymbol{\omega}_i$

$\boldsymbol{a} = \dot{\boldsymbol{v}}_i, \boldsymbol{\alpha} = \dot{\boldsymbol{\omega}}_i$

$\boldsymbol{f}_j^{\text{ext}}, \boldsymbol{\tau}_j^{\text{ext}}$

$\boldsymbol{f}_j^{\text{ext}}, \boldsymbol{\tau}_j^{\text{ext}}$

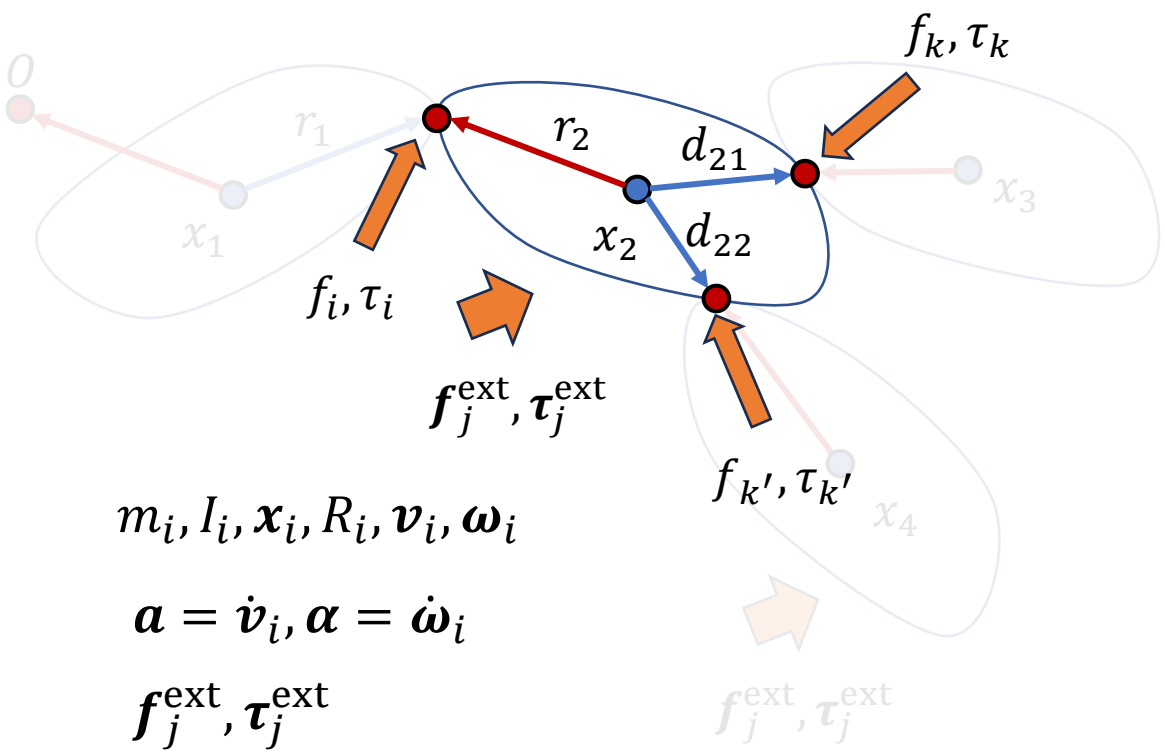
Recursive Newton Euler Algorithm (RNEA)

$$m_i \dot{\boldsymbol{v}}_i = \boldsymbol{f}_i + \sum_j \boldsymbol{f}_j^{\text{ext}}$$

$$I_i \dot{\boldsymbol{\omega}}_i + \boldsymbol{\omega}_i \times I_i \boldsymbol{\omega}_i = \boldsymbol{\tau}_i + \boldsymbol{r}_i \times \boldsymbol{f}_i + \sum_j \boldsymbol{\tau}_j^{\text{ext}}$$

Inverse Dynamics

$$[\boldsymbol{x}, \boldsymbol{v}, \dot{\boldsymbol{v}}] \mapsto \boldsymbol{f}$$



$m_i, I_i, \boldsymbol{x}_i, R_i, \boldsymbol{v}_i, \boldsymbol{\omega}_i$

$\boldsymbol{a} = \dot{\boldsymbol{v}}_i, \boldsymbol{\alpha} = \dot{\boldsymbol{\omega}}_i$

$\boldsymbol{f}_j^{\text{ext}}, \boldsymbol{\tau}_j^{\text{ext}}$

Recursive Newton Euler Algorithm (RNEA)

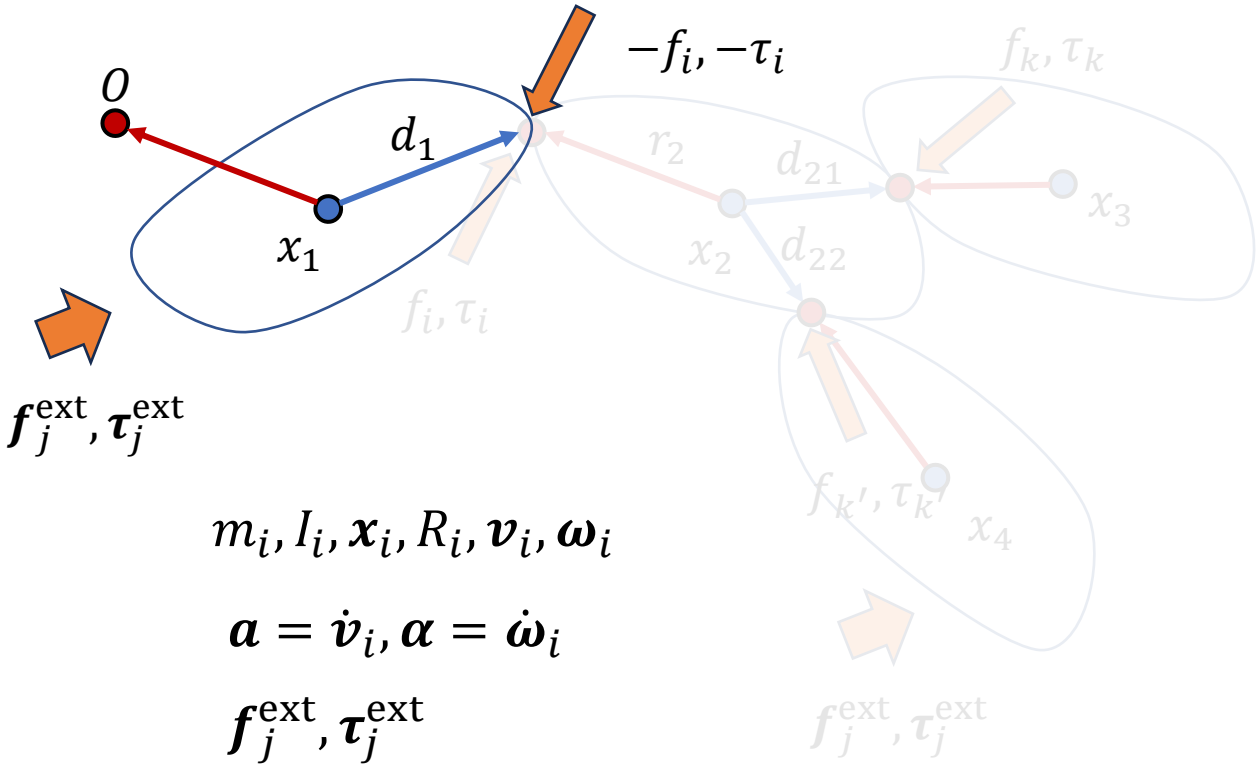
$$m_i \dot{\boldsymbol{v}}_i = \boldsymbol{f}_i - \sum_{k \in c(i)} \boldsymbol{f}_k + \sum_j \boldsymbol{f}_j^{\text{ext}}$$

$$I_i \dot{\boldsymbol{\omega}}_i + \boldsymbol{\omega}_i \times I_i \boldsymbol{\omega}_i = \boldsymbol{\tau}_i + \boldsymbol{r}_i \times \boldsymbol{f}_i$$

$$- \sum_{k \in c(i)} (\boldsymbol{\tau}_k + \boldsymbol{d}_{ik} \times \boldsymbol{f}_k) + \sum_j \boldsymbol{\tau}_j^{\text{ext}}$$

Inverse Dynamics

$$[\boldsymbol{x}, \boldsymbol{v}, \dot{\boldsymbol{v}}] \mapsto \boldsymbol{f}$$



$m_i, I_i, \boldsymbol{x}_i, R_i, \boldsymbol{v}_i, \boldsymbol{\omega}_i$

$\boldsymbol{a} = \dot{\boldsymbol{v}}_i, \boldsymbol{\alpha} = \dot{\boldsymbol{\omega}}_i$

$\boldsymbol{f}_j^{\text{ext}}, \boldsymbol{\tau}_j^{\text{ext}}$

Recursive Newton Euler Algorithm (RNEA)

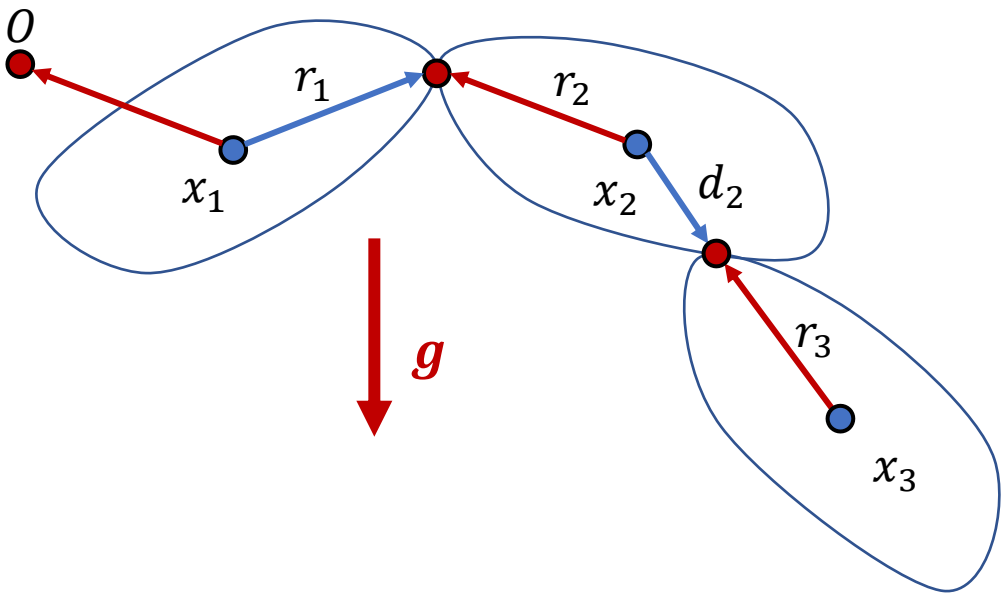
$$m_i \dot{\boldsymbol{v}}_i = \boldsymbol{f}_i - \sum_{k \in c(i)} \boldsymbol{f}_k + \sum_j \boldsymbol{f}_j^{\text{ext}}$$

$$I_i \dot{\boldsymbol{\omega}}_i + \boldsymbol{\omega}_i \times I_i \boldsymbol{\omega}_i = \boldsymbol{\tau}_i + \boldsymbol{r}_i \times \boldsymbol{f}_i$$

$$- \sum_{k \in c(i)} (\boldsymbol{\tau}_k + \boldsymbol{d}_{ik} \times \boldsymbol{f}_k) + \sum_j \boldsymbol{\tau}_j^{\text{ext}}$$

Example: Gravity Compensation

$$[\boldsymbol{x}, \mathbf{0}, \mathbf{0}] \mapsto \boldsymbol{f}$$



$(\boldsymbol{t}_0, r_0, r_1, r_2, \dots\dots) \Rightarrow \boldsymbol{x}_i, R_i$

$(\dot{\boldsymbol{t}}_0, \dot{r}_0, \dot{r}_1, \dot{r}_2, \dots\dots) \Rightarrow \boldsymbol{v}_i, \boldsymbol{\omega}_i$

Given: $m_i, I_i, \boldsymbol{x}_i, R_i, \boldsymbol{v}_i = 0, \boldsymbol{\omega}_i = 0$

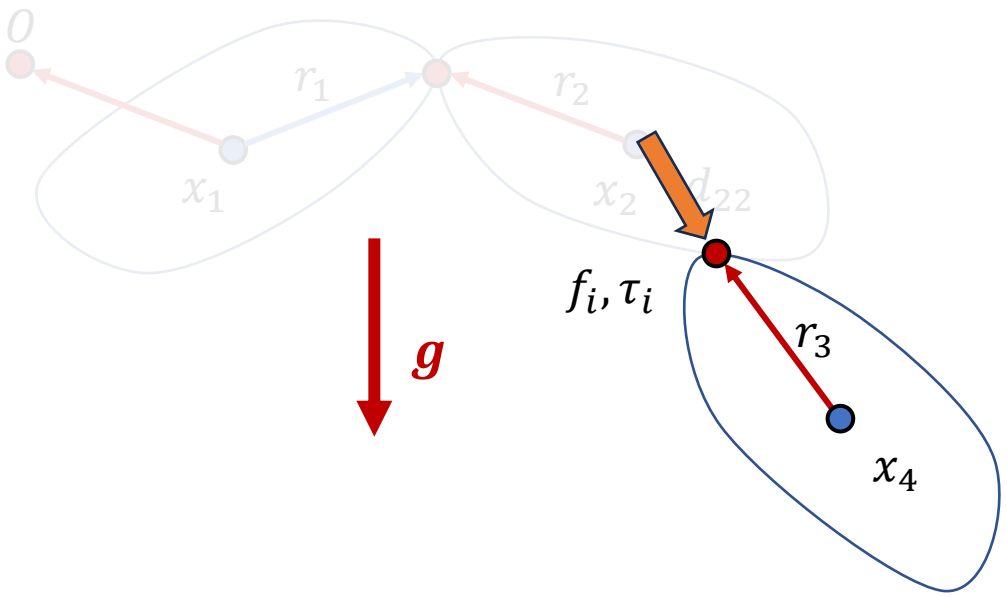
$\boldsymbol{a} = 0, \boldsymbol{\alpha} = 0$

Gravity: $\boldsymbol{g}$

Compute: $\boldsymbol{f}_k, \boldsymbol{\tau}_k$

Example: Gravity Compensation

$$[\boldsymbol{x}, \mathbf{0}, \mathbf{0}] \mapsto \boldsymbol{f}$$



$(\boldsymbol{t}_0, r_0, r_1, r_2, \dots\dots) \Rightarrow \boldsymbol{x}_i, R_i$

$(\dot{\boldsymbol{t}}_0, \dot{r}_0, \dot{r}_1, \dot{r}_2, \dots\dots) \Rightarrow \boldsymbol{v}_i, \boldsymbol{\omega}_i$

Given: $m_i, I_i, \boldsymbol{x}_i, R_i, \boldsymbol{v}_i = 0, \boldsymbol{\omega}_i = 0$

$\boldsymbol{a} = 0, \boldsymbol{\alpha} = 0$

Gravity: $\boldsymbol{g}$

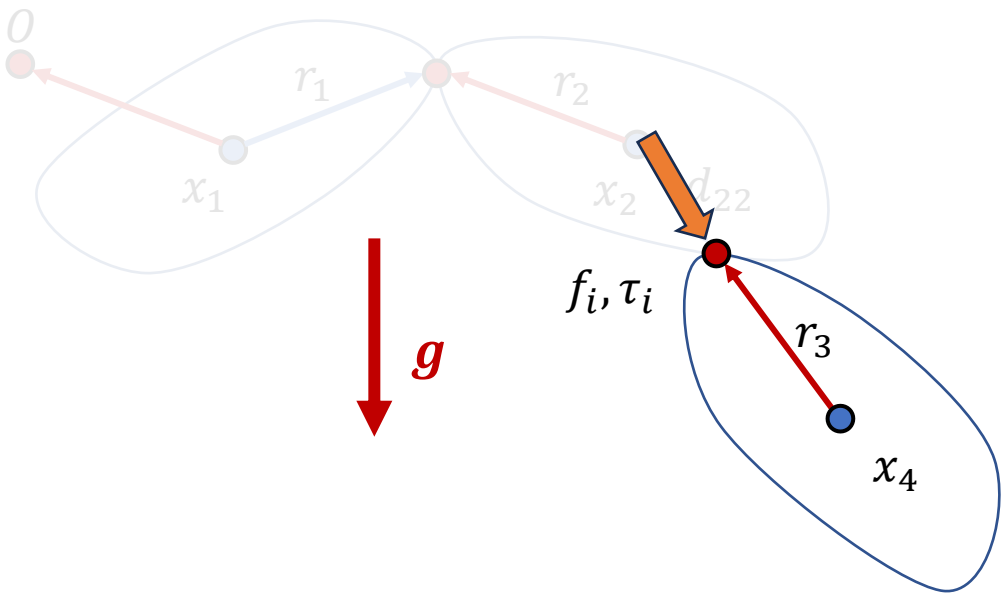
Compute: $\boldsymbol{f}_k, \boldsymbol{\tau}_k$

$$\boldsymbol{f}_3 = -m_3 \boldsymbol{g}$$

$$m_i \dot{\boldsymbol{v}}_i = \boldsymbol{f}_i + \sum_j \boldsymbol{f}_j^{\text{ext}}$$

Example: Gravity Compensation

$$[\boldsymbol{x}, \mathbf{0}, \mathbf{0}] \mapsto \boldsymbol{f}$$



$(\boldsymbol{t}_0, r_0, r_1, r_2, \dots\dots) \Rightarrow \boldsymbol{x}_i, R_i$

$(\dot{\boldsymbol{t}}_0, \dot{r}_0, \dot{r}_1, \dot{r}_2, \dots\dots) \Rightarrow \boldsymbol{v}_i, \boldsymbol{\omega}_i$

Given: $m_i, I_i, \boldsymbol{x}_i, R_i, \boldsymbol{v}_i = 0, \boldsymbol{\omega}_i = 0$

$\boldsymbol{a} = 0, \boldsymbol{\alpha} = 0$

Gravity: $\boldsymbol{g}$

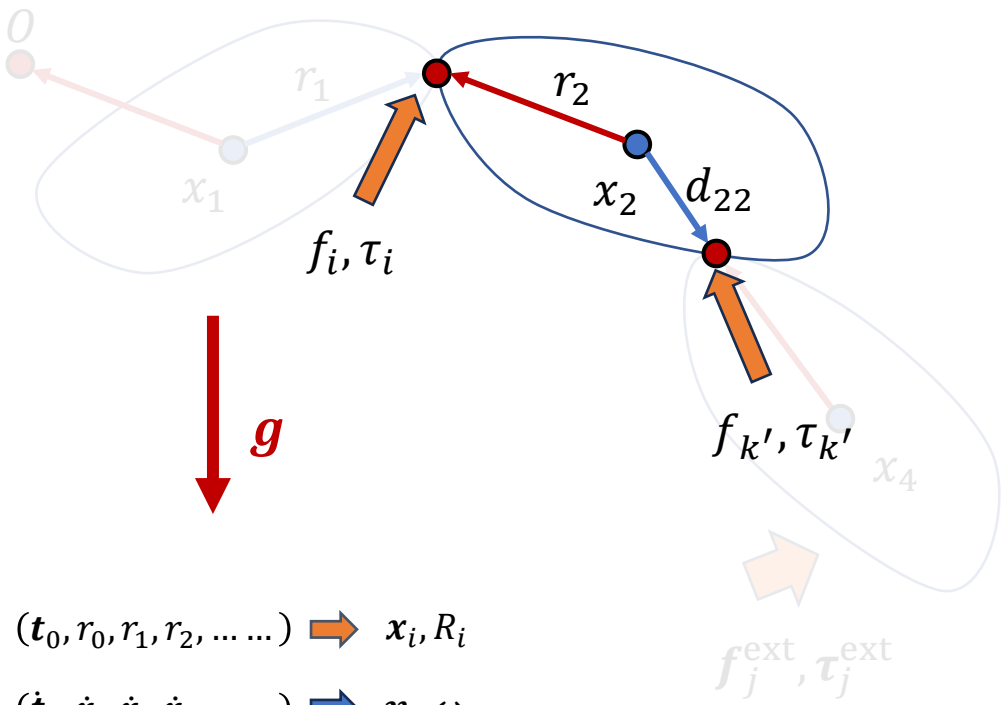
Compute: $\boldsymbol{f}_k, \boldsymbol{\tau}_k$

$$\boldsymbol{f}_3 = -m_3 \boldsymbol{g} \qquad \boldsymbol{\tau}_3 = \boldsymbol{r}_3 \times m_3 \boldsymbol{g}$$

$$I_i \dot{\boldsymbol{\omega}}_i + \boldsymbol{\omega}_i \times I_i \boldsymbol{\omega}_i = \boldsymbol{\tau}_i + \boldsymbol{r}_i \times \boldsymbol{f}_i + \sum_j \boldsymbol{\tau}_j^{\text{ext}}$$

Example: Gravity Compensation

$$[\boldsymbol{x}, \mathbf{0}, \mathbf{0}] \mapsto \boldsymbol{f}$$



$(\boldsymbol{t}_0, r_0, r_1, r_2, \dots\dots) \Rightarrow \boldsymbol{x}_i, R_i$

$(\dot{\boldsymbol{t}}_0, \dot{r}_0, \dot{r}_1, \dot{r}_2, \dots\dots) \Rightarrow \boldsymbol{v}_i, \boldsymbol{\omega}_i$

Given: $m_i, I_i, \boldsymbol{x}_i, R_i, \boldsymbol{v}_i = 0, \boldsymbol{\omega}_i = 0$

$\boldsymbol{a} = 0, \boldsymbol{\alpha} = 0$

Gravity: $\boldsymbol{g}$

Compute: $\boldsymbol{f}_k, \boldsymbol{\tau}_k$

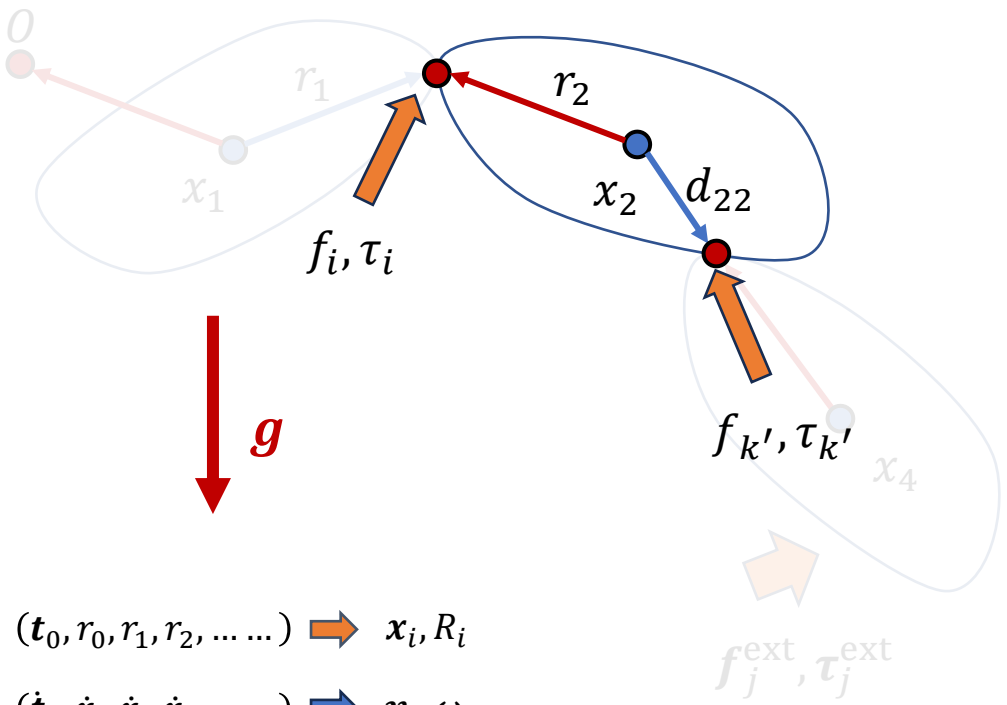
$$\boldsymbol{f}_3 = -m_3\boldsymbol{g} \qquad \boldsymbol{\tau}_3 = \boldsymbol{r}_3 \times m_3\boldsymbol{g}$$

$$\boldsymbol{f}_2 = -m_2\boldsymbol{g} - m_3\boldsymbol{g}$$

$$m_i\dot{\boldsymbol{v}}_i = \boldsymbol{f}_i - \sum_{k \in c(i)} \boldsymbol{f}_k + \sum_j \boldsymbol{f}_j^{\text{ext}}$$

Example: Gravity Compensation

$$[\boldsymbol{x}, \mathbf{0}, \mathbf{0}] \mapsto \boldsymbol{f}$$



$(\boldsymbol{t}_0, r_0, r_1, r_2, \dots\dots) \Rightarrow \boldsymbol{x}_i, R_i$

$(\dot{\boldsymbol{t}}_0, \dot{r}_0, \dot{r}_1, \dot{r}_2, \dots\dots) \Rightarrow \boldsymbol{v}_i, \boldsymbol{\omega}_i$

Given: $m_i, I_i, \boldsymbol{x}_i, R_i, \boldsymbol{v}_i = 0, \boldsymbol{\omega}_i = 0$

$\boldsymbol{a} = 0, \boldsymbol{\alpha} = 0$

Gravity: $\boldsymbol{g}$

Compute: $\boldsymbol{f}_k, \boldsymbol{\tau}_k$

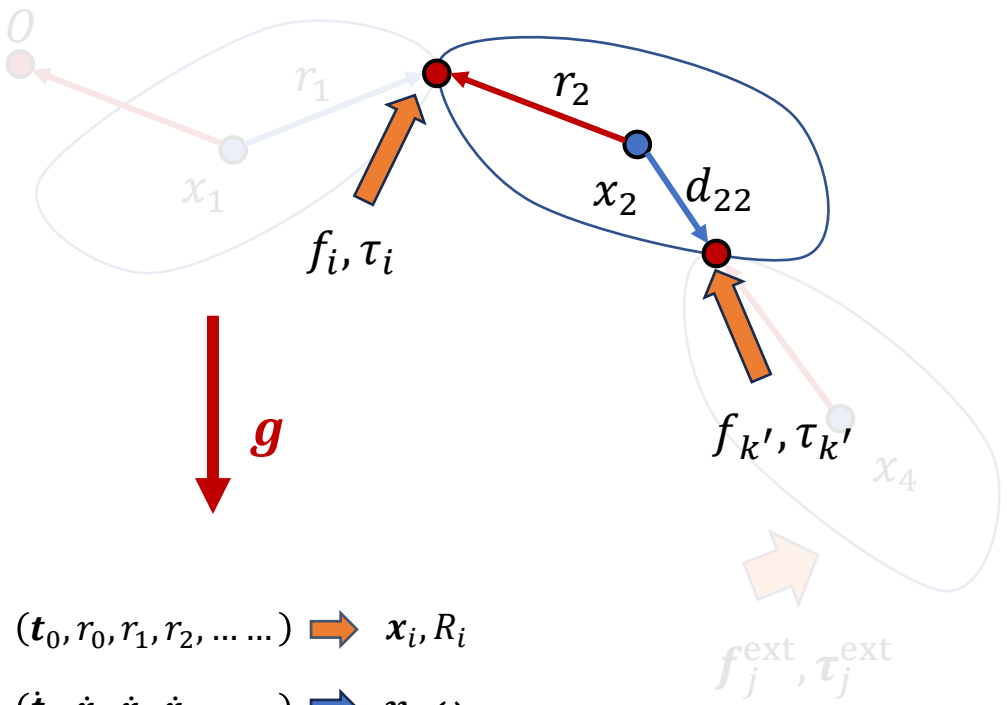
$$\boldsymbol{f}_3 = -m_3\boldsymbol{g} \qquad \boldsymbol{\tau}_3 = \boldsymbol{r}_3 \times m_3\boldsymbol{g}$$

$$\boldsymbol{f}_2 = -m_2\boldsymbol{g} - m_3\boldsymbol{g} \qquad \boldsymbol{\tau}_2 = ?$$

$$I_i\dot{\boldsymbol{\omega}}_i + \boldsymbol{\omega}_i \times I_i\boldsymbol{\omega}_i = \boldsymbol{\tau}_i + \boldsymbol{r}_i \times \boldsymbol{f}_i - \sum_{k \in c(i)} (\boldsymbol{\tau}_k + \boldsymbol{d}_{ik} \times \boldsymbol{f}_k) + \sum_j \boldsymbol{\tau}_j^{\text{ext}}$$

Example: Gravity Compensation

$$[\boldsymbol{x}, \mathbf{0}, \mathbf{0}] \mapsto \boldsymbol{f}$$



$(\boldsymbol{t}_0, r_0, r_1, r_2, \dots\dots) \Rightarrow \boldsymbol{x}_i, R_i$

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Given: $m_i, I_i, \boldsymbol{x}_i, R_i, \boldsymbol{v}_i = 0, \boldsymbol{\omega}_i = 0$

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Compute: $\boldsymbol{f}_k, \boldsymbol{\tau}_k$

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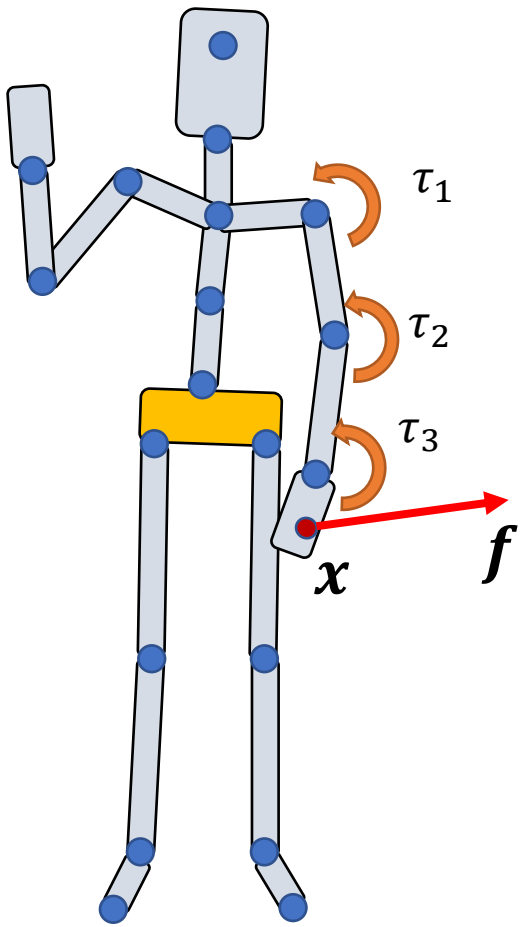
$$\boldsymbol{f}_2 = -m_2\boldsymbol{g} - m_3\boldsymbol{g} \qquad \boldsymbol{\tau}_2 = \boldsymbol{r}_2 \times m_2\boldsymbol{g} + (\boldsymbol{r}_2 - \boldsymbol{d}_2 + \boldsymbol{r}_3) \times m_3\boldsymbol{g}$$

$$I_i\dot{\boldsymbol{\omega}}_i + \boldsymbol{\omega}_i \times I_i\boldsymbol{\omega}_i = \boldsymbol{\tau}_i + \boldsymbol{r}_i \times \boldsymbol{f}_i - \sum_{k \in c(i)} (\boldsymbol{\tau}_k + \boldsymbol{d}_{ik} \times \boldsymbol{f}_k) + \sum_j \boldsymbol{\tau}_j^{\text{ext}}$$

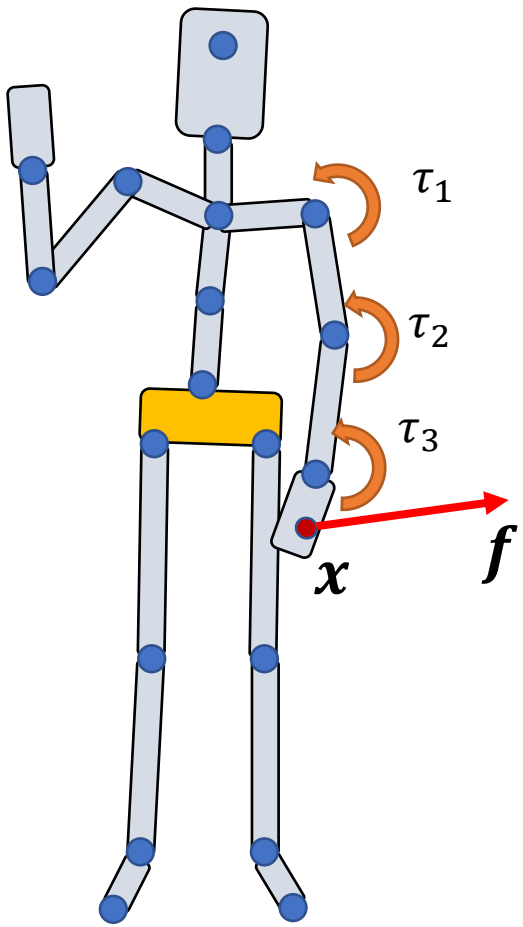
Jacobian Transpose Control



Can we use joint torques $\boldsymbol{\tau}_i$ to mimic the effect of a force $\boldsymbol{f}$ applied at $\boldsymbol{x}$

- Note that the desired force $\boldsymbol{f}$ is not actually applied
- Also called "virtual force"

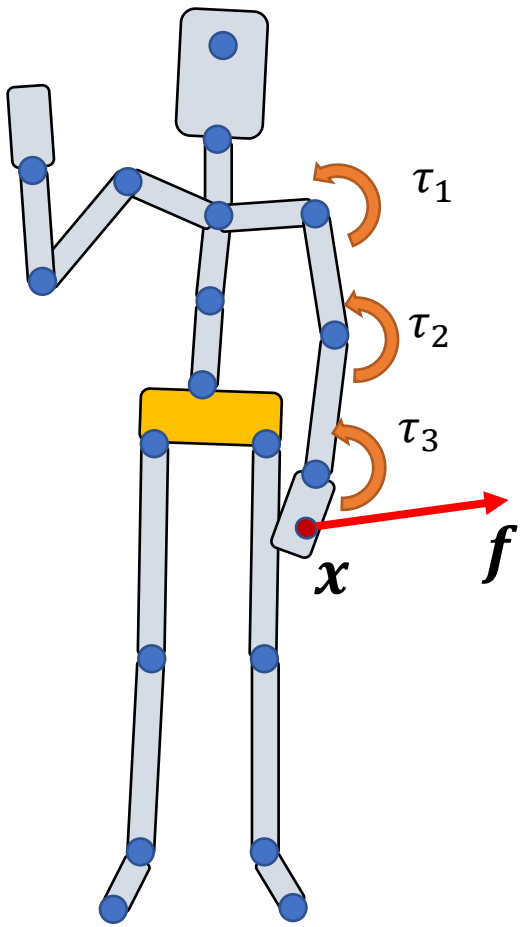
Jacobian Transpose Control



Can we use joint torques $\boldsymbol{\tau}_i$ to mimic the **effect** of a force $\boldsymbol{f}$ applied at $\boldsymbol{x}$

Make $\boldsymbol{f}$ and $\boldsymbol{\tau}_i$ done the same work

Jacobian Transpose Control

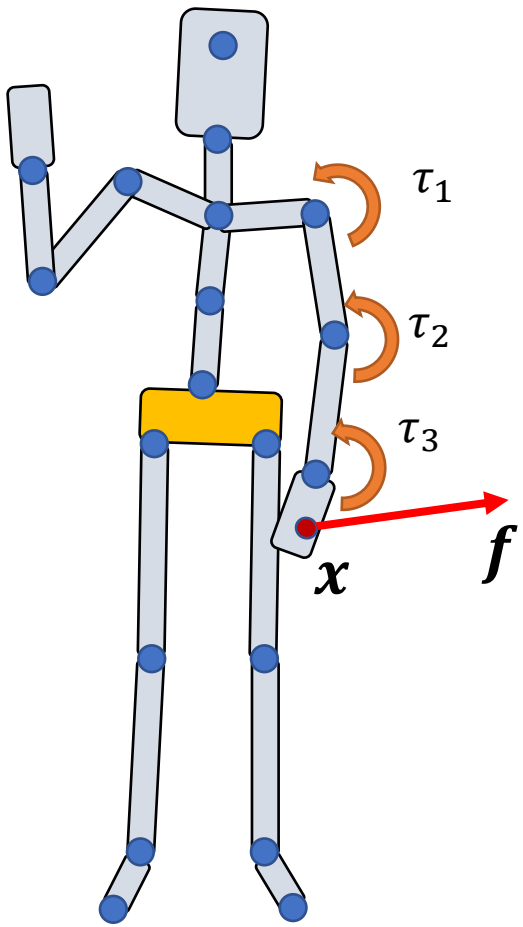


Can we use joint torques $\boldsymbol{\tau}_i$ to mimic the **effect** of a force $\boldsymbol{f}$ applied at $\boldsymbol{x}$

Make $\boldsymbol{f}$ and $\boldsymbol{\tau}_i$ done the same power

$$P = \boldsymbol{f}^T \dot{\boldsymbol{x}} = \boldsymbol{\tau}^T \dot{\boldsymbol{\theta}}$$

Jacobian Transpose Control



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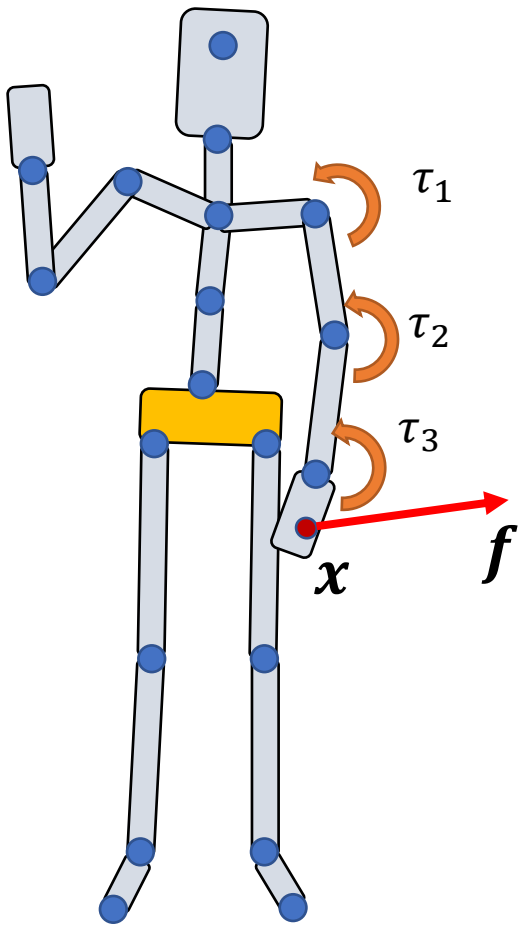
Forward kinematics $\boldsymbol{x} = g(\boldsymbol{\theta}) \Rightarrow \dot{\boldsymbol{x}} = J\dot{\boldsymbol{\theta}}$

$$J = \frac{\partial g}{\partial \boldsymbol{\theta}}$$

Jacobian Transpose Control



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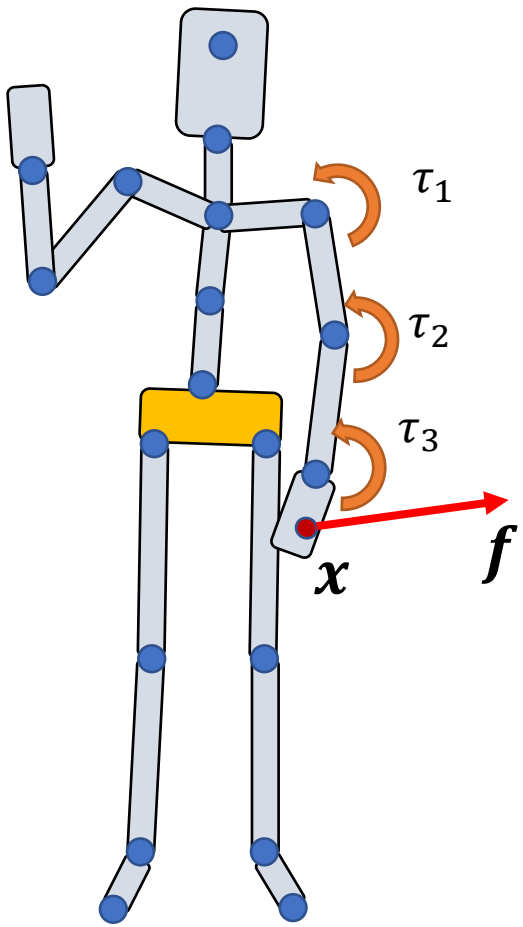
$$\boldsymbol{f}^T J \dot{\boldsymbol{\theta}} = \boldsymbol{\tau}^T \dot{\boldsymbol{\theta}}$$

$$J = \frac{\partial g}{\partial \boldsymbol{\theta}}$$

Jacobian Transpose Control



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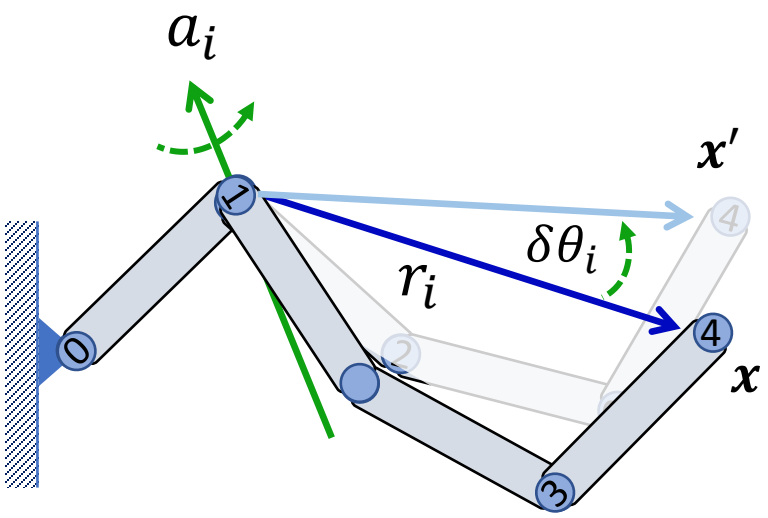
$$\boldsymbol{\tau} = J^T \boldsymbol{f}$$

$$J = \frac{\partial g}{\partial \boldsymbol{\theta}}$$

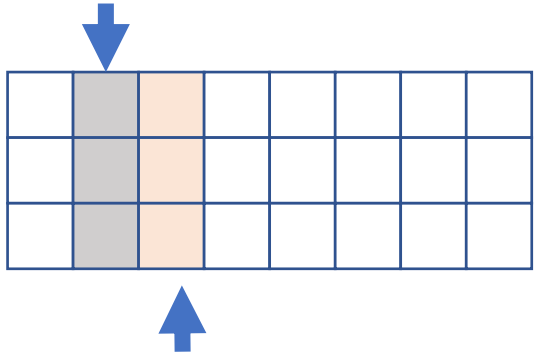
Recall: Geometric Approach

$$J = \frac{\partial g}{\partial \boldsymbol{\theta}} = \begin{pmatrix} \frac{\partial g}{\partial \theta_0} & \frac{\partial g}{\partial \theta_1} & \cdots & \frac{\partial g}{\partial \theta_n} \end{pmatrix}$$

Assuming all joints are hinge joint

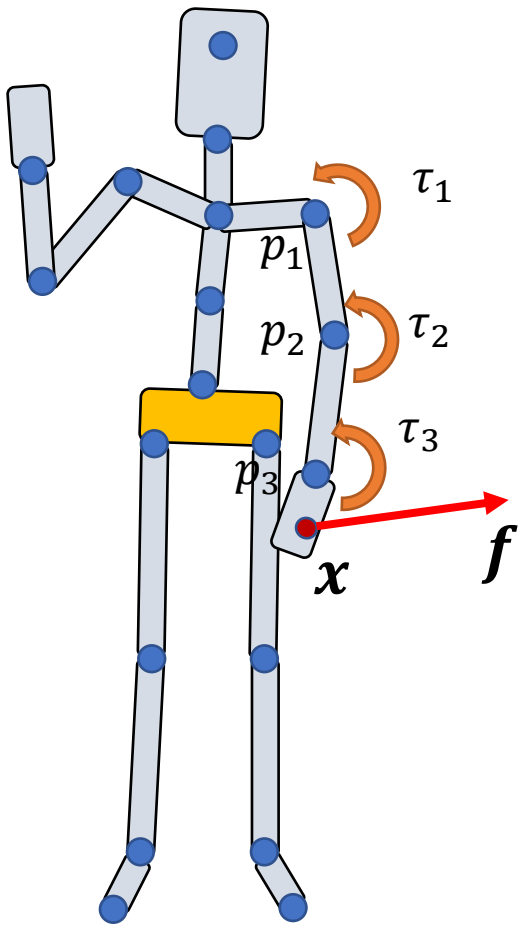


$$\frac{\partial g}{\partial \theta_1} = \boldsymbol{a}_1 \times \boldsymbol{r}_1$$



$$\frac{\partial f}{\partial \theta_2} = \boldsymbol{a}_2 \times \boldsymbol{r}_2$$

Jacobian Transpose Control



Can we use joint torques $\boldsymbol{\tau}_i$ to mimic the **effect** of a force $\boldsymbol{f}$ applied at $\boldsymbol{x}$

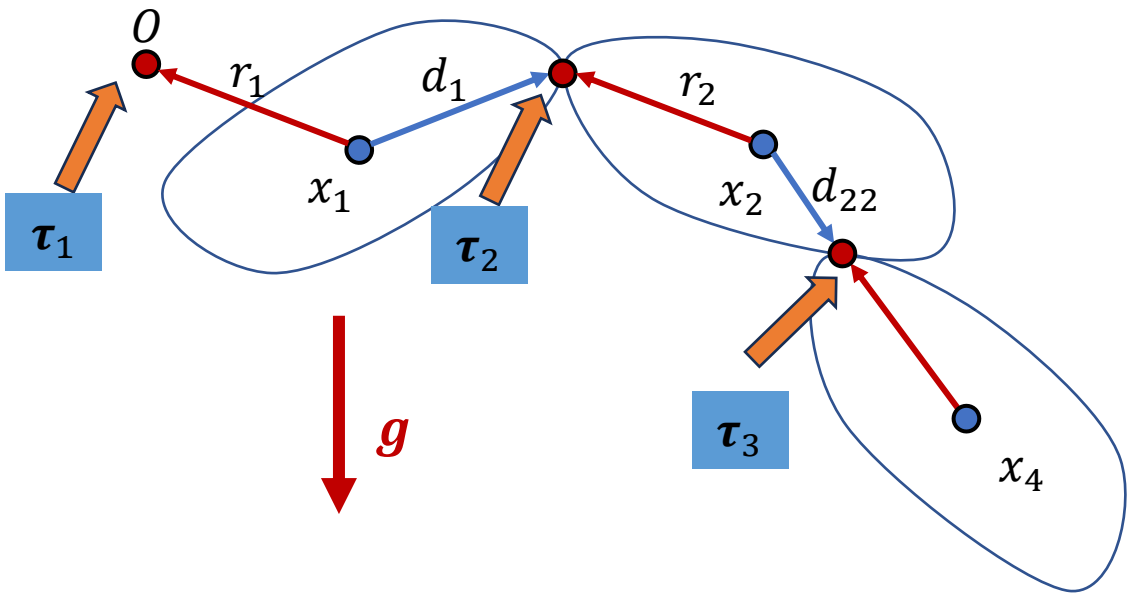
Make $\boldsymbol{f}$ and $\boldsymbol{\tau}_i$ done the same power

$$\boldsymbol{\tau} = J^T \boldsymbol{f} \qquad J = \frac{\partial g}{\partial \boldsymbol{\theta}}$$

$$\boldsymbol{\tau}_i = (\boldsymbol{x} - \boldsymbol{p}_i) \times \boldsymbol{f}$$

Example: Gravity Compensation

$$[\boldsymbol{x}, \mathbf{0}, \mathbf{0}] \mapsto \boldsymbol{f}$$



$(\boldsymbol{t}_0, r_0, r_1, r_2, \dots\dots) \Rightarrow \boldsymbol{x}_i, R_i$

$(\dot{\boldsymbol{t}}_0, \dot{r}_0, \dot{r}_1, \dot{r}_2, \dots\dots) \Rightarrow \boldsymbol{v}_i, \boldsymbol{\omega}_i$

Given: $m_i, I_i, \boldsymbol{x}_i, R_i, \boldsymbol{v}_i = 0, \boldsymbol{\omega}_i = 0$

$\boldsymbol{a} = 0, \boldsymbol{\alpha} = 0$

Gravity: $\boldsymbol{g}$

Compute: $\boldsymbol{\tau}_k$

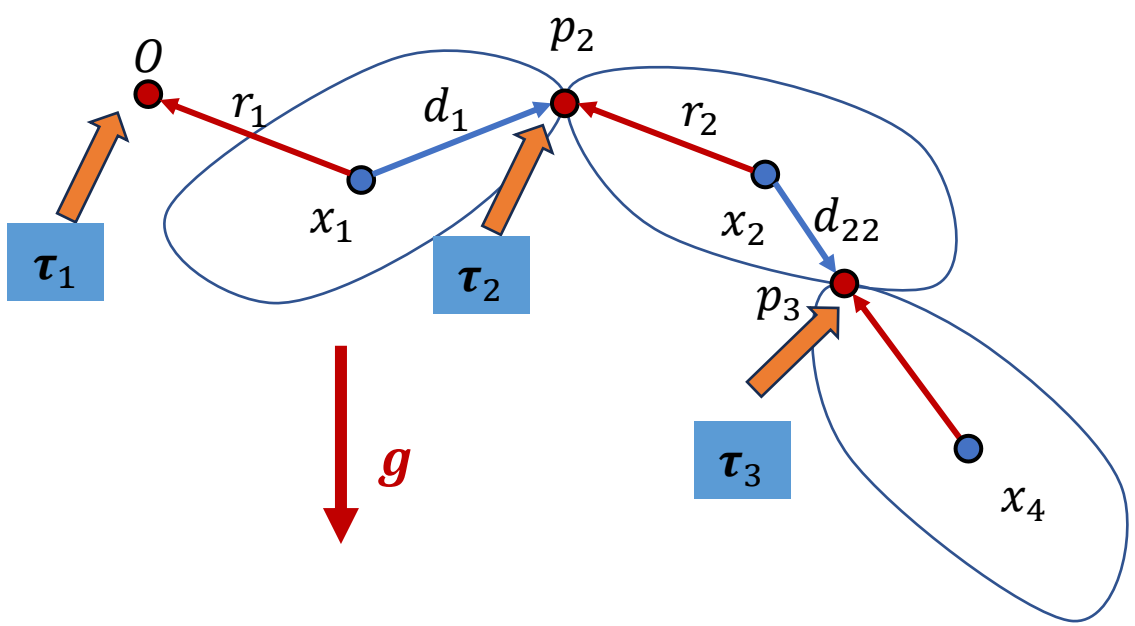
$$\boldsymbol{\tau}_3 = \boldsymbol{r}_3 \times m_3 \boldsymbol{g}$$

$$\boldsymbol{\tau}_2 = \boldsymbol{r}_2 \times m_2 \boldsymbol{g} + (\boldsymbol{r}_2 - \boldsymbol{d}_2 + \boldsymbol{r}_3) \times m_3 \boldsymbol{g}$$

$$\boldsymbol{\tau}_1 = \boldsymbol{r}_1 \times m_1 \boldsymbol{g} + (\boldsymbol{r}_1 - \boldsymbol{d}_1 + \boldsymbol{r}_2) \times m_2 \boldsymbol{g} + (\boldsymbol{r}_1 - \boldsymbol{d}_1 + \boldsymbol{r}_2 - \boldsymbol{d}_2 + \boldsymbol{r}_3) \times m_3 \boldsymbol{g}$$

Example: Gravity Compensation

$$[\boldsymbol{x}, \mathbf{0}, \mathbf{0}] \mapsto \boldsymbol{f}$$



$(\boldsymbol{t}_0, r_0, r_1, r_2, \dots\dots) \Rightarrow \boldsymbol{x}_i, R_i$

$(\dot{\boldsymbol{t}}_0, \dot{r}_0, \dot{r}_1, \dot{r}_2, \dots\dots) \Rightarrow \boldsymbol{v}_i, \boldsymbol{\omega}_i$

Given: $m_i, I_i, \boldsymbol{x}_i, R_i, \boldsymbol{v}_i = 0, \boldsymbol{\omega}_i = 0$

$\boldsymbol{a} = 0, \boldsymbol{\alpha} = 0$

Gravity: $\boldsymbol{g}$

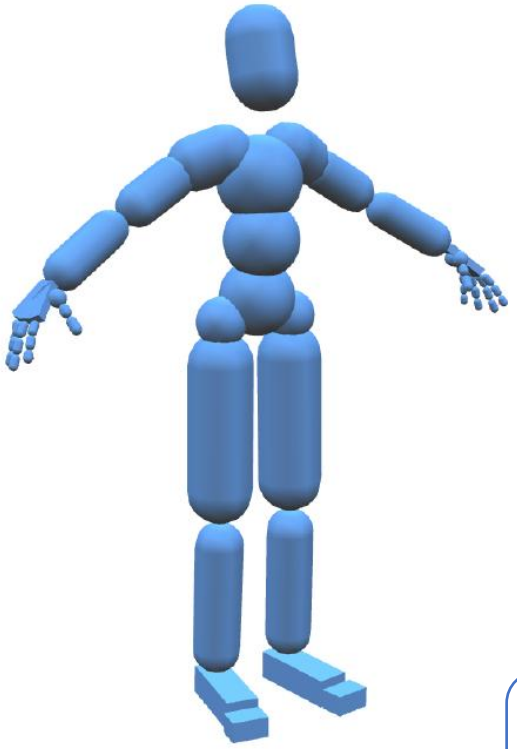
Compute: $\boldsymbol{\tau}_k$

$$\boldsymbol{\tau}_3 = (\boldsymbol{x}_4 - \boldsymbol{p}_3) \times (-m_3 \boldsymbol{g})$$

$$\boldsymbol{\tau}_2 = (\boldsymbol{x}_2 - \boldsymbol{p}_2) \times (-m_2 \boldsymbol{g}) + (\boldsymbol{x}_4 - \boldsymbol{p}_2) \times (-m_3 \boldsymbol{g})$$

$$\boldsymbol{\tau}_1 = (\boldsymbol{x}_1 - \boldsymbol{o}) \times (-m_1 \boldsymbol{g}) + (\boldsymbol{x}_2 - \boldsymbol{o}) \times (-m_2 \boldsymbol{g}) + (\boldsymbol{x}_4 - \boldsymbol{o}) \times (-m_3 \boldsymbol{g})$$

Simulating + Control



Current state:
$\boldsymbol{s}_t = (\boldsymbol{x}_t, \boldsymbol{v}_t, R_t, \boldsymbol{\omega}_t)$

Controller
π
$\pi: s_t, t \mapsto f, \tau$

f, τ
Joint torques

Dynamic Simulator

Next state:
$\boldsymbol{s}_{t+1} = (\boldsymbol{x}_{t+1}, \boldsymbol{v}_{t+1}, R_{t+1}, \boldsymbol{\omega}_{t+1})$

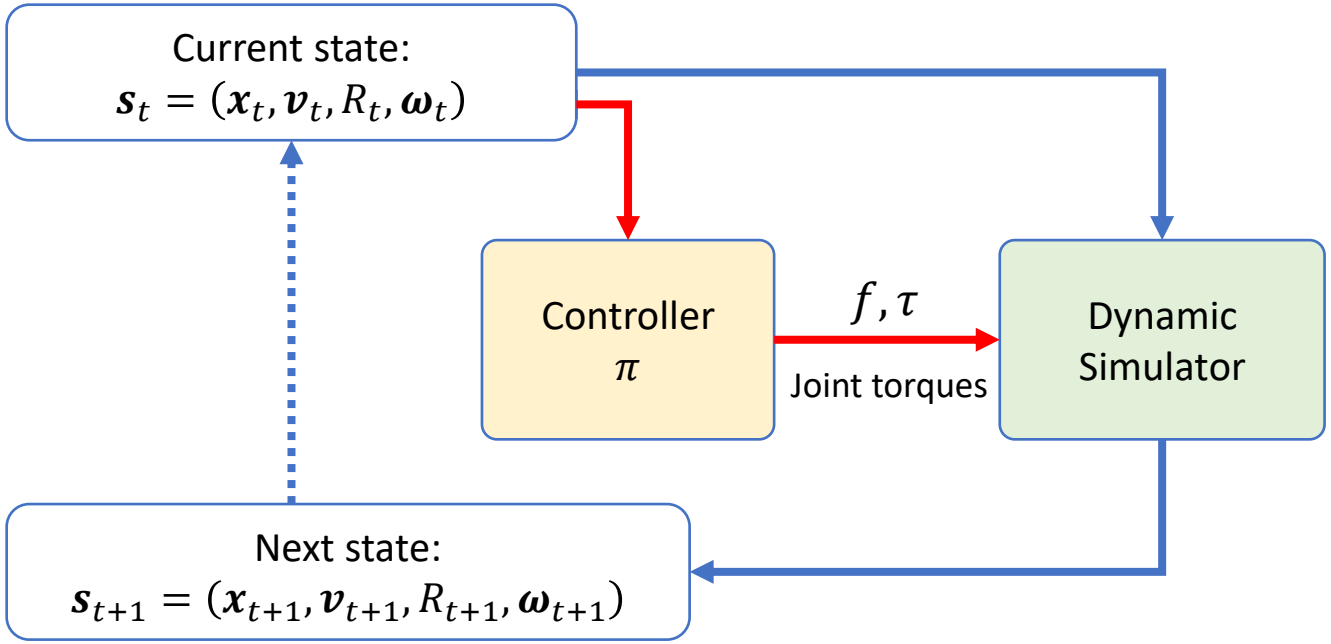
Feedforward vs. Feedback

Feedforward (Open-loop) control:

$$f, \tau = \pi(t)$$

- Apply predefined control signals without considering the current state of the system
- Assuming unchanging system. Perturbations may lead to unpredicted results

$\pi: s_t, t \mapsto f, \tau$



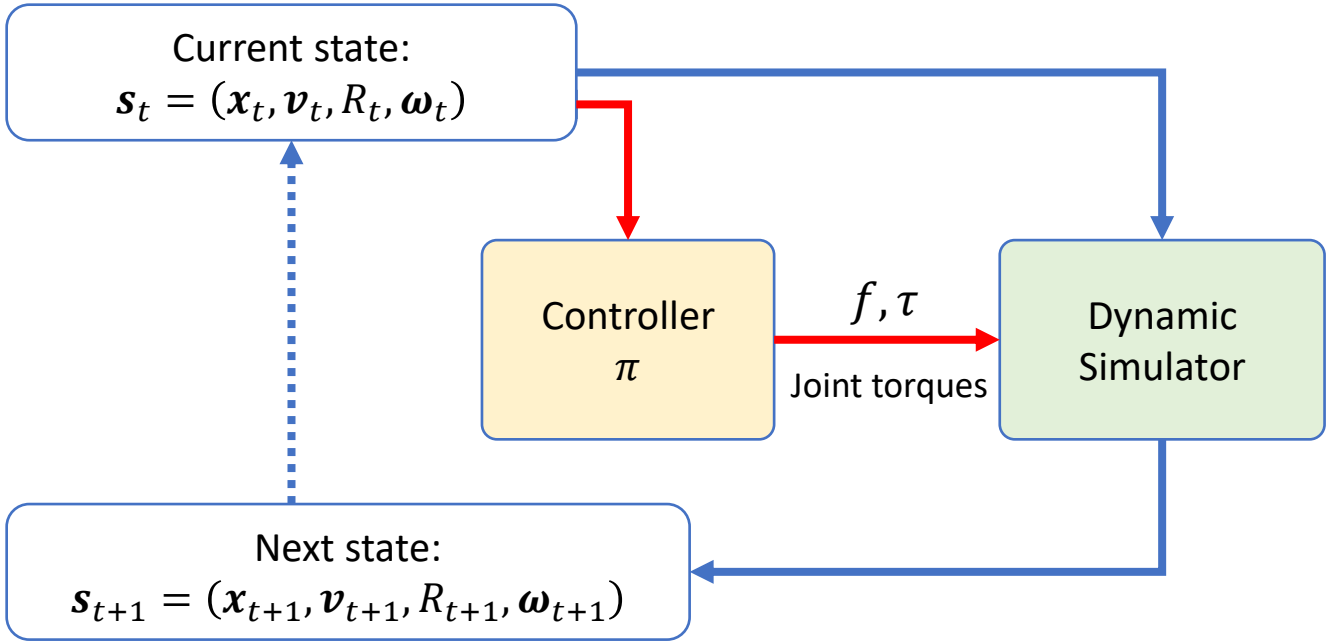
Feedforward vs. Feedback

Feedback control:

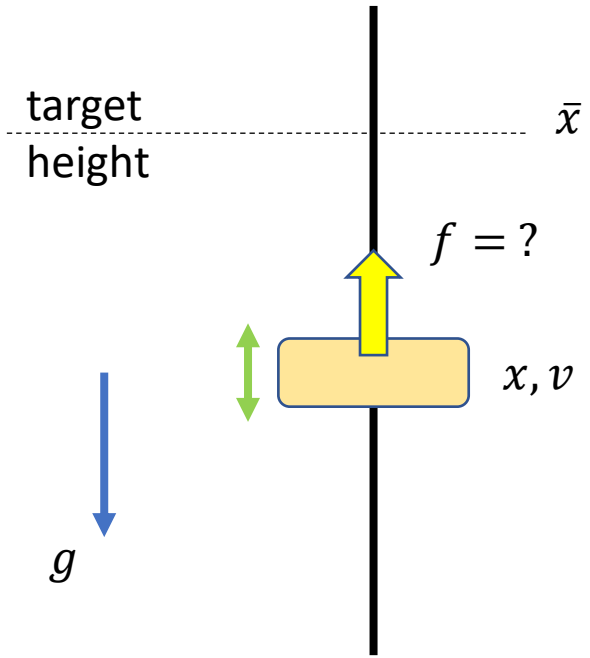
$$f, \tau = \pi(s_t, t)$$

- Adjust control signals based on the current state of the system
- Certain perturbations are expected. The feedback signal will be used to improves the performance at the next state.

$\pi: s_t, t \mapsto f, \tau$

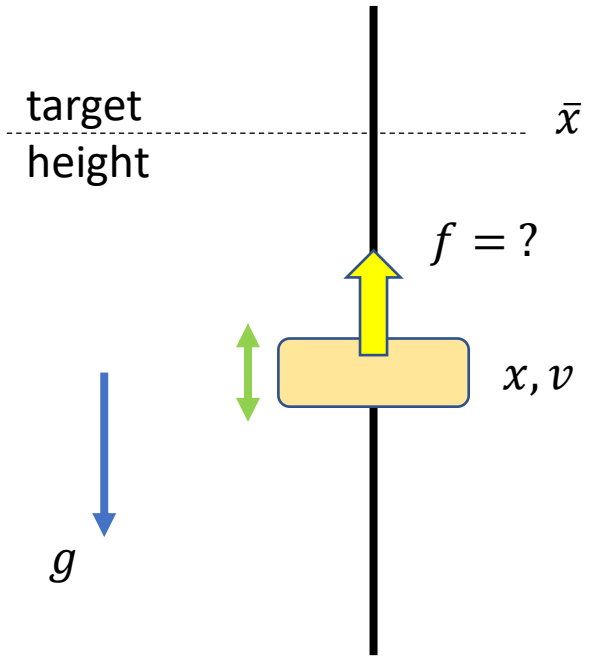


Proportional-Derivative Control



Compute force f to move
the object to the target height

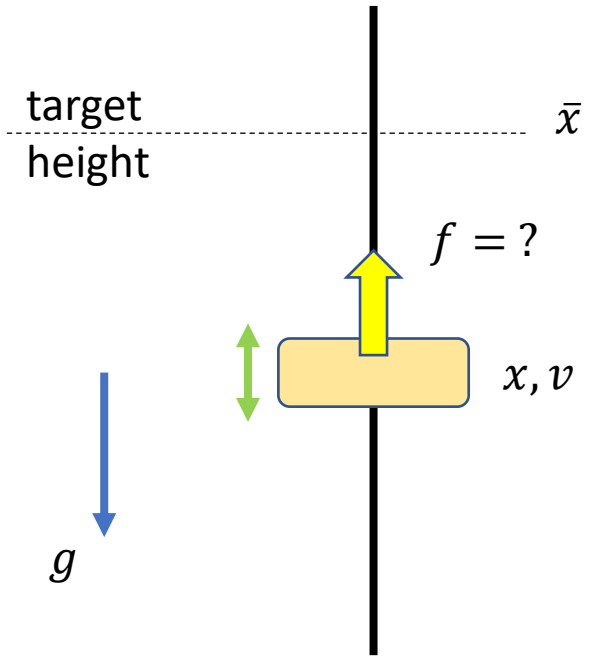
Proportional-Derivative Control



$$f = k_p(\bar{x} - x)$$

Compute force f to move
the object to the target height

Proportional-Derivative Control



Compute force f to move
the object to the target height

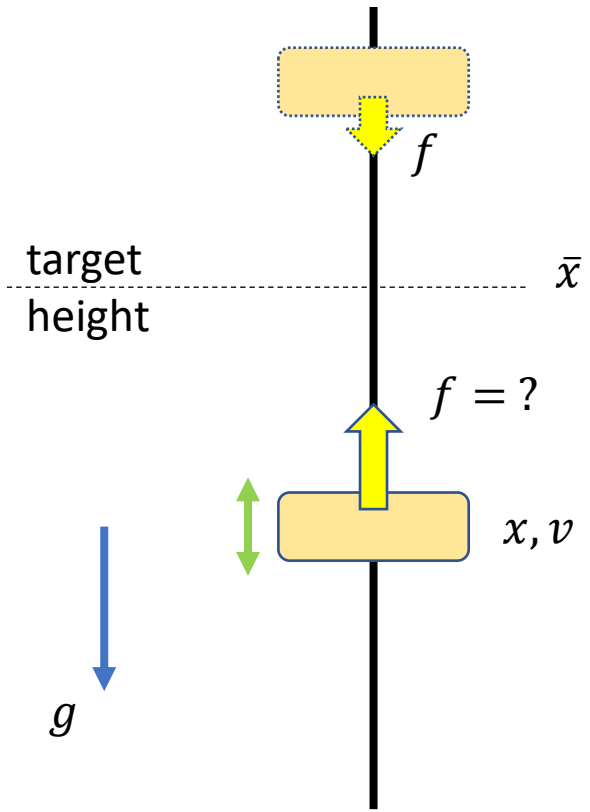
target
state

$$f = k_p(\bar{x} - x)$$

stiffness
(gain)

current
state

Proportional-Derivative Control



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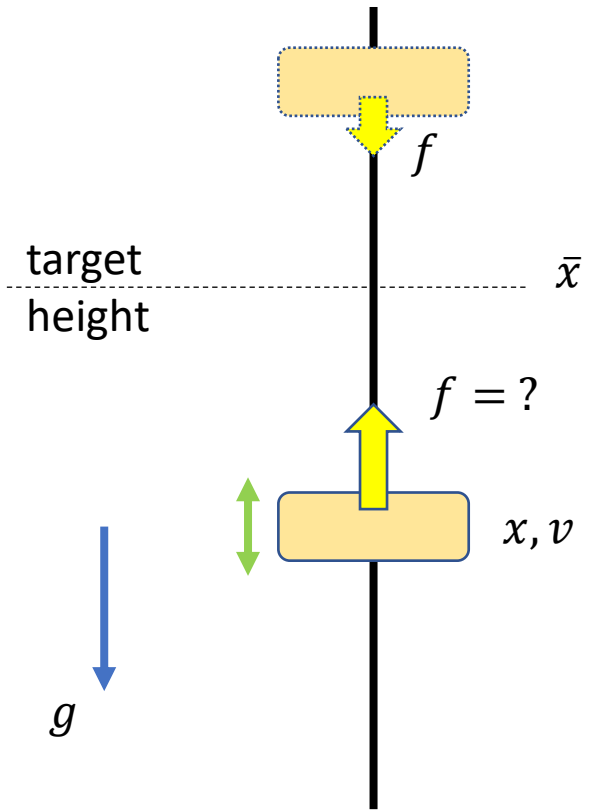
Proportional-Derivative Control



Compute force f to move
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$$f = k_p(\bar{x} - x)$$

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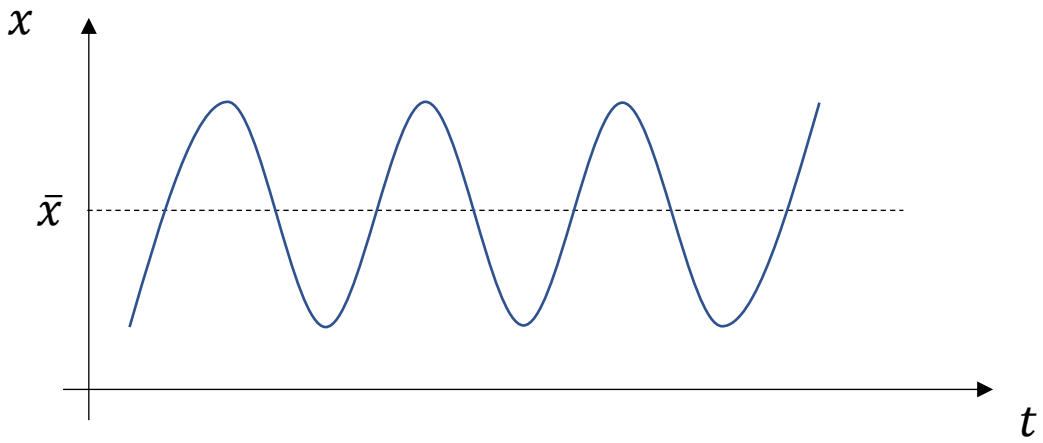
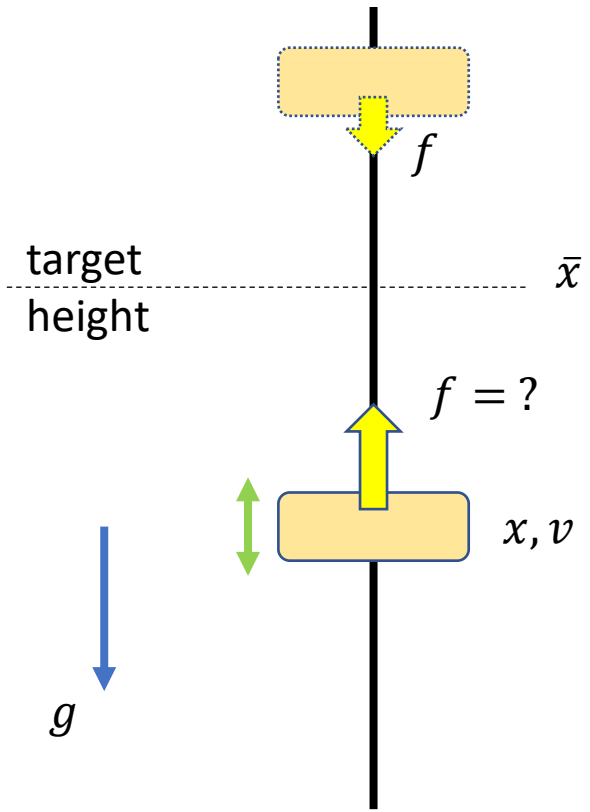


Proportional-Derivative Control



Compute force f to move
the object to the target height

target state

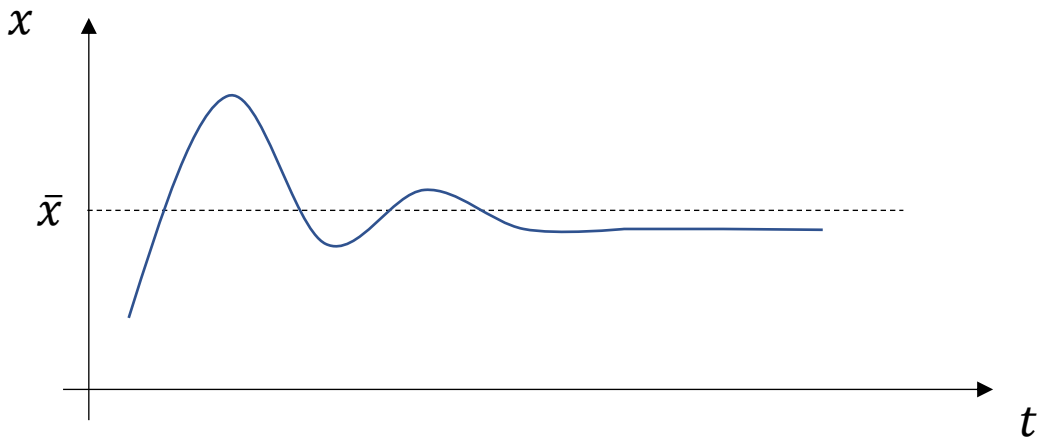
damping

$$f = k_p(\bar{x} - x) - k_d v$$

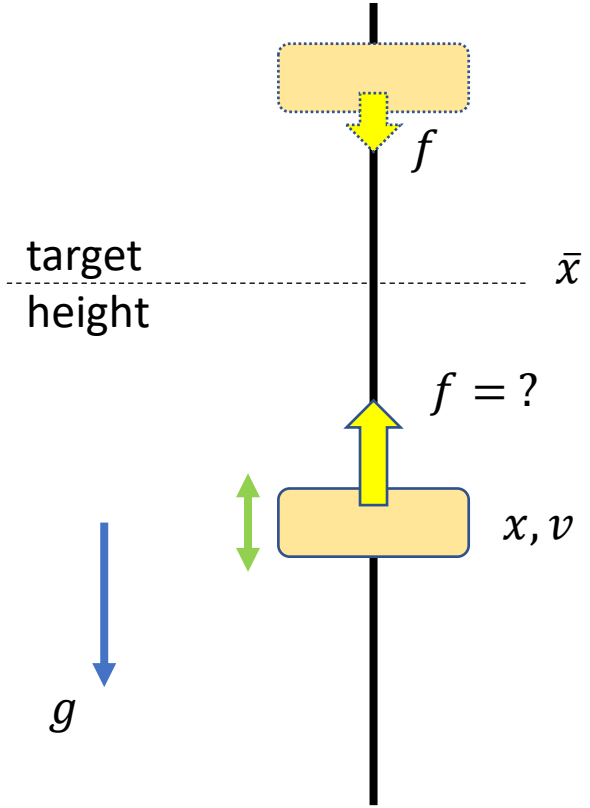
stiffness (gain)

current state

current velocity



Proportional-Derivative Control

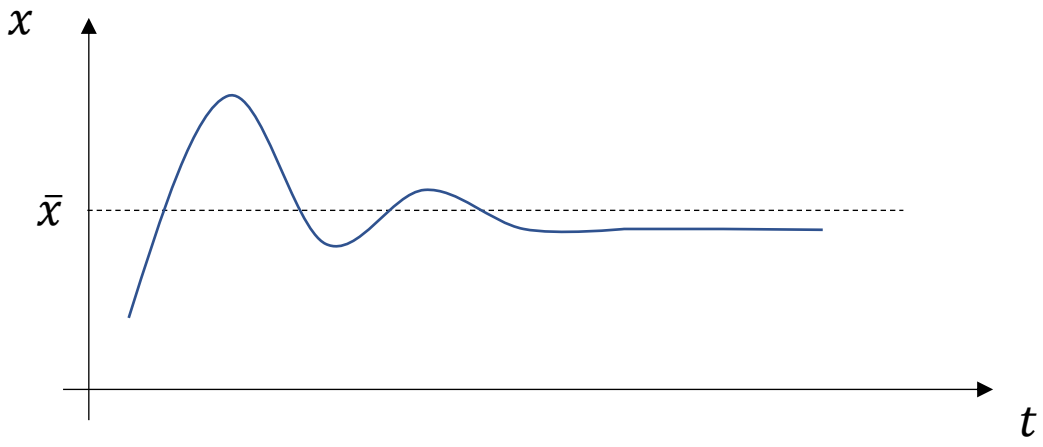


Compute force f to move
the object to the target height

Let $e = \bar{x} - x$ be the error

$$f = k_p e + k_d \dot{e}$$

proportional control — derivative control



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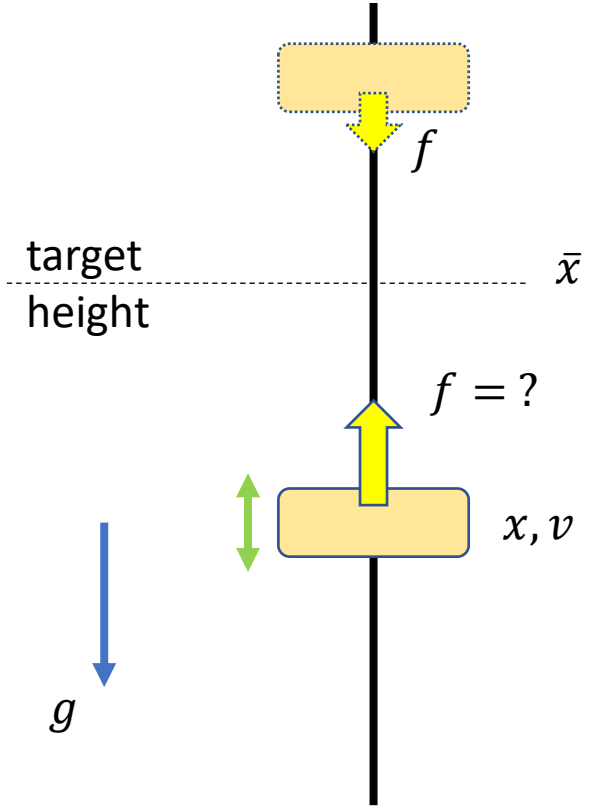

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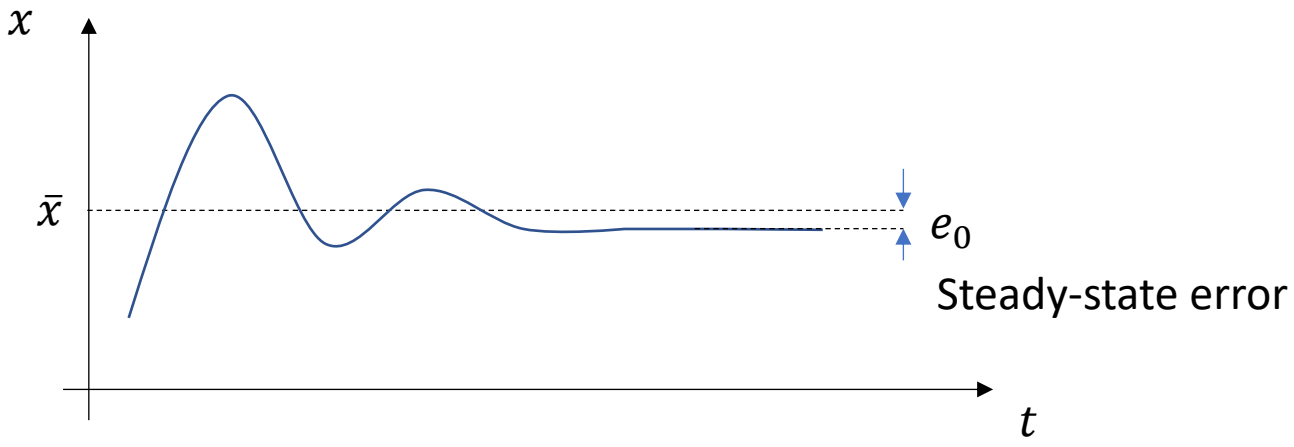
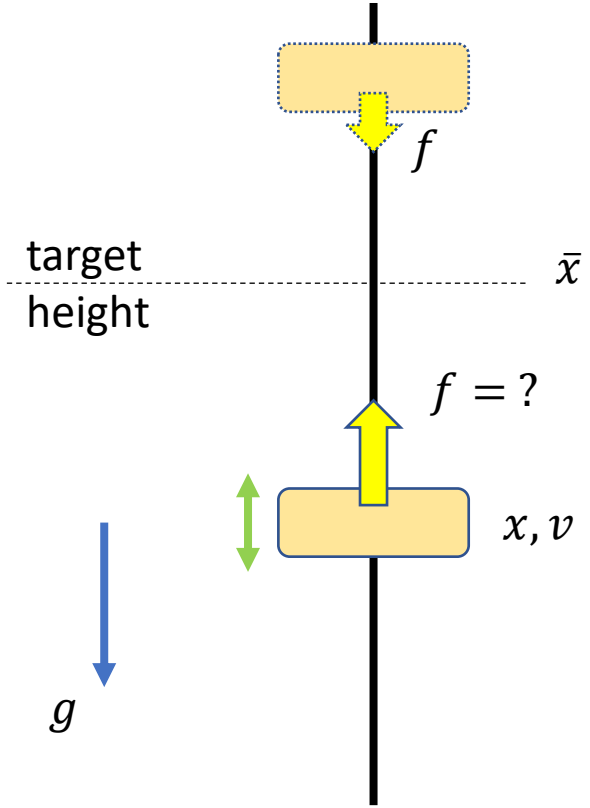


Proportional-Derivative Control



Compute force f to move
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Let $e = \bar{x} - x$ be the error

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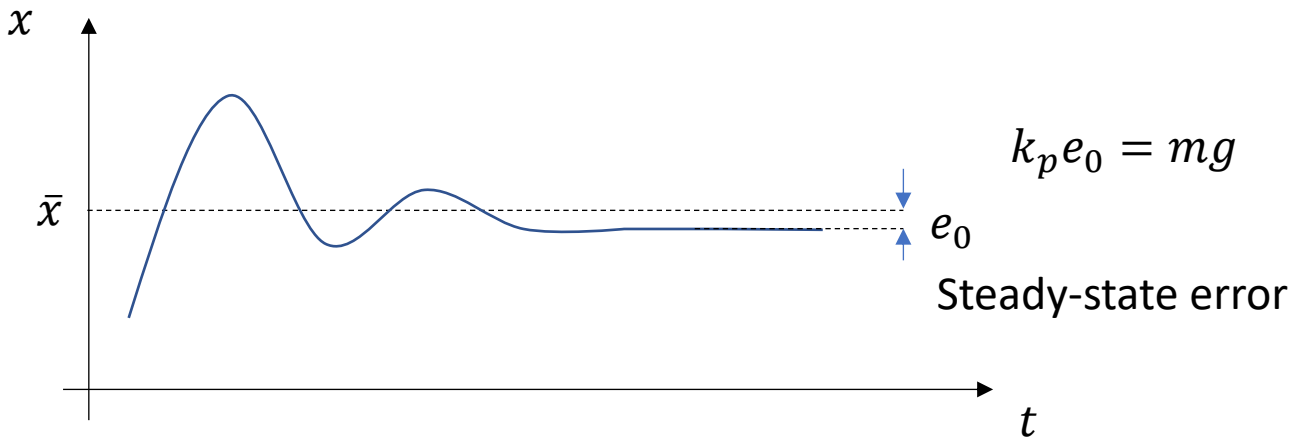
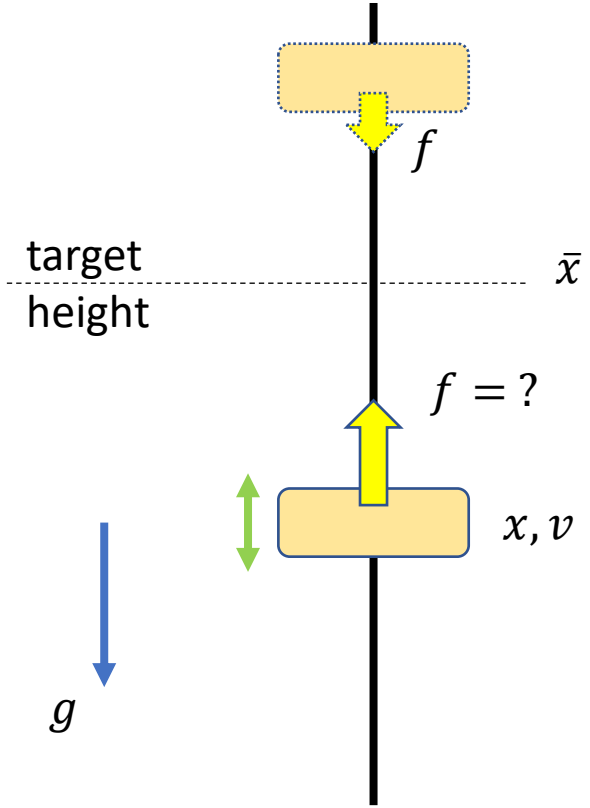
proportional control — derivative control



$k_p e_0 = mg$

Steady-state error

Proportional-Derivative Control



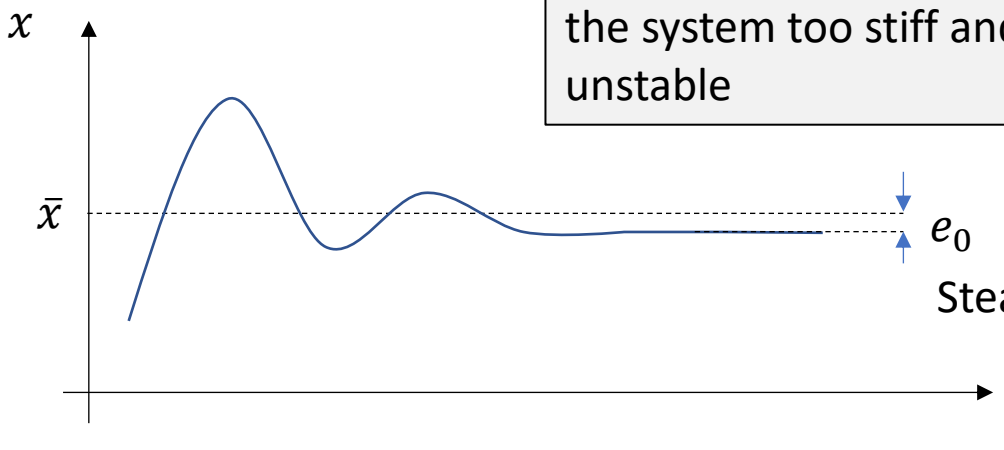
Compute force f to move the object to the target height

Let $e = \bar{x} - x$ be the error

$$f = k_p e + k_d \dot{e}$$

proportional control — derivative control

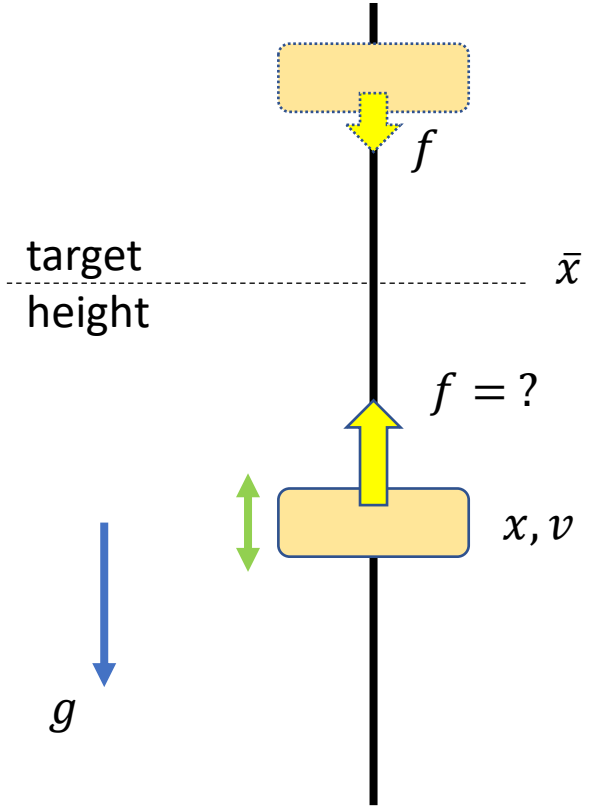
Increase stiffness k_p reduces the steady-state error, but can make the system too stiff and numerically unstable



$k_p e_0 = mg$

Steady-state error

Proportional-Integral-Derivative controller



Compute force f to move the object to the target height

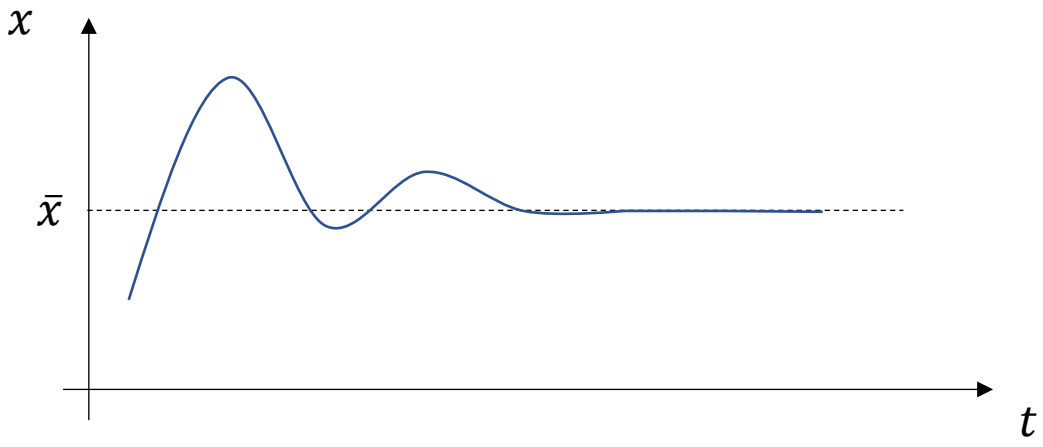
Let $e = \bar{x} - x$ be the error

$$f = k_p e + k_d \dot{e} + k_i \int_t e dt$$

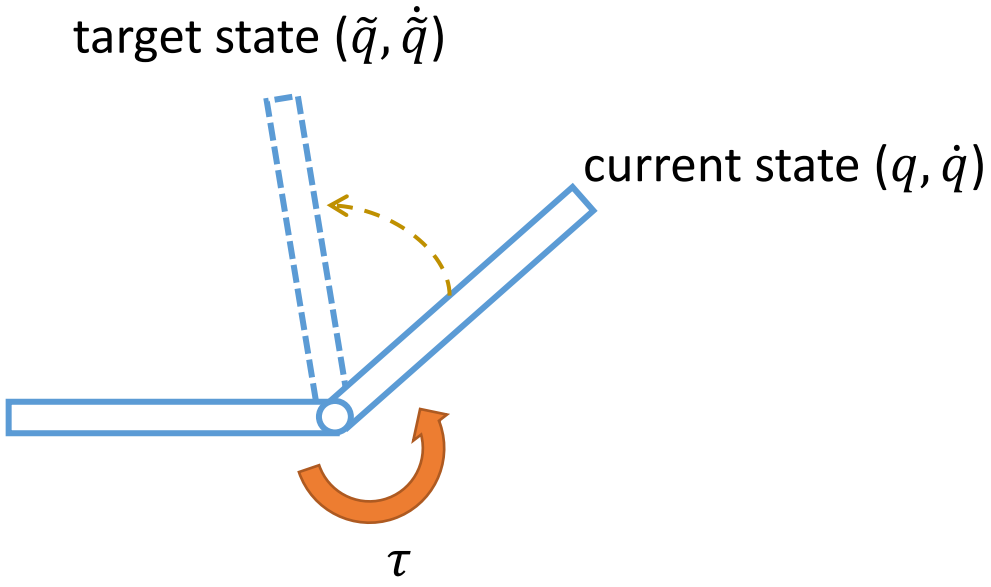
proportional control

derivative control

Integral control (rarely used in animation)



PD Control for Characters



target velocity
often zero

target pose

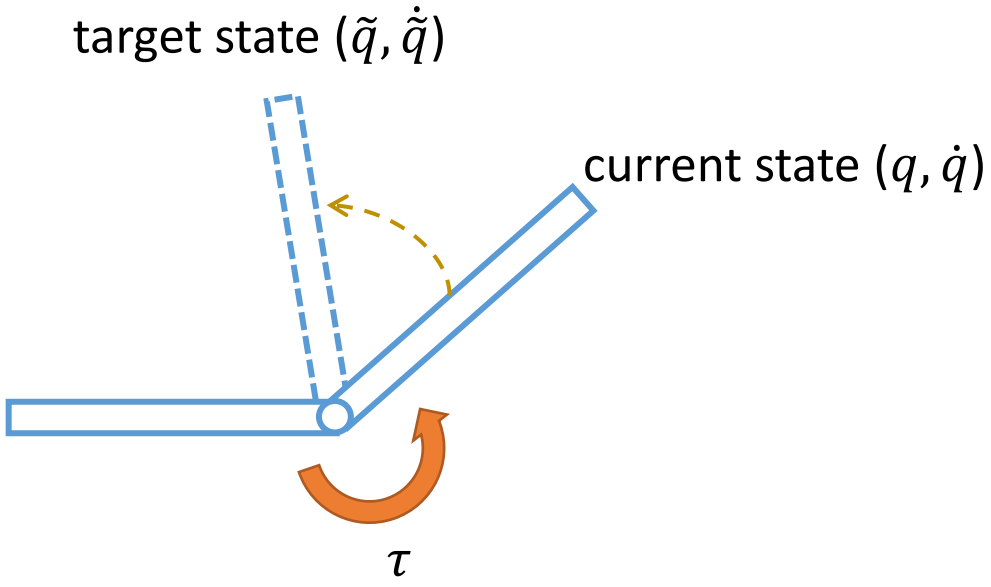
$$\tau = k_p(\bar{q} - q) + k_d(\dot{\bar{q}} - \dot{q})$$

joint torque

proportional control

derivative control

PD Control for Characters



target pose

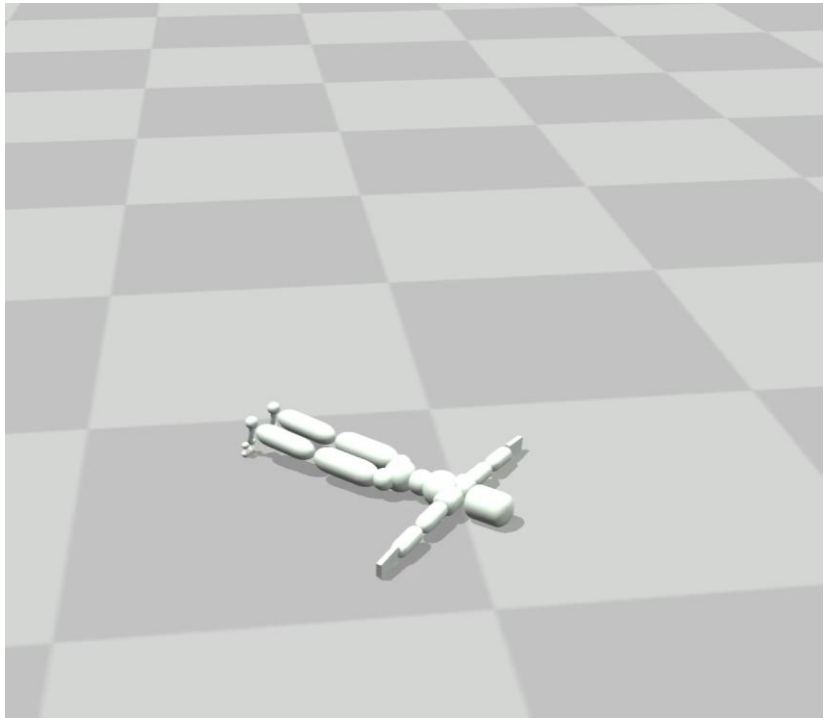
$$\tau = k_p(\bar{q} - q) - k_d\dot{q}$$

joint torque

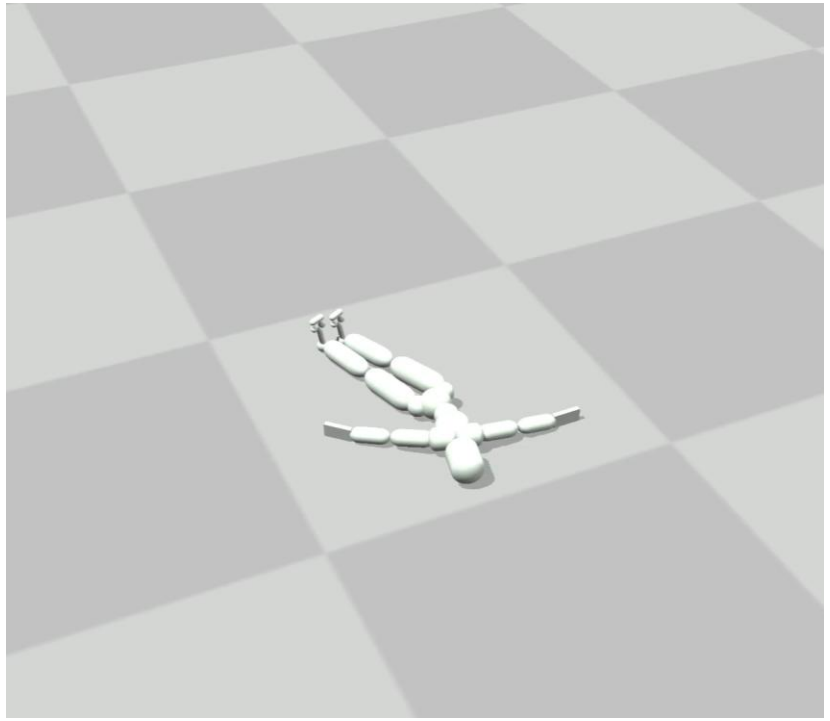
proportional control

derivative control

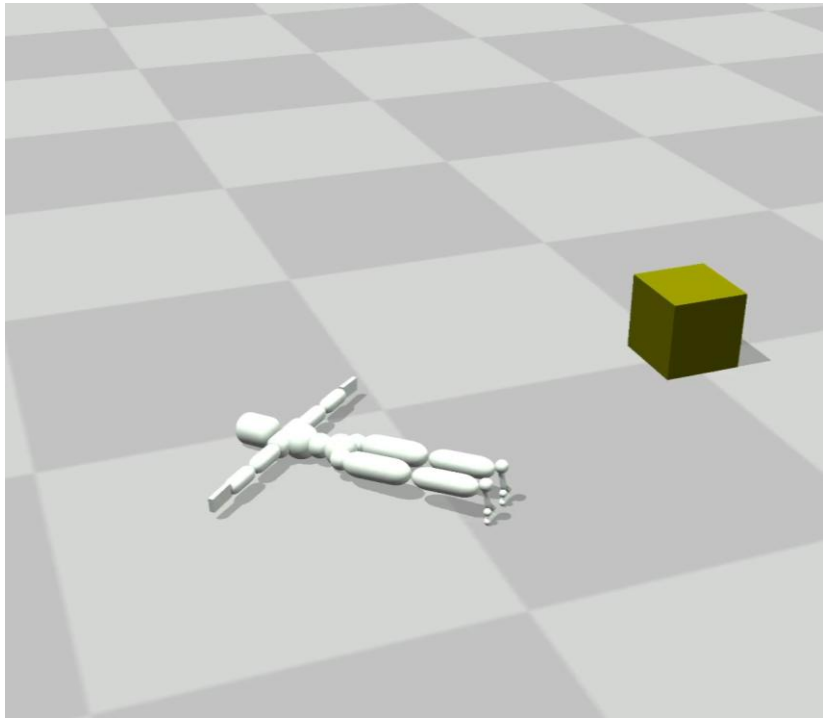
PD Control for Characters



Normal stiffness k_p

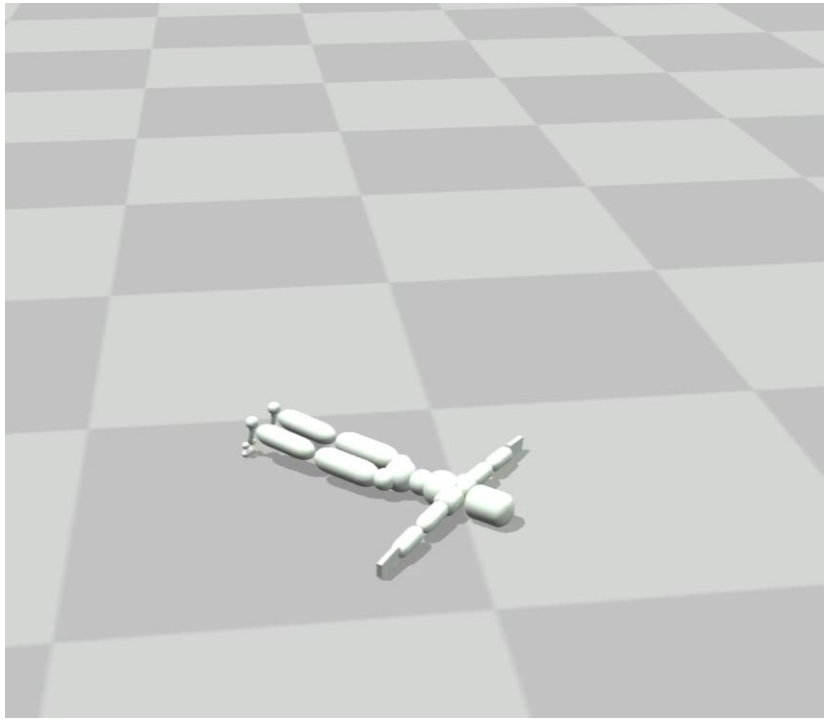


Small stiffness k_p

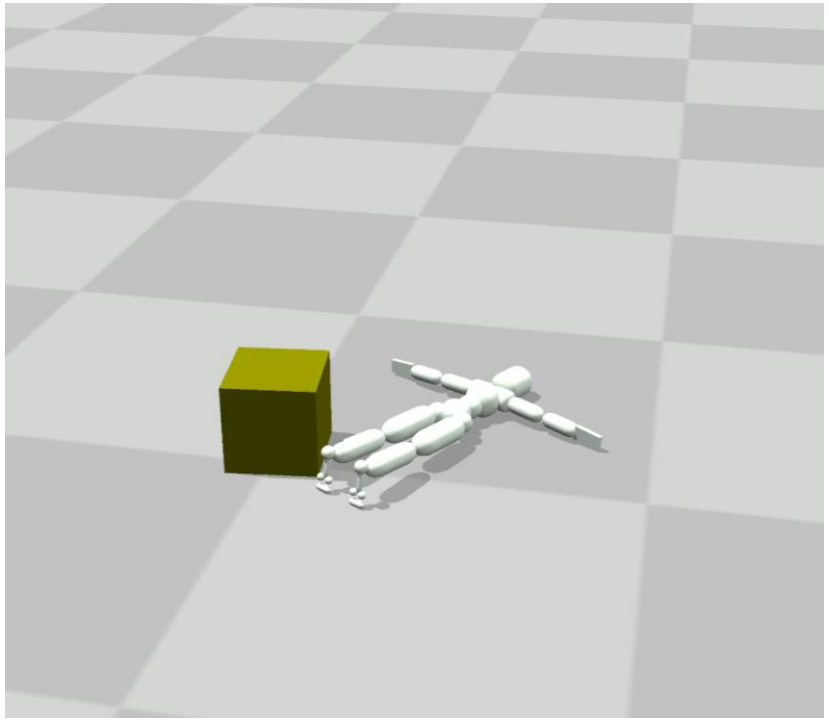


Large stiffness k_p

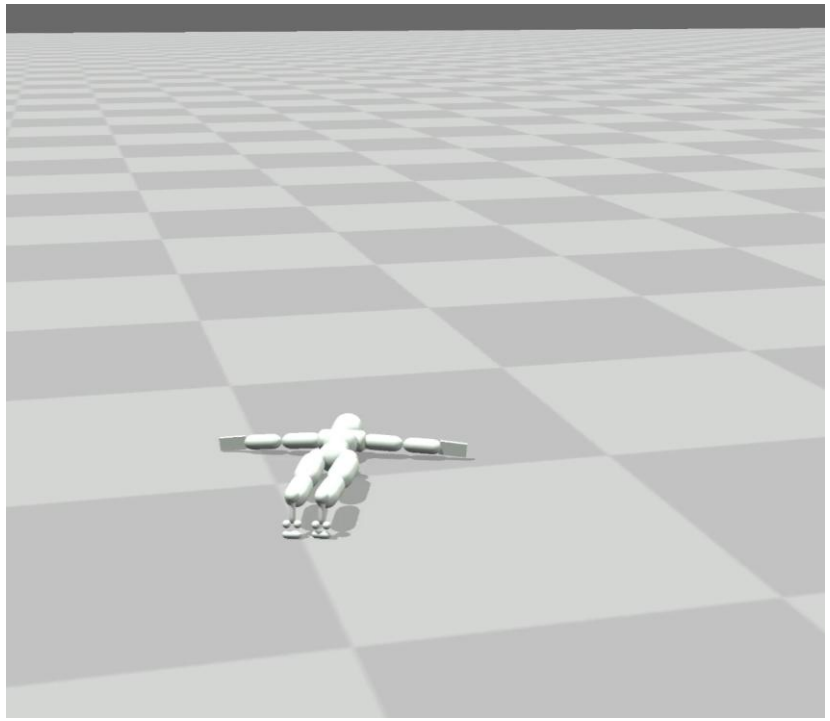
PD Control for Characters



Normal damping k_d

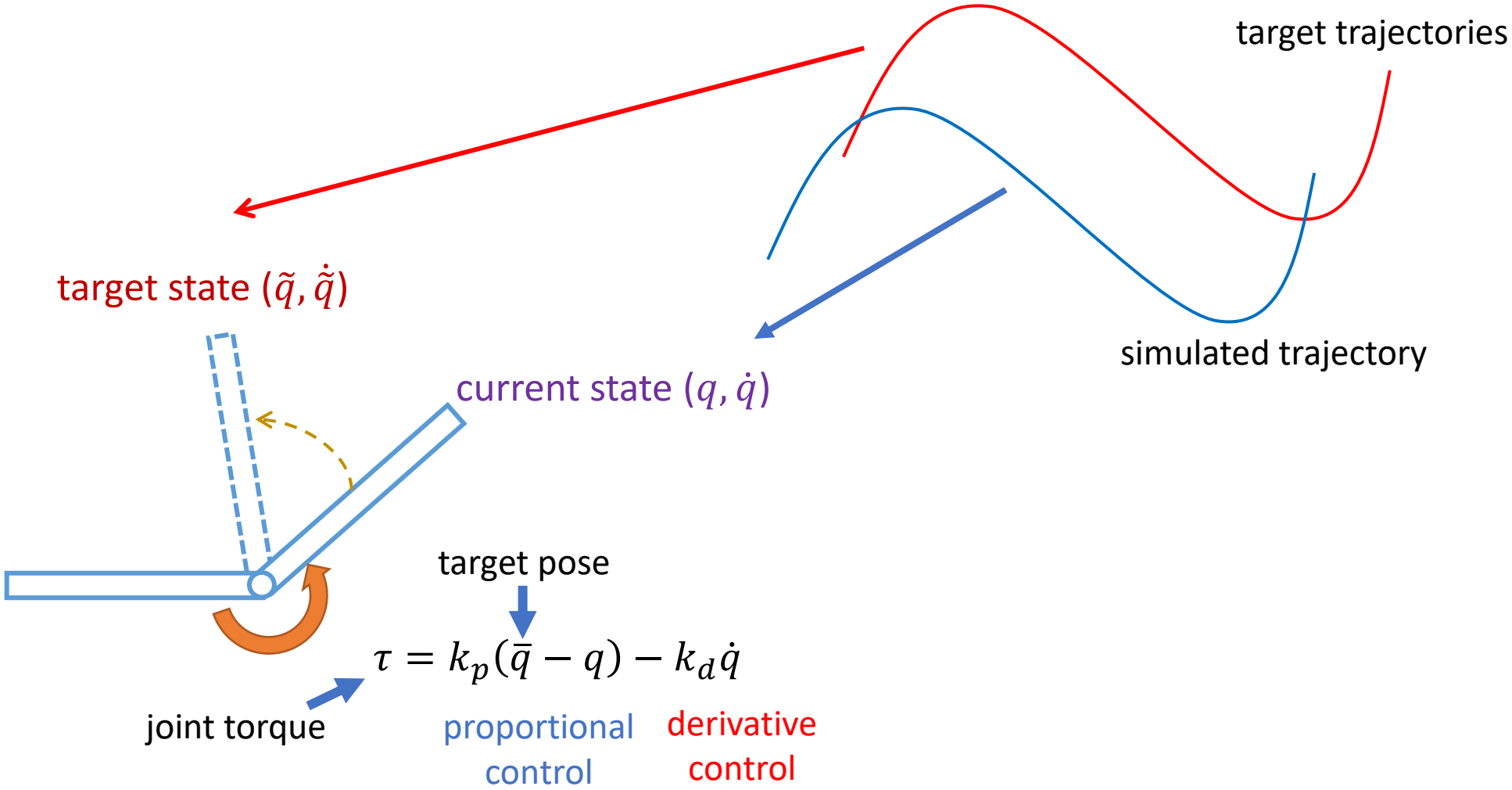


Small damping k_d

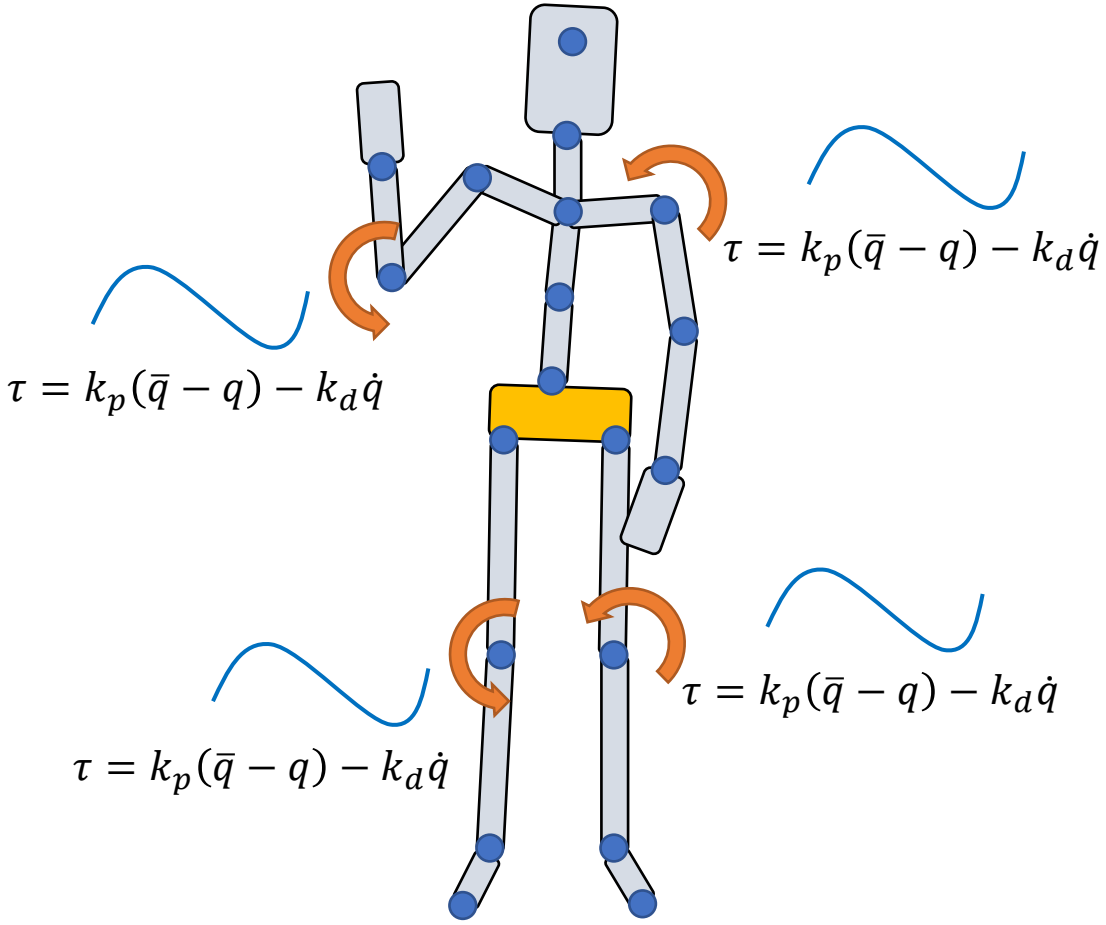


Large damping k_d

Tracking Controllers



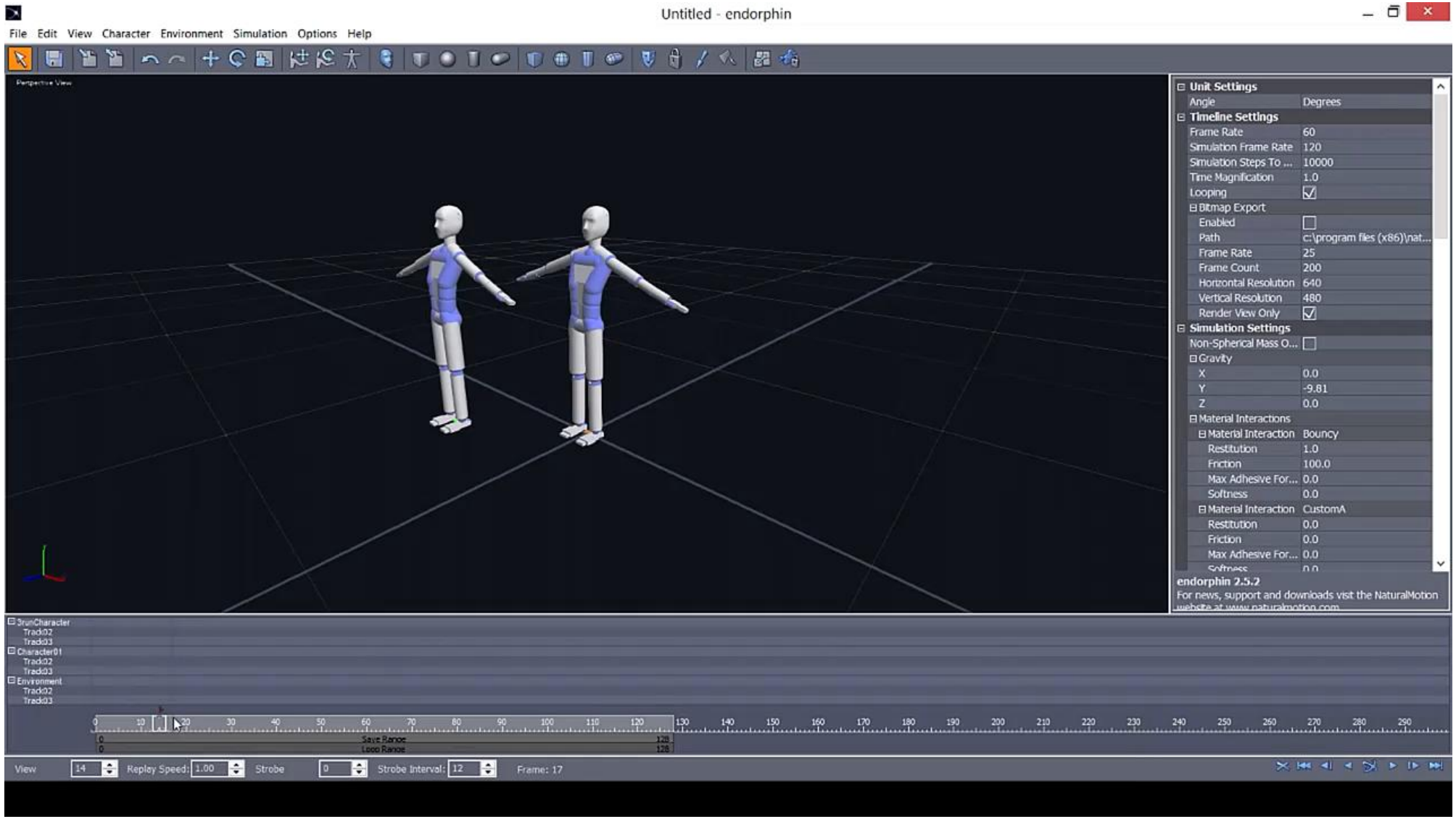
Full-body Tracking Controllers



Tracking a Trajectory

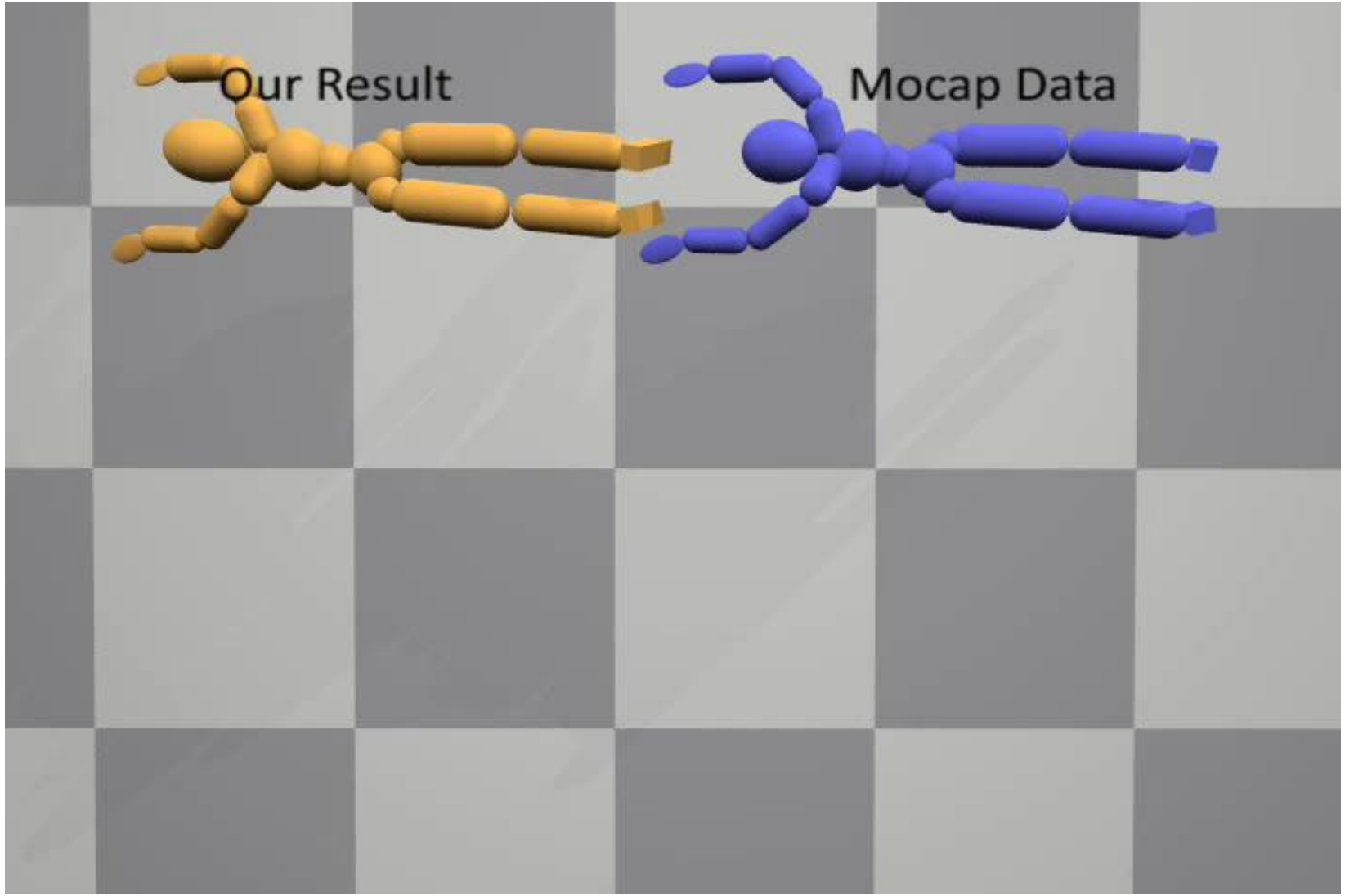
[Hodgins and Wooten 1995, Animating Human Athletics]

Trajectory Creation



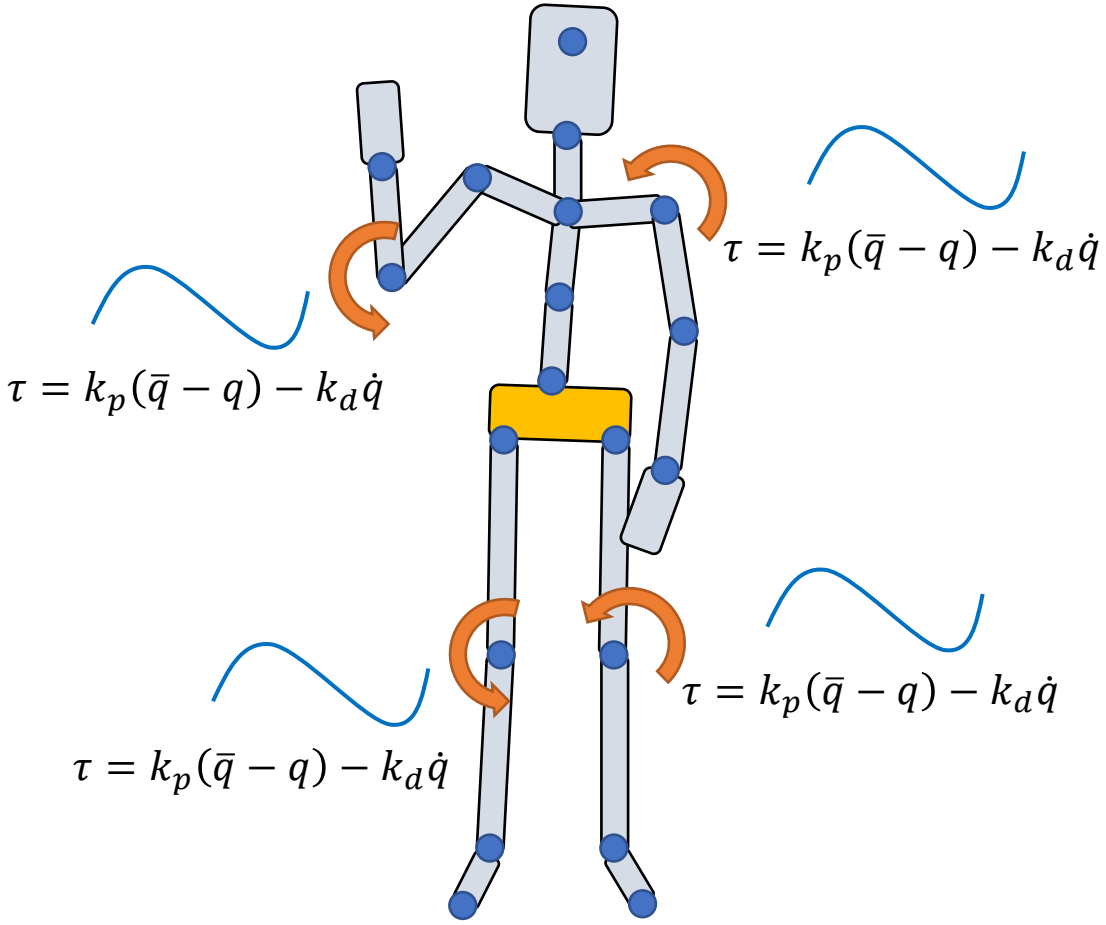
NaturalMotion - Endorphin

Tracking Mocap



[SAMCON – Liu et al 2010]

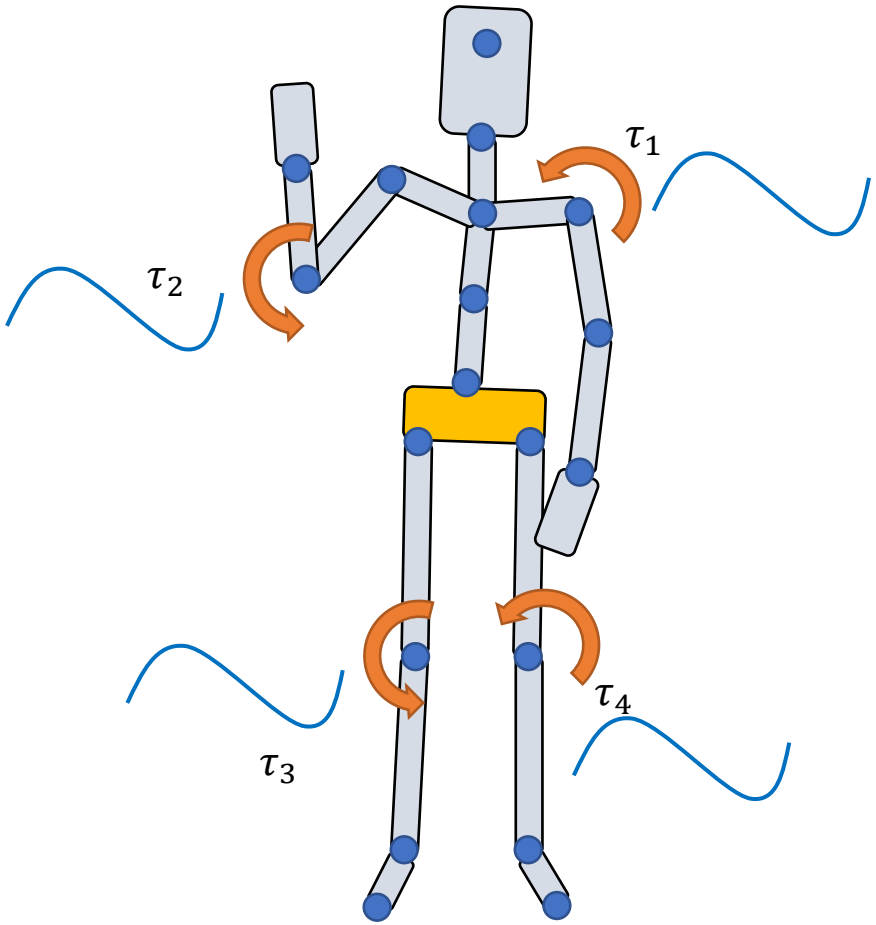
Full-body Tracking Controllers



Is PD control a feedforward control?

a feedback control?

Tracking Mocap with Joint Torques



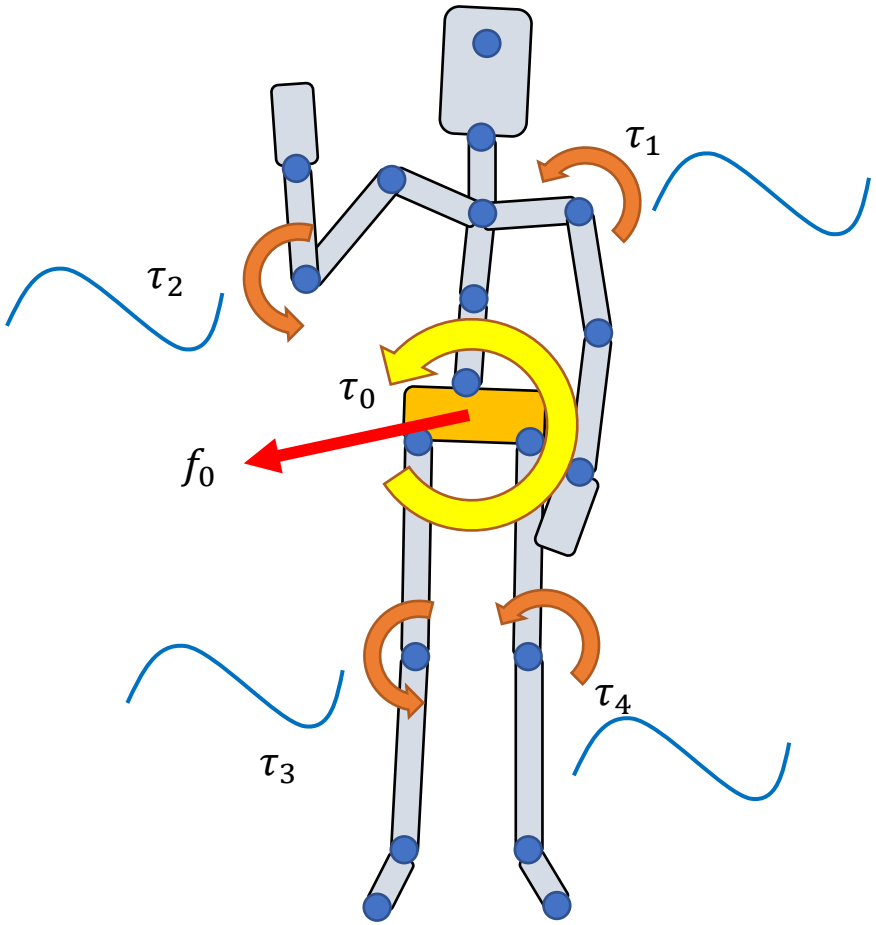
τ_j: joint torques

Apply τ_j to "child" body

Apply $-\tau_j$ to "parent" body

All forces/torques sum up to zero

Tracking Mocap with Residual Forces/Torques



τ_j: joint torques

Apply τ_j to "child" body

Apply $-\tau_j$ to "parent" body

All forces/torques sum up to zero

f_0, τ_0: root force / torque

Apply f_0 to the root body

Apply τ_0 to the root body

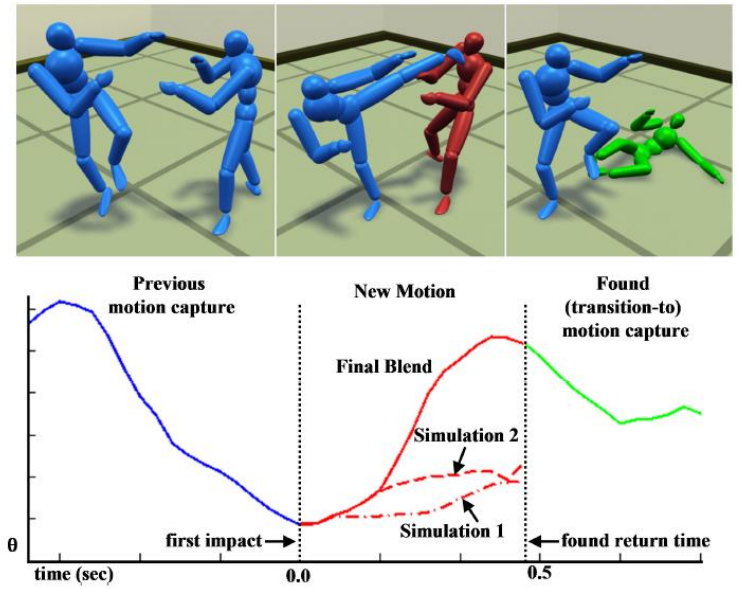
Non-zero net force/torque on the character!

Physically Plausible Animation

Party Animals

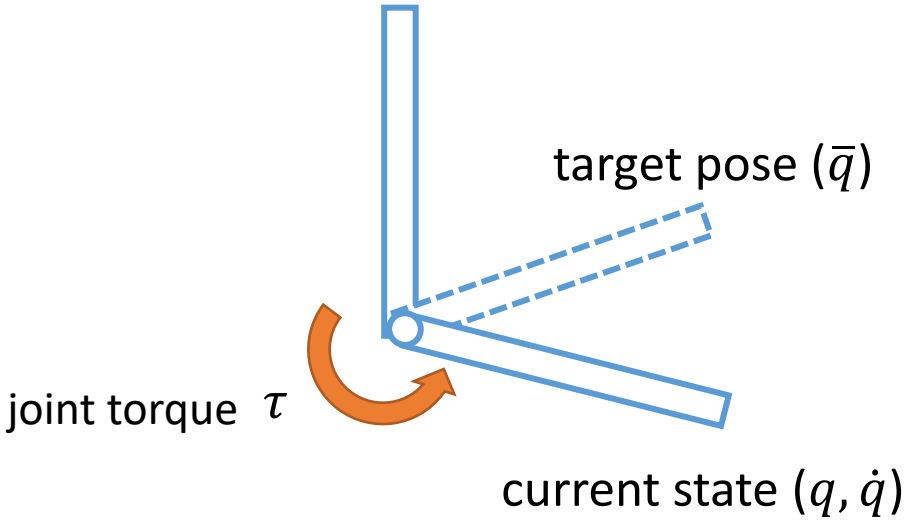
Totally Accurate Battle Simulator
https://www.youtube.com/watch?v=WFKGWfdG3bU

Mixture Simulation and Mocap



Zordan et al. 2005
Dynamic response for motion capture animation

PD Control for Characters



target pose

$$\tau = k_p(\bar{q} - q) - k_d\dot{q}$$

proportional control

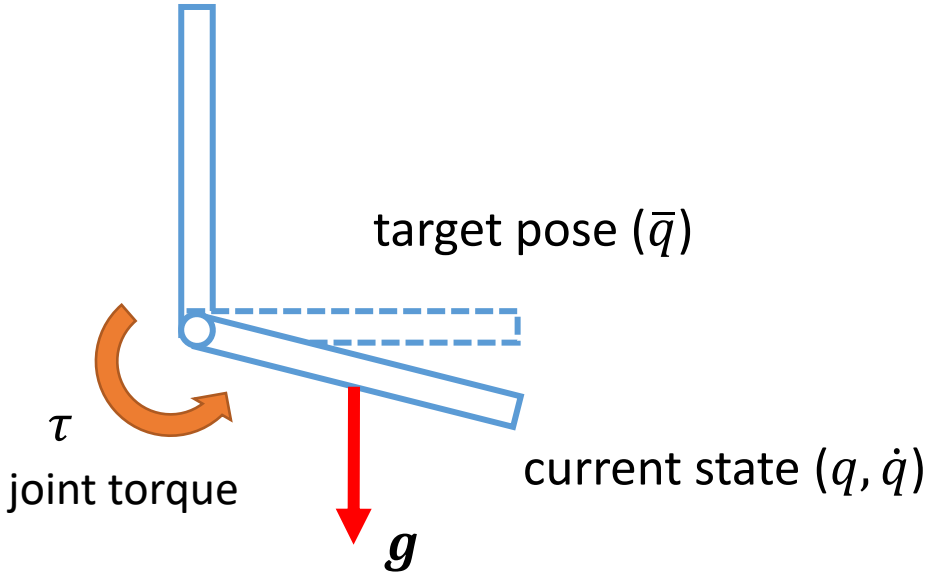
derivative control

Problems with PD Control

PD control computes torques based on errors

- Steady state error

This arm never reaches the target angle under gravity

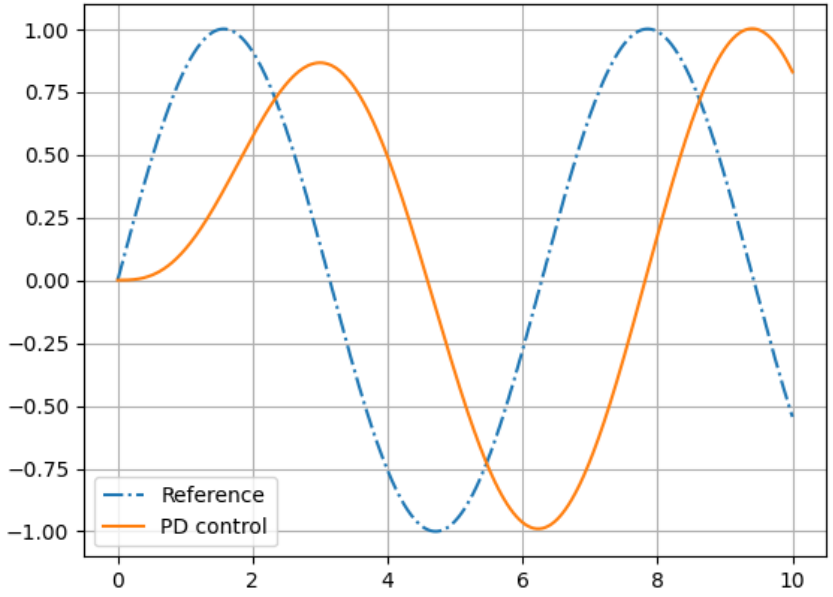


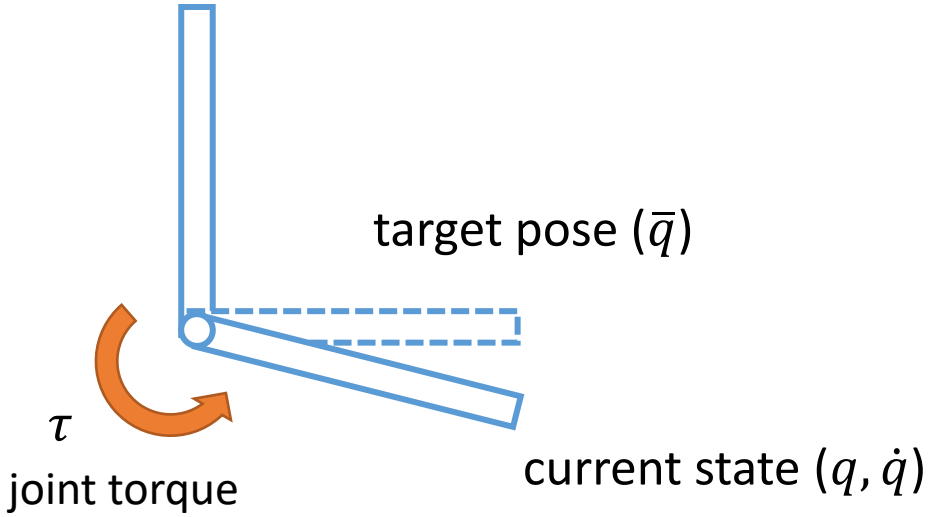
$$\tau = k_p(\bar{q} - q) - k_d\dot{q}$$

Problems with PD Control

PD control computes torques based on errors

- Steady state error
- Motion falls behind the reference





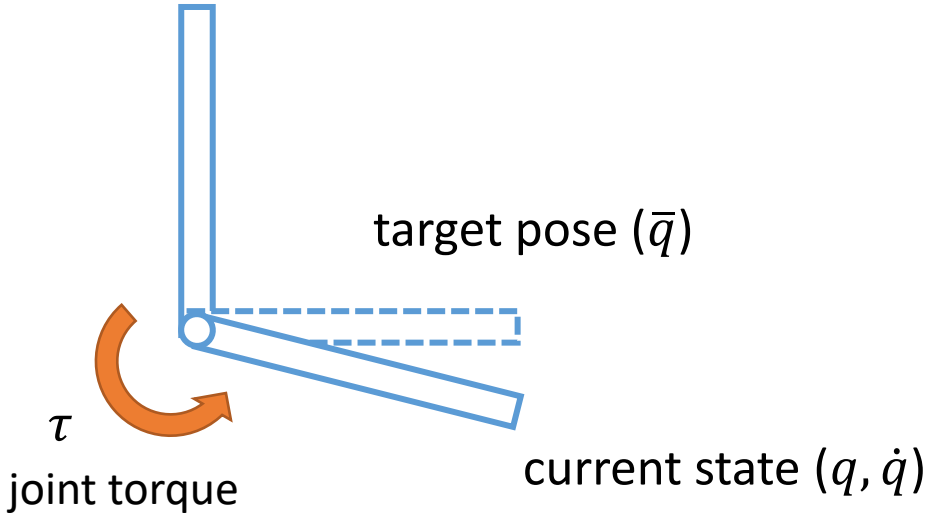
$$\tau = k_p(\bar{q} - q) - k_d\dot{q}$$

Problems with PD Control

PD control computes torques based on errors

- Steady state error
- Motion falls behind the reference

High-gain (k_p) control is more precise…



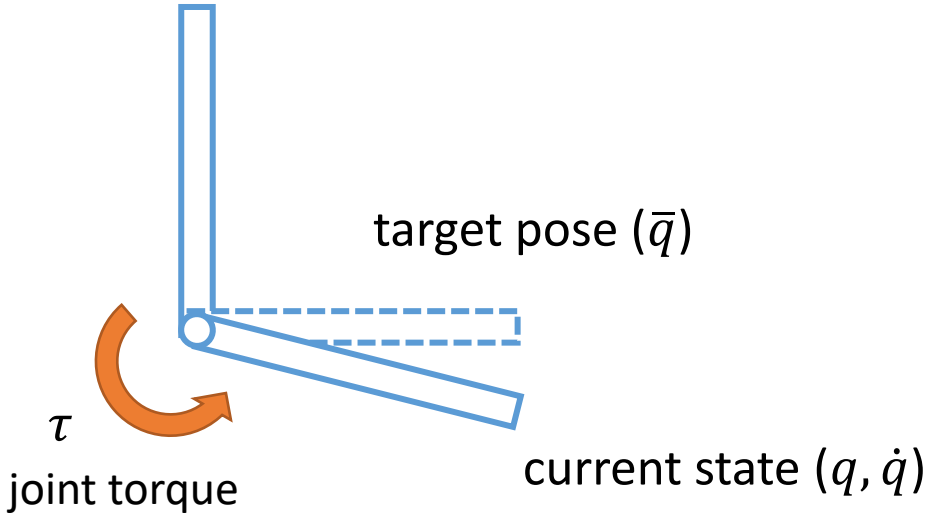
$$\tau = k_p(\bar{q} - q) - k_d\dot{q}$$

Problems with PD Control

PD control computes torques based on errors

- Steady state error
- Motion falls behind the reference

High-gain (k_p) control is more precise but less stable…



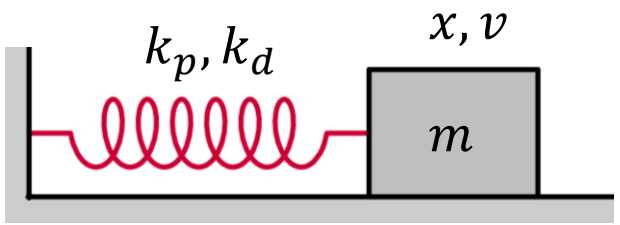
$$\tau = k_p(\bar{q} - q) - k_d\dot{q}$$

Stability of PD Control

Semi-implicit Euler Integration

$$v_{n+1} = v_n + h\frac{f}{m}$$

$$x_{n+1} = x_n + hv_{n+1}$$



$$f = -k_p x - k_d v$$

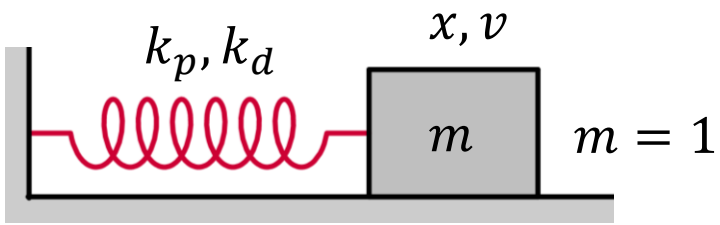
h: simulation time step

Stability of PD Control

Semi-implicit Euler Integration

$$v_{n+1} = v_n + hf$$

$$x_{n+1} = x_n + hv_{n+1}$$



$$f = -k_p x - k_d v$$

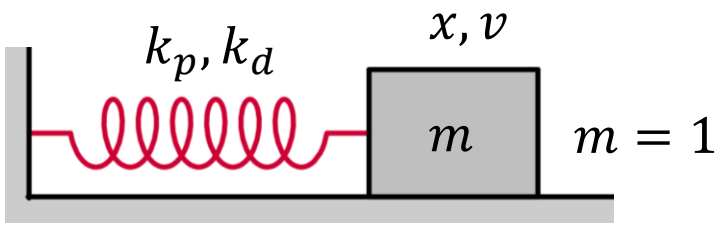
h: simulation time step

Stability of PD Control

Semi-implicit Euler Integration

$$v_{n+1} = v_n + h\left(-k_p x_n - k_d v_n\right)$$

$$x_{n+1} = x_n + h v_{n+1}$$



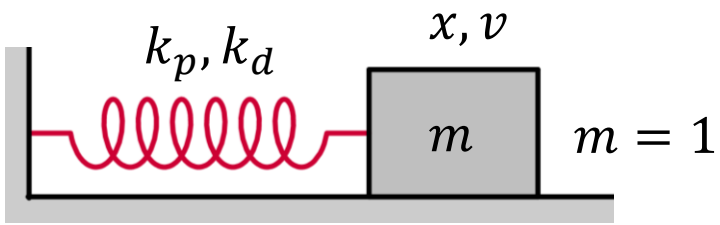
$$f = -k_p x - k_d v$$

h: simulation time step

Stability of PD Control

Semi-implicit Euler Integration

$$\begin{bmatrix} v_{n+1} \\ x_{n+1} \end{bmatrix} = \begin{bmatrix} 1 - k_d h & -k_p h \\ h(1 - k_d h) & 1 - k_p h^2 \end{bmatrix} \begin{bmatrix} v_n \\ x_n \end{bmatrix}$$



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Stability of PD Control

Semi-implicit Euler Integration

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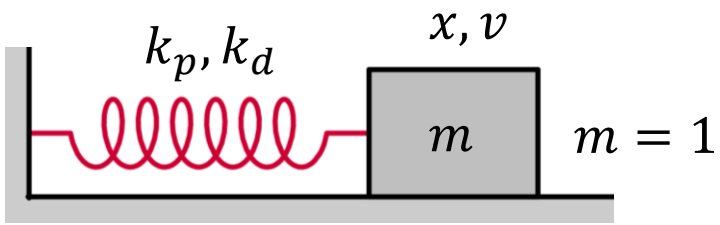
$$\boldsymbol{s}_{n+1} = A\boldsymbol{s}_n$$

$$\boldsymbol{s}_{n+1} = A^n \boldsymbol{s}_1$$



$$f = -k_p x - k_d v$$

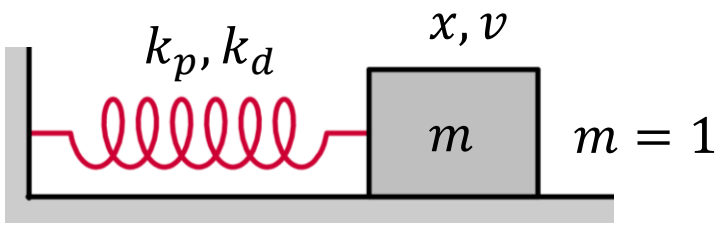
h: simulation time step

Stability of PD Control

$$\boldsymbol{s}_{n+1} = A\boldsymbol{s}_n \quad \Rightarrow \quad \boldsymbol{s}_{n+1} = A^n \boldsymbol{s}_1$$

Assume A has two eigenvalues λ_1, λ_2 and two eigen vectors $\boldsymbol{v}_1, \boldsymbol{v}_2$. Note $\lambda_i \in \mathbb{C}$

$$A = P \begin{bmatrix} \lambda_1 & \\ & \lambda_2 \end{bmatrix} P^{-1} \qquad P = \begin{bmatrix} \vdots & \vdots \\ \boldsymbol{v}_1 & \boldsymbol{v}_2 \\ \vdots & \vdots \end{bmatrix}$$



$$f = -k_p x - k_d v$$

h: simulation time step

$$A = \begin{bmatrix} 1 - k_d h & -k_p h \\ h(1 - k_d h) & 1 - k_p h^2 \end{bmatrix}$$

Stability of PD Control

$$\boldsymbol{s}_{n+1} = A\boldsymbol{s}_n \quad \Rightarrow \quad \boldsymbol{s}_{n+1} = A^n \boldsymbol{s}_1$$

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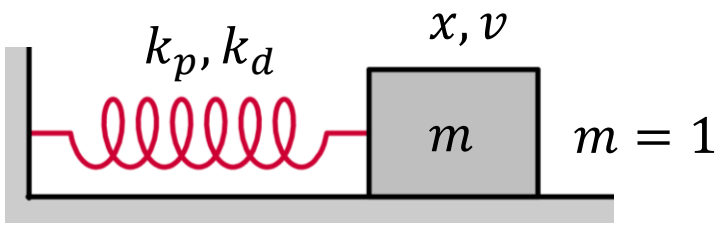
$$\Downarrow$$

$$\boldsymbol{s}_{n+1} = P \begin{bmatrix} \lambda_1 & \\ & \lambda_2 \end{bmatrix} P^{-1} \boldsymbol{s}_n$$



$$f = -k_p x - k_d v$$

h: simulation time step

$$A = \begin{bmatrix} 1 - k_d h & -k_p h \\ h(1 - k_d h) & 1 - k_p h^2 \end{bmatrix}$$

Stability of PD Control

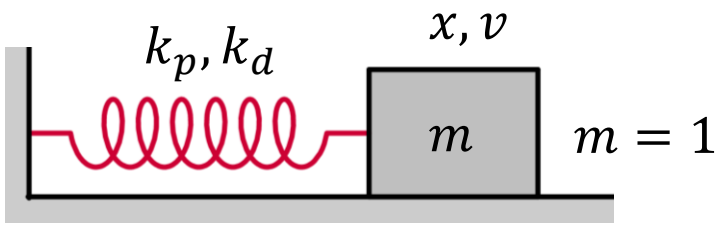
$$\boldsymbol{s}_{n+1} = A\boldsymbol{s}_n \quad \Rightarrow \quad \boldsymbol{s}_{n+1} = A^n \boldsymbol{s}_1$$

Assume A has two eigenvalues λ_1, λ_2 and two eigen vectors $\boldsymbol{v}_1, \boldsymbol{v}_2$. Note $\lambda_i \in \mathbb{C}$

$$A = P \begin{bmatrix} \lambda_1 & \\ & \lambda_2 \end{bmatrix} P^{-1} \qquad P = \begin{bmatrix} \vdots & \vdots \\ \boldsymbol{v}_1 & \boldsymbol{v}_2 \\ \vdots & \vdots \end{bmatrix}$$

$$\Downarrow$$

$$\boldsymbol{z}_{n+1} = \begin{bmatrix} \lambda_1 & \\ & \lambda_2 \end{bmatrix} \boldsymbol{z}_n \qquad \boldsymbol{z} = P^{-1}\boldsymbol{s}$$



$$f = -k_p x - k_d v$$

h: simulation time step

$$A = \begin{bmatrix} 1 - k_d h & -k_p h \\ h(1 - k_d h) & 1 - k_p h^2 \end{bmatrix}$$

Stability of PD Control

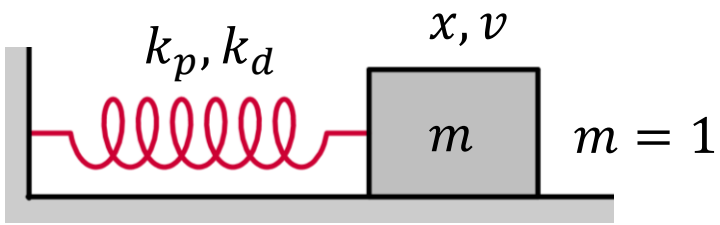
$$\boldsymbol{s}_{n+1} = A\boldsymbol{s}_n \quad \Rightarrow \quad \boldsymbol{s}_{n+1} = A^n \boldsymbol{s}_1$$

Assume A has two eigenvalues λ_1, λ_2 and two eigen vectors $\boldsymbol{v}_1, \boldsymbol{v}_2$. Note $\lambda_i \in \mathbb{C}$

$$A = P \begin{bmatrix} \lambda_1 & \\ & \lambda_2 \end{bmatrix} P^{-1} \qquad P = \begin{bmatrix} \vdots & \vdots \\ \boldsymbol{v}_1 & \boldsymbol{v}_2 \\ \vdots & \vdots \end{bmatrix}$$

$$\Downarrow$$

$$\boldsymbol{z}_{n+1} = \begin{bmatrix} \lambda_1 & \\ & \lambda_2 \end{bmatrix} \boldsymbol{z}_n \quad \Rightarrow \quad \boldsymbol{z}_{n+1} = \begin{bmatrix} \lambda_1 & \\ & \lambda_2 \end{bmatrix}^n \boldsymbol{z}_1$$



$$f = -k_p x - k_d v$$

h: simulation time step

$$A = \begin{bmatrix} 1 - k_d h & -k_p h \\ h(1 - k_d h) & 1 - k_p h^2 \end{bmatrix}$$

Stability of PD Control

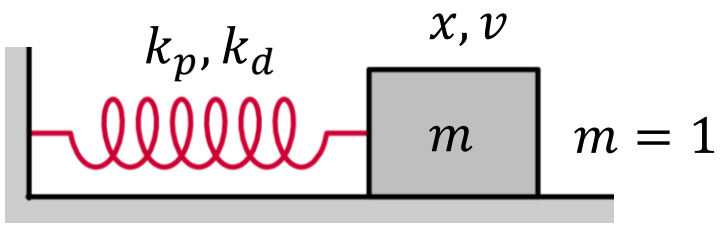
$$\boldsymbol{s}_{n+1} = A\boldsymbol{s}_n \quad \Rightarrow \quad \boldsymbol{s}_{n+1} = A^n \boldsymbol{s}_1$$

Assume A has two eigenvalues λ_1, λ_2 and two eigen vectors $\boldsymbol{v}_1, \boldsymbol{v}_2$. Note $\lambda_i \in \mathbb{C}$

$$A = P \begin{bmatrix} \lambda_1 & \\ & \lambda_2 \end{bmatrix} P^{-1} \qquad P = \begin{bmatrix} \vdots & \vdots \\ \boldsymbol{v}_1 & \boldsymbol{v}_2 \\ \vdots & \vdots \end{bmatrix}$$

$$\Downarrow$$

$$\boldsymbol{z}_{n+1} = \begin{bmatrix} \lambda_1 & \\ & \lambda_2 \end{bmatrix} \boldsymbol{z}_n \quad \Rightarrow \quad \boldsymbol{z}_{n+1} = \begin{bmatrix} \lambda_1^n & \\ & \lambda_2^n \end{bmatrix} \boldsymbol{z}_1$$



$$f = -k_p x - k_d v$$

h: simulation time step

$$A = \begin{bmatrix} 1 - k_d h & -k_p h \\ h(1 - k_d h) & 1 - k_p h^2 \end{bmatrix}$$

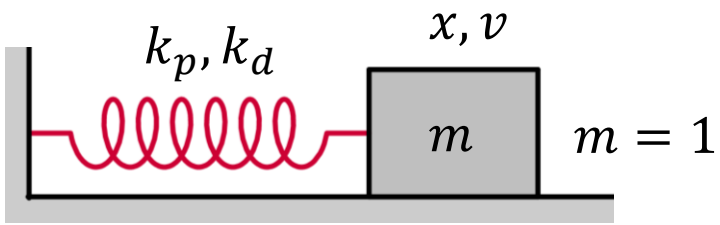
Stability of PD Control

$$\boldsymbol{z}_{n+1} = \begin{bmatrix} \lambda_1 & \\ & \lambda_2 \end{bmatrix} \boldsymbol{z}_n \Rightarrow \boldsymbol{z}_{n+1} = \begin{bmatrix} \lambda_1^n & \\ & \lambda_2^n \end{bmatrix} \boldsymbol{z}_1$$

$\lambda_1, \lambda_2 \in \mathbb{C}$ are eigenvalues of A

if $|\lambda_1| > 1$ $\Rightarrow$ $\lim_{n\to\infty} \|\boldsymbol{z}_n\| \to \infty$

The system is unstable!



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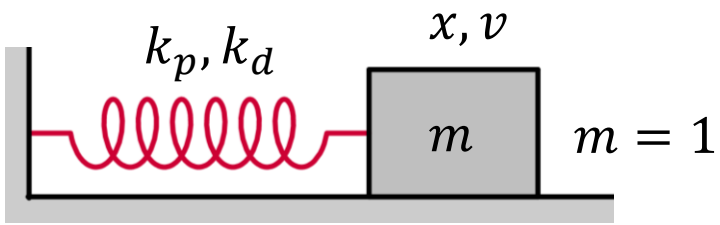
Condition of stability: $|\lambda_i| \leq 1$ for all λ_i



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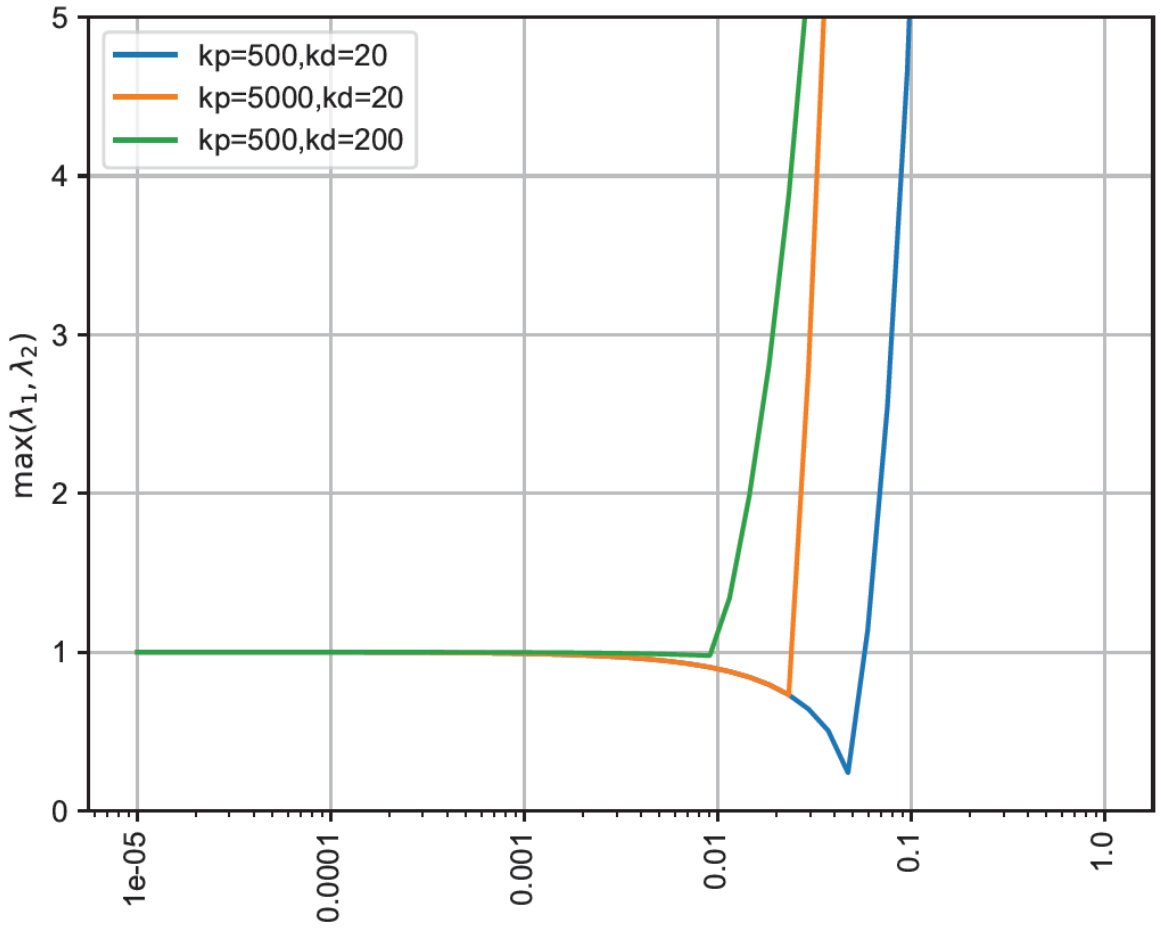

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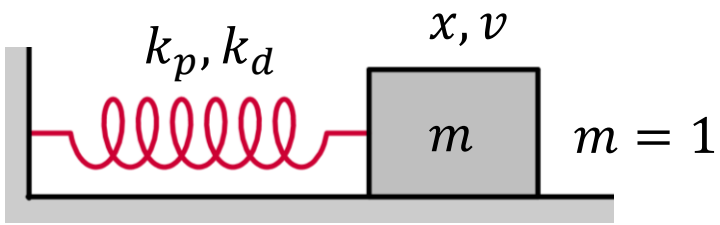

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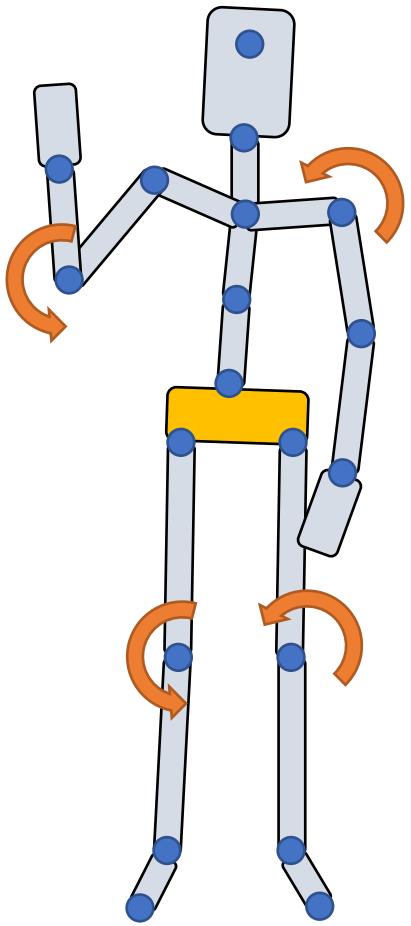
PD Control for Characters



$$\tau = k_p(\bar{q} - q) - k_d\dot{q}$$

- Determining gain and damping coefficients can be difficult…
  - A typical setting $k_p = 200$, $k_d = 20$ for a 50kg character
  - Light body requires smaller gains
  - Dynamic motions need larger gains
- High-gain/high-damping control can be unstable, so small times is necessary
  - $h = 0.5\sim1\text{ms}$ is often used, or $1000\sim2000\text{Hz}$
  - Higher gain/damping requires smaller time step

A More Stable PD Control

Semi-implicit Euler Integration

$$v_{n+1} = v_n + h\left(-k_p x_n - k_d v_n\right)$$

$$x_{n+1} = x_n + h v_{n+1}$$

$$v_{n+1} = v_n + h\left(-k_p x_n - k_d \boldsymbol{v_{n+1}}\right)$$

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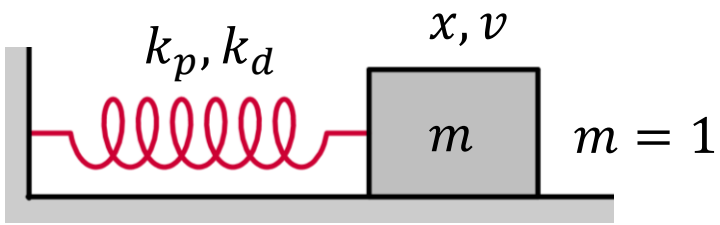


$$f = -k_p x - k_d v$$

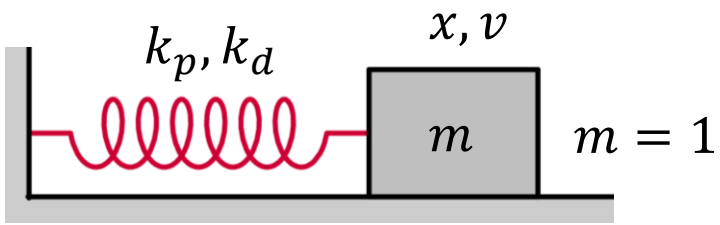
h: simulation time step

A More Stable PD Control

$$v_{n+1} = v_n + h\left(-k_p x_n - k_d \boldsymbol{v_{n+1}}\right)$$

$$x_{n+1} = x_n + h v_{n+1}$$

$$\begin{bmatrix} v_{n+1} \\ x_{n+1} \end{bmatrix} = \frac{1}{1 + hk_d} \begin{bmatrix} 1 & -k_p h \\ h & 1 + k_d h - k_p h^2 \end{bmatrix} \begin{bmatrix} v_n \\ x_n \end{bmatrix}$$

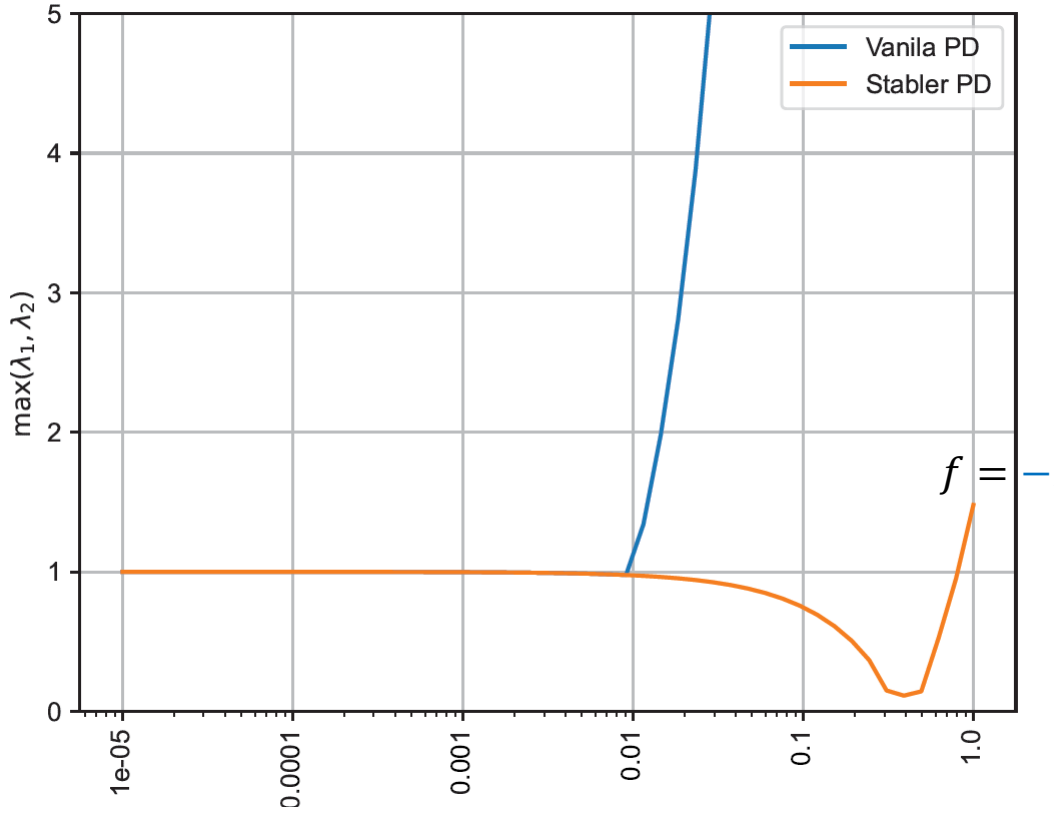


$$f = -k_p x - k_d v$$

h: simulation time step

A More Stable PD Control

$$f = -k_p x_n - k_d v_{\boldsymbol{n}}$$



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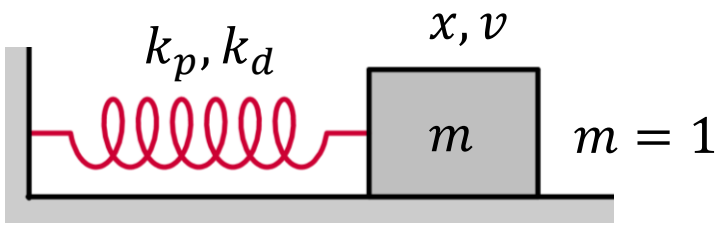
h: simulation time step



$$f = -k_p x - k_d v$$

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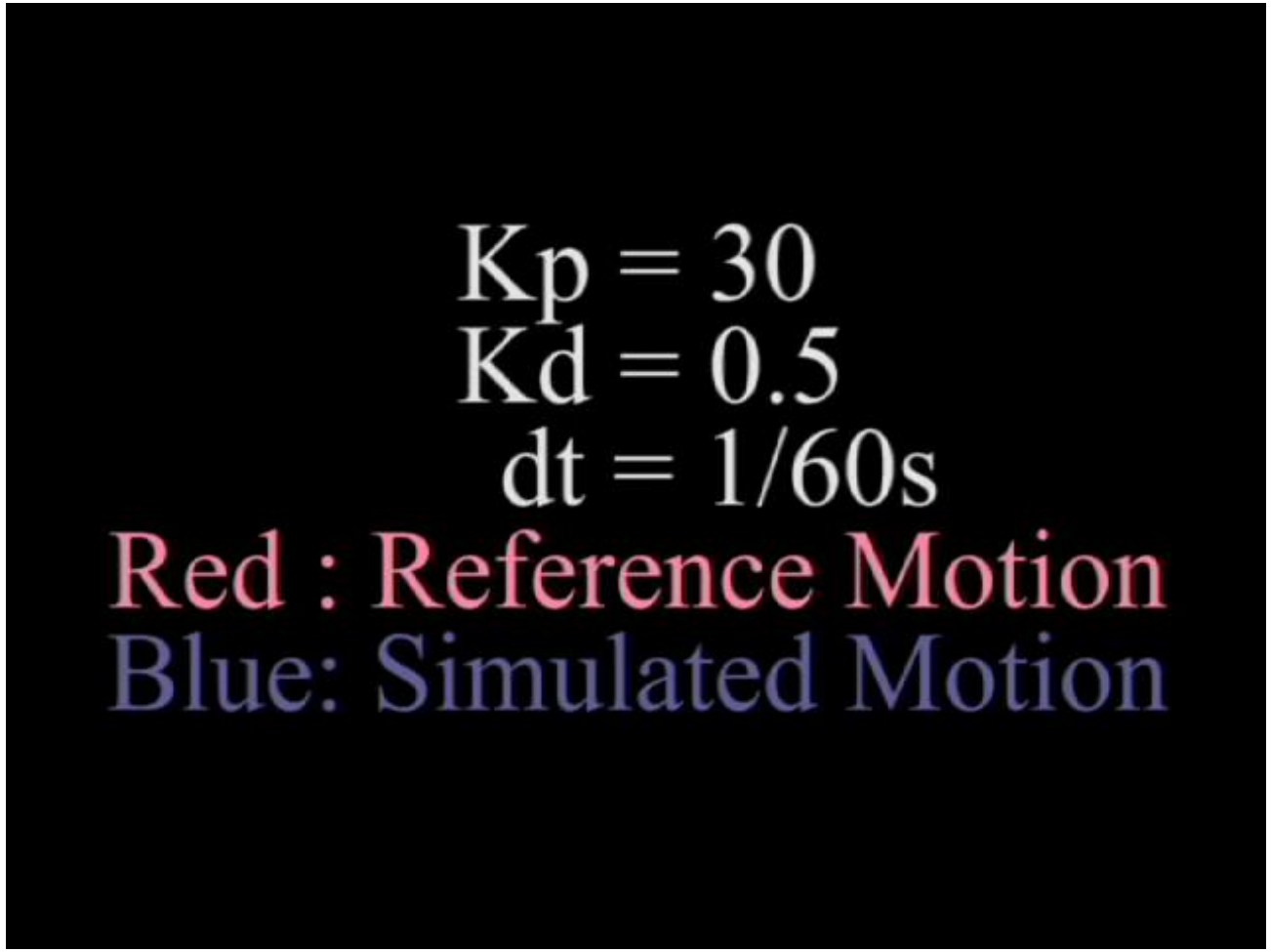
Stable PD Control

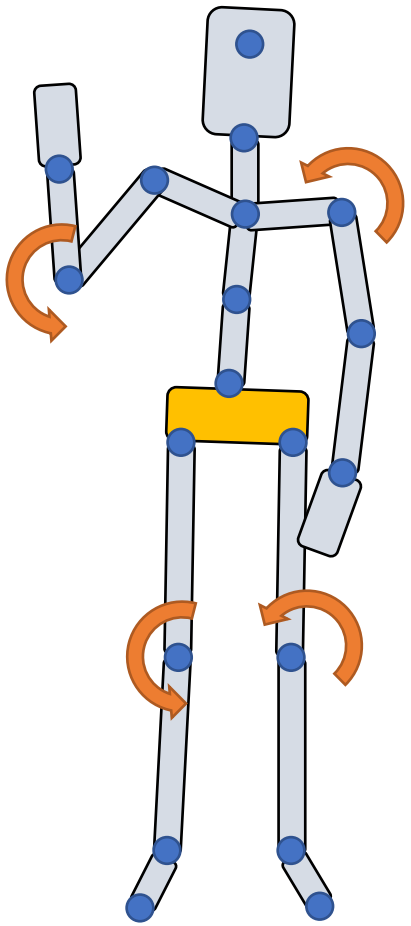


$$\tau_{\mathbf{int}} = -\mathbf{K}_p(\mathbf{q}^n + \dot{\mathbf{q}}^n \Delta t - \bar{\mathbf{q}}^{n+1}) - \mathbf{K}_d(\dot{\mathbf{q}}^n + \ddot{\mathbf{q}}^n \Delta t)$$

Stable Proportional-Derivative Controllers

Jie Tan* Karen Liu† Greg Turk‡
Georgia Institute of Technology

PD Control for Characters



$$\tau = k_p(\bar{q} - q) - k_d\dot{q}$$

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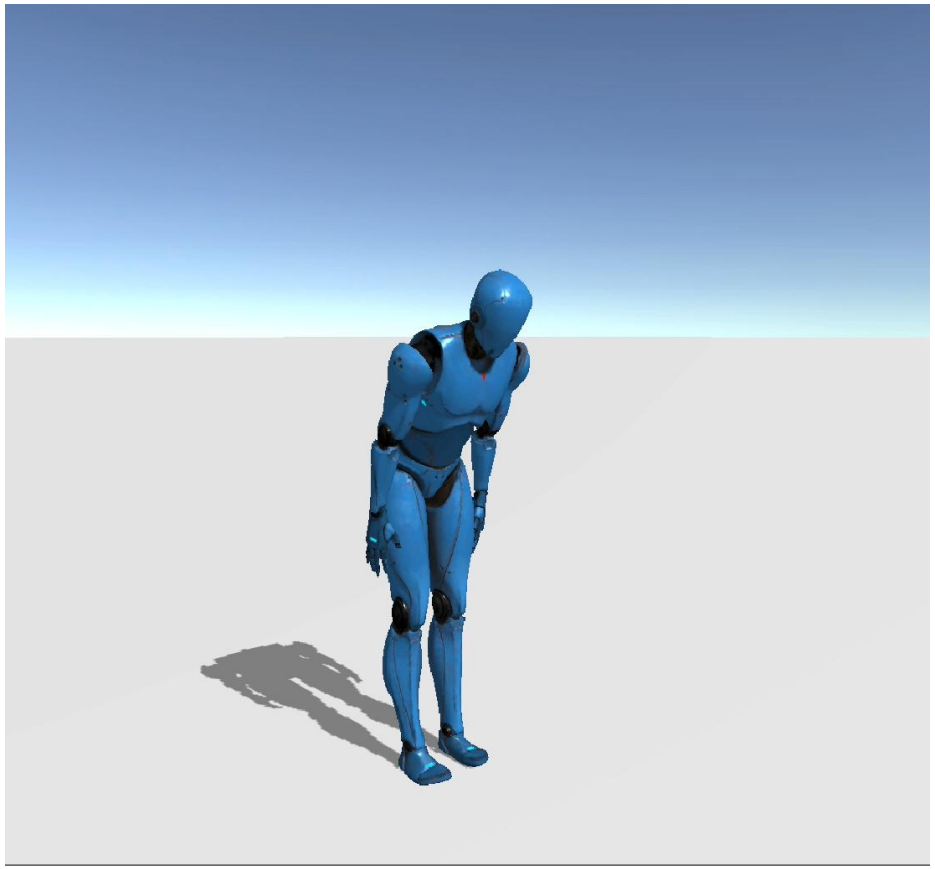
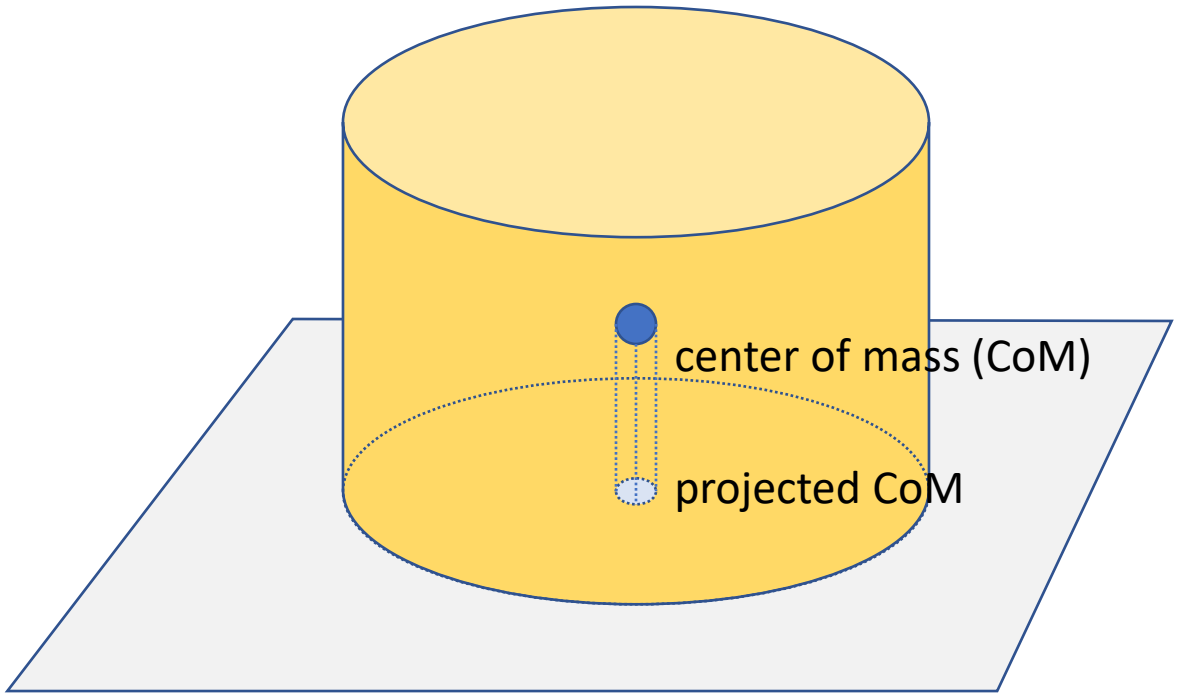
- High-gain/high-damping control can be unstable, so small times is necessary
  - $h = 0.5\sim1\text{ms}$ is often used, or $1000\sim2000\text{Hz}$
  - $h = 1/120\text{s}\sim1/60\text{s}$, or 120Hz/60Hz with Stable PD
  - Higher gain/damping requires smaller time step

Static Balance



Static Balance

What is balance?



Static Balance

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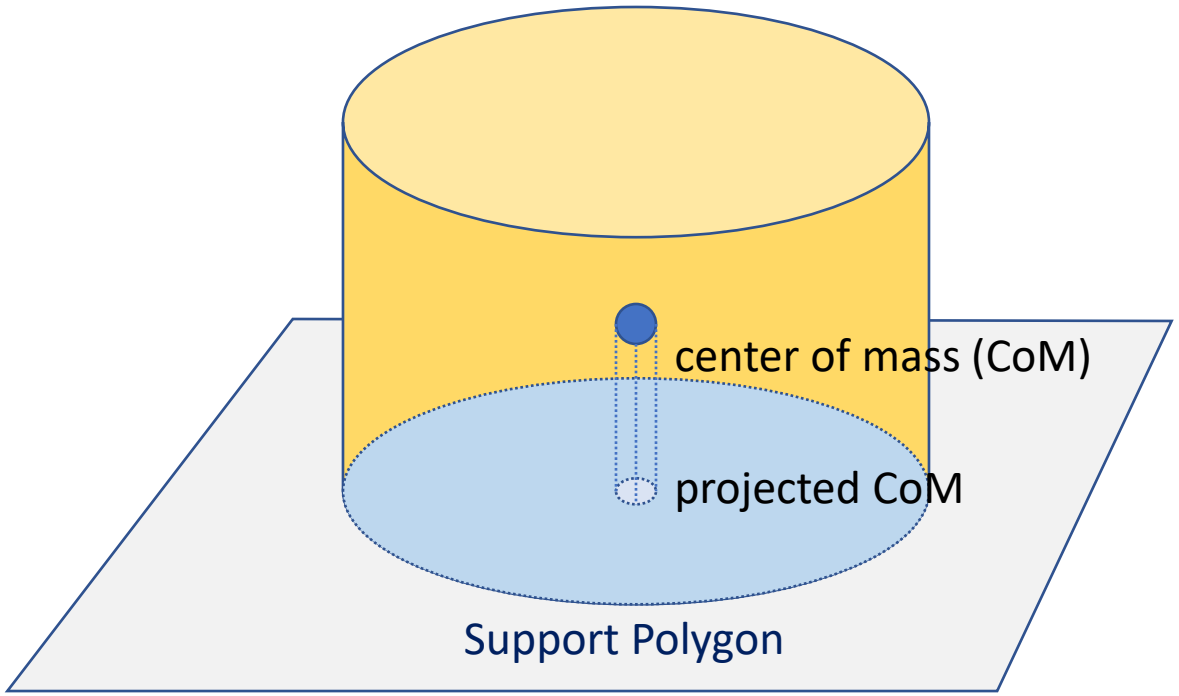
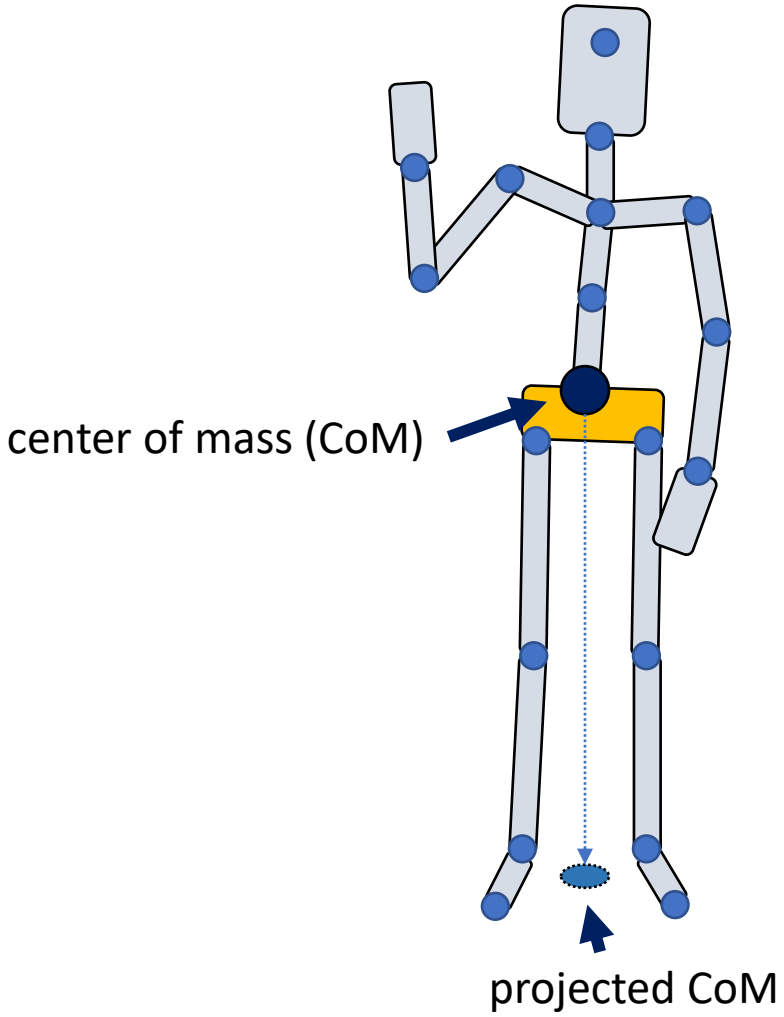


Static Balance

What is balance?

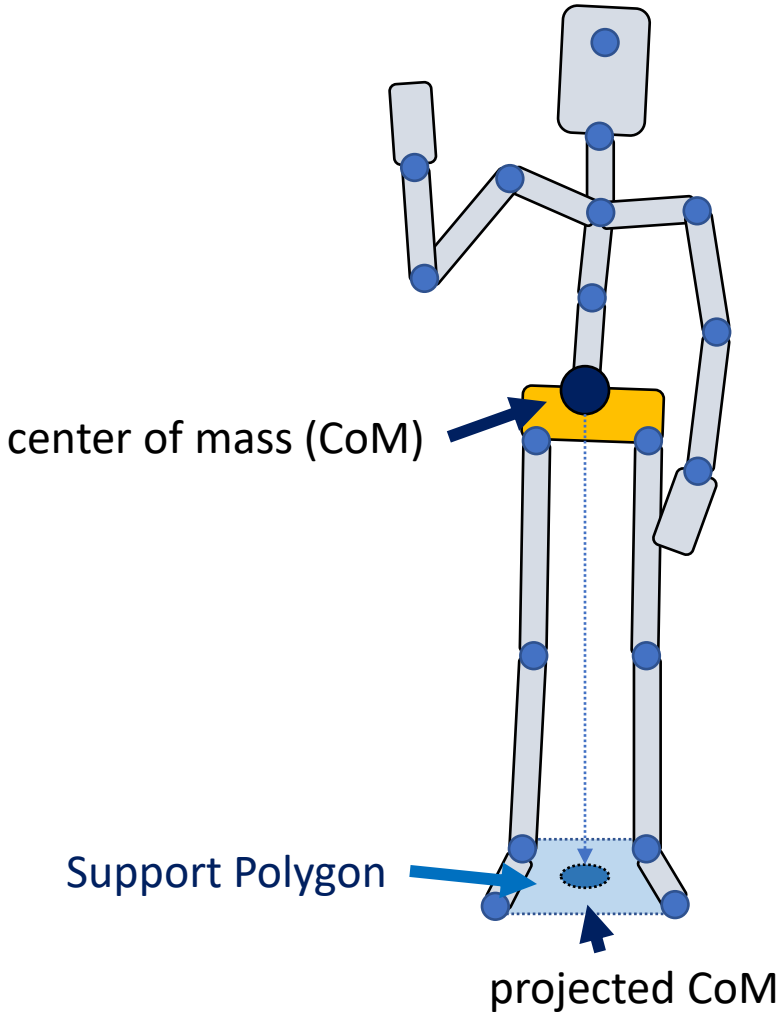
静平衡条件



support：两个脚构成的凸包

Static Balance

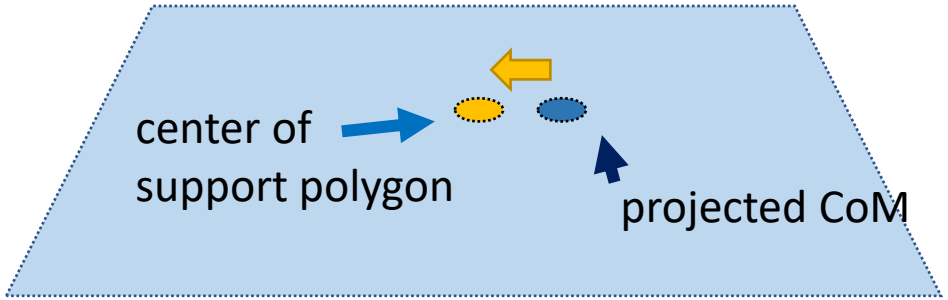
What is balance?

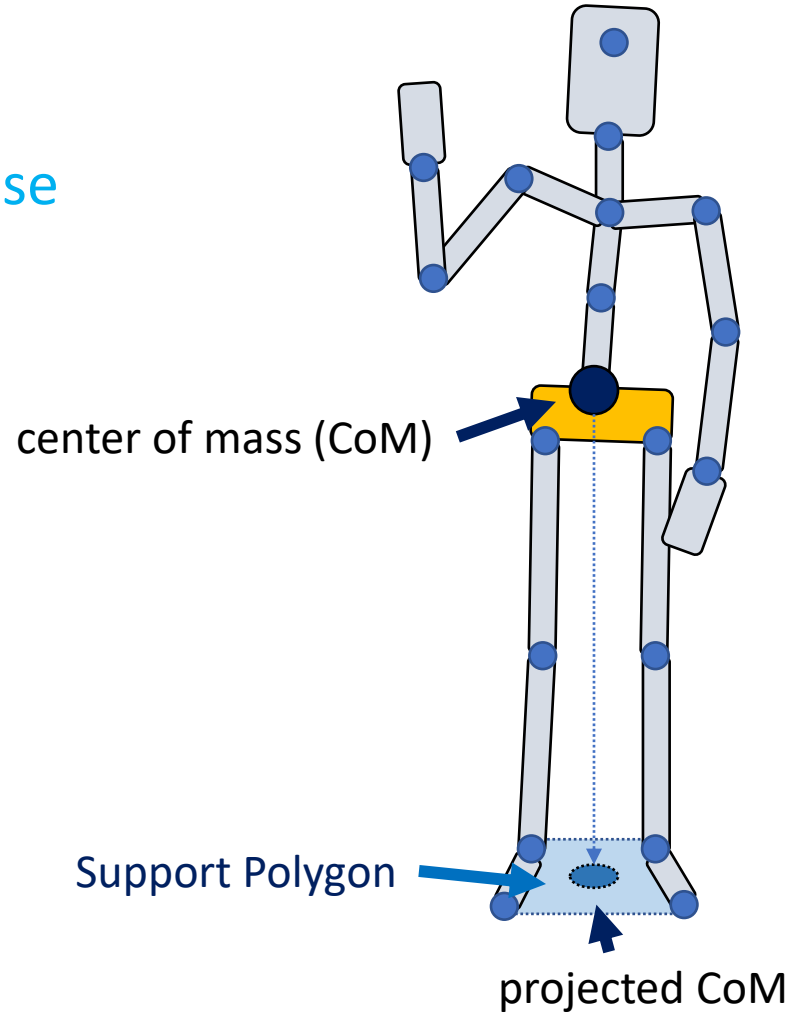


Static Balance

A simple strategy to maintain balance:

- Keep projected CoM close to the center of support polygon while tracking a standing pose





Static Balance

A simple strategy to maintain balance:

- Keep projected CoM close to the center of support polygon while tracking a standing pose
- Use PD control to compute feedback torque

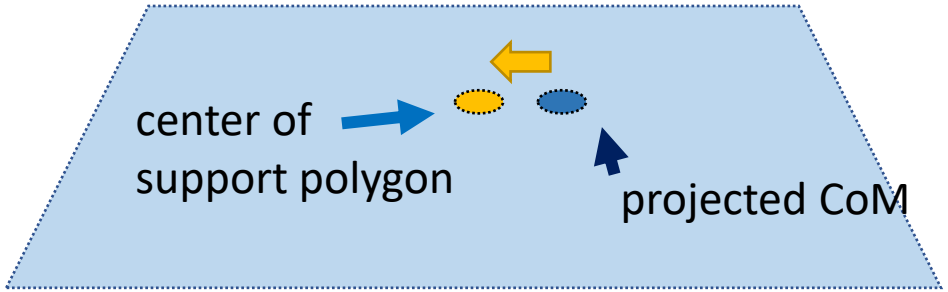
$$\tau = k_p(\bar{\boldsymbol{c}} - \boldsymbol{c}) - k_d\dot{\boldsymbol{c}}$$

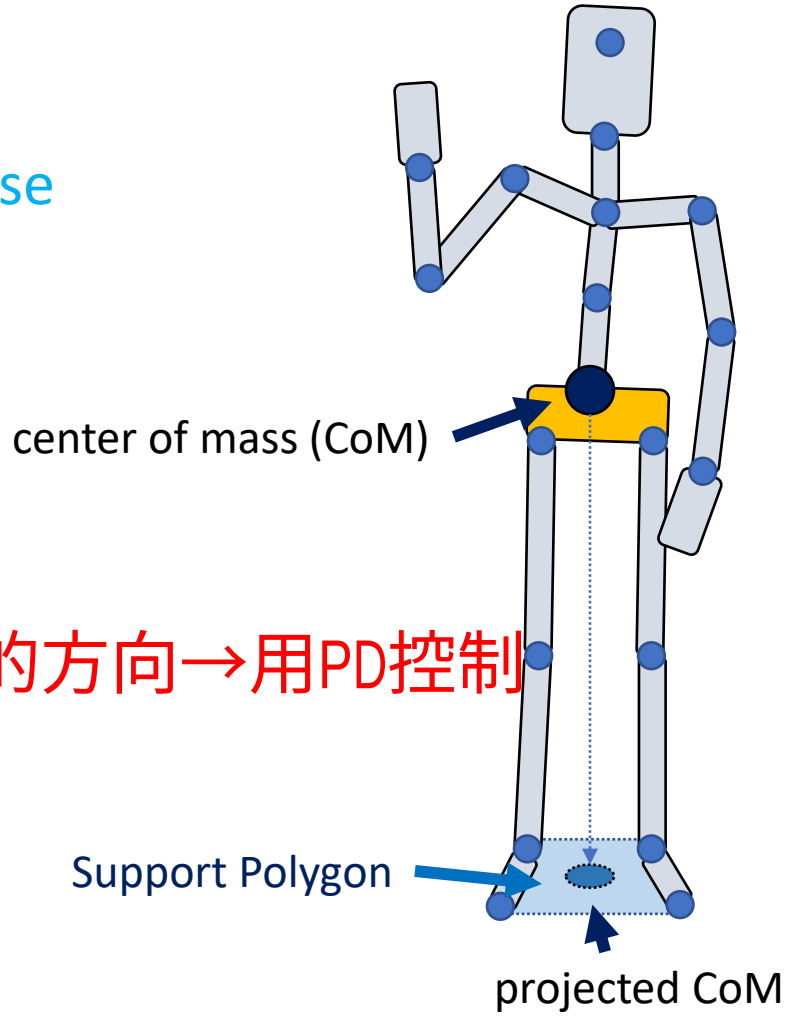
CoM velocity

desired CoM position (e.g. center of support polygon)

CoM position of the character

指定一个关节（如脚腕），和质心移动是相反的方向→用PD控制





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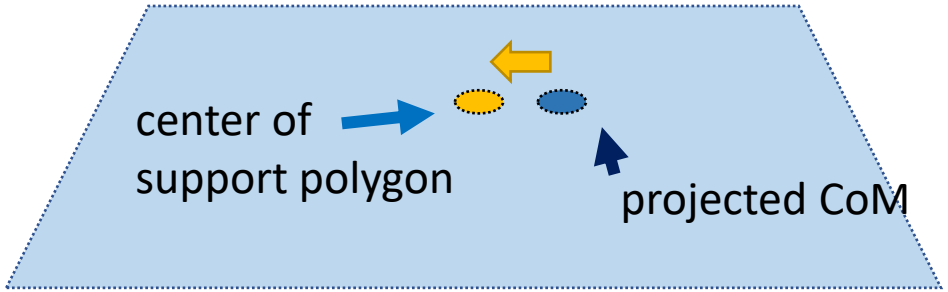
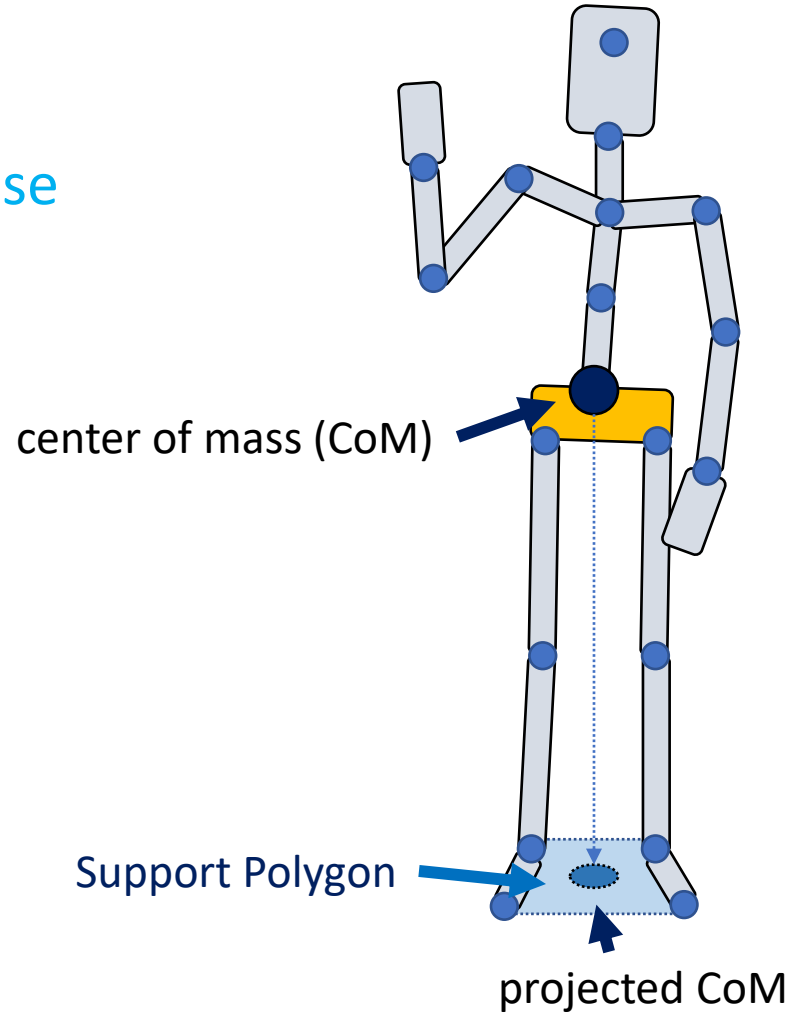
gain coefficient

damping coefficient

need to tune manually…





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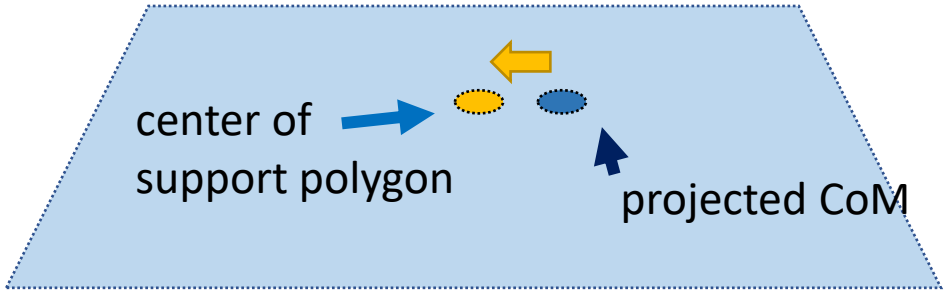
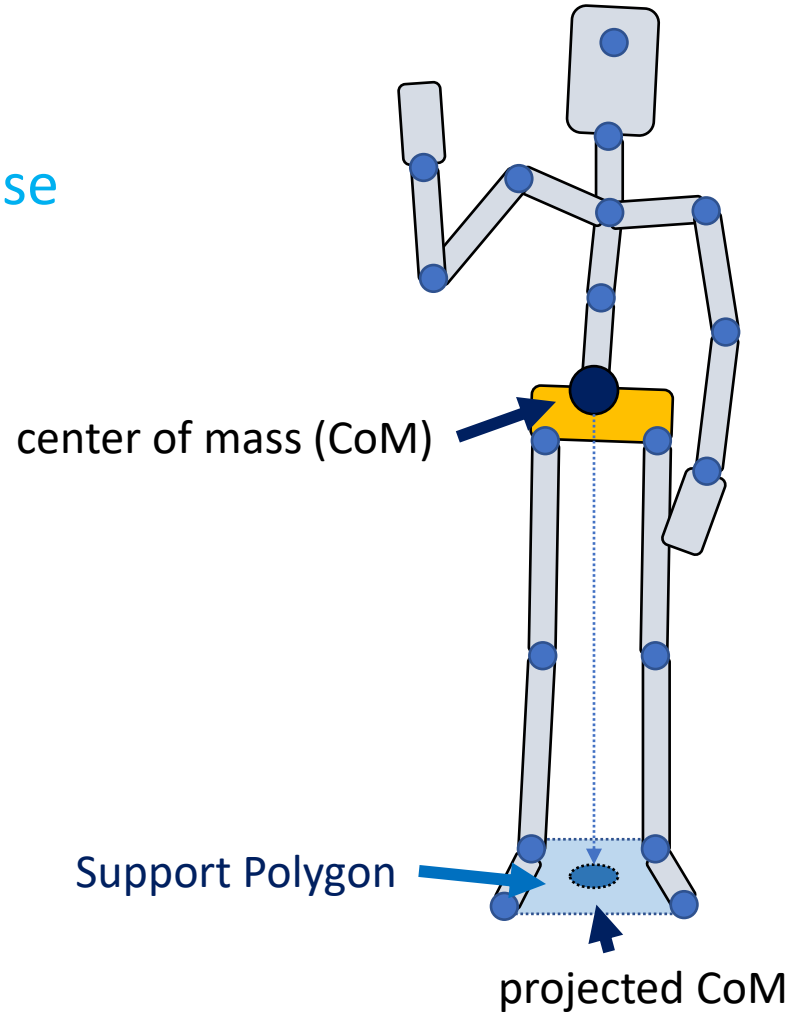
- Keep projected CoM close to the center of support polygon while tracking a standing pose
- Use PD control to compute feedback torque

$$\tau = k_p(\bar{\boldsymbol{c}} - \boldsymbol{c}) - k_d\dot{\boldsymbol{c}}$$

- Apply the feedback torque at ankles (ankle strategy) or hips (hip strategy)



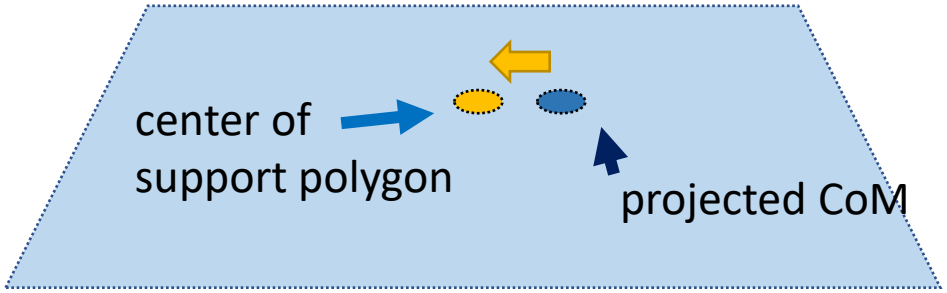


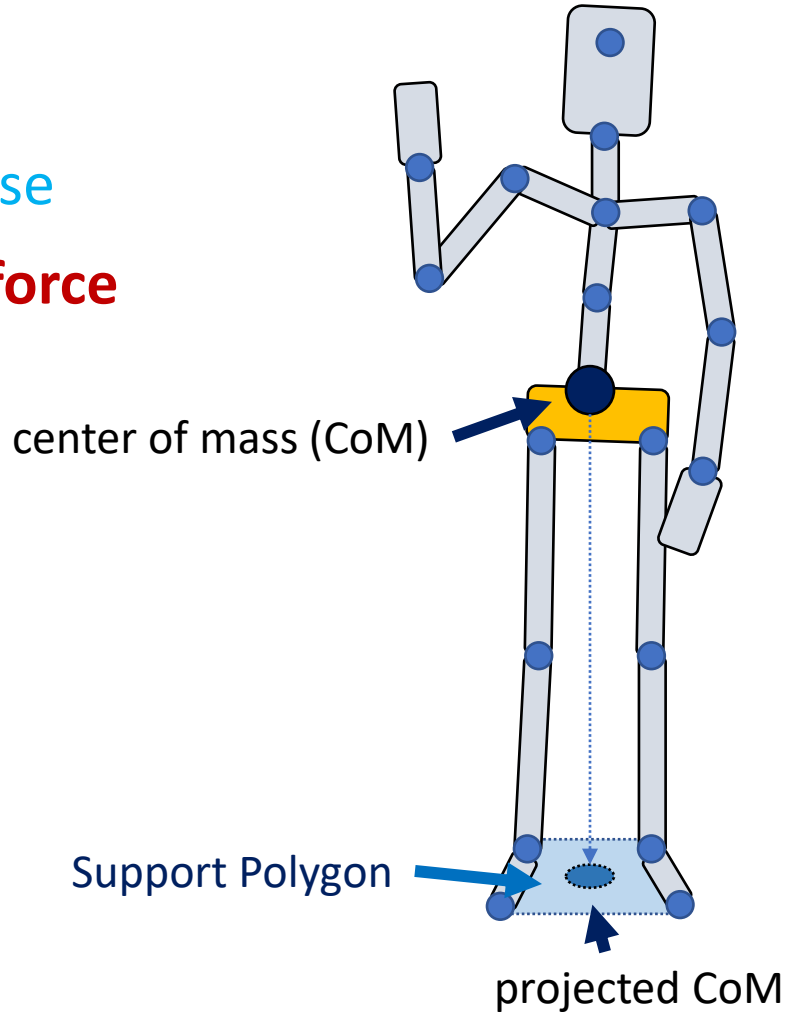
Static Balance

A simple strategy to maintain balance:

- Keep projected CoM close to the center of support polygon while tracking a standing pose
- Use PD control to compute feedback **virtual force**

$$f = k_p(\bar{\boldsymbol{c}} - \boldsymbol{c}) - k_d\dot{\boldsymbol{c}}$$





Static Balance

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- Keep projected CoM close to the center of support polygon while tracking a standing pose
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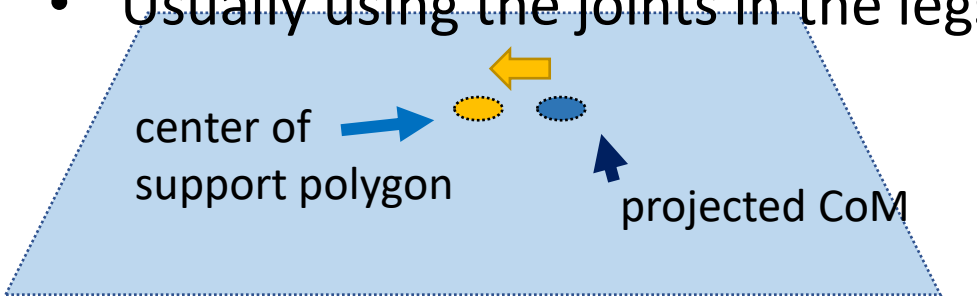
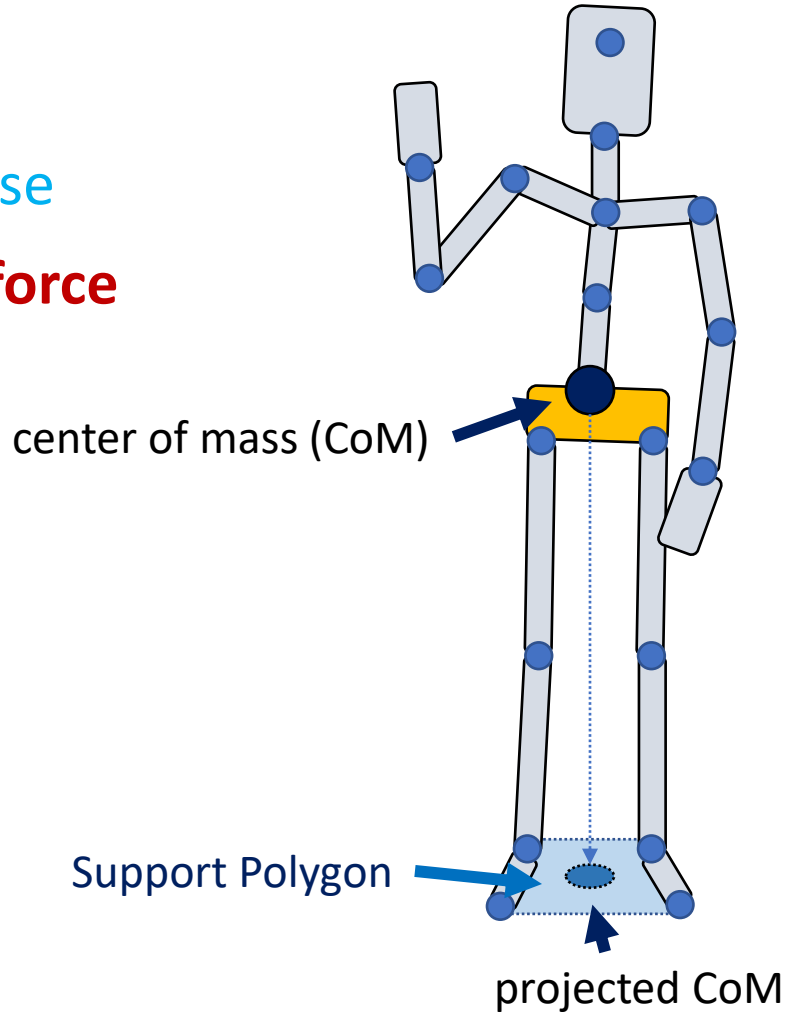
$$f = k_p(\bar{\boldsymbol{c}} - \boldsymbol{c}) - k_d\dot{\boldsymbol{c}}$$

- Assuming f is applied to the CoM, compute necessary joint torques using Jacobian transpose control to achieve it
  - Usually using the joints in the legs

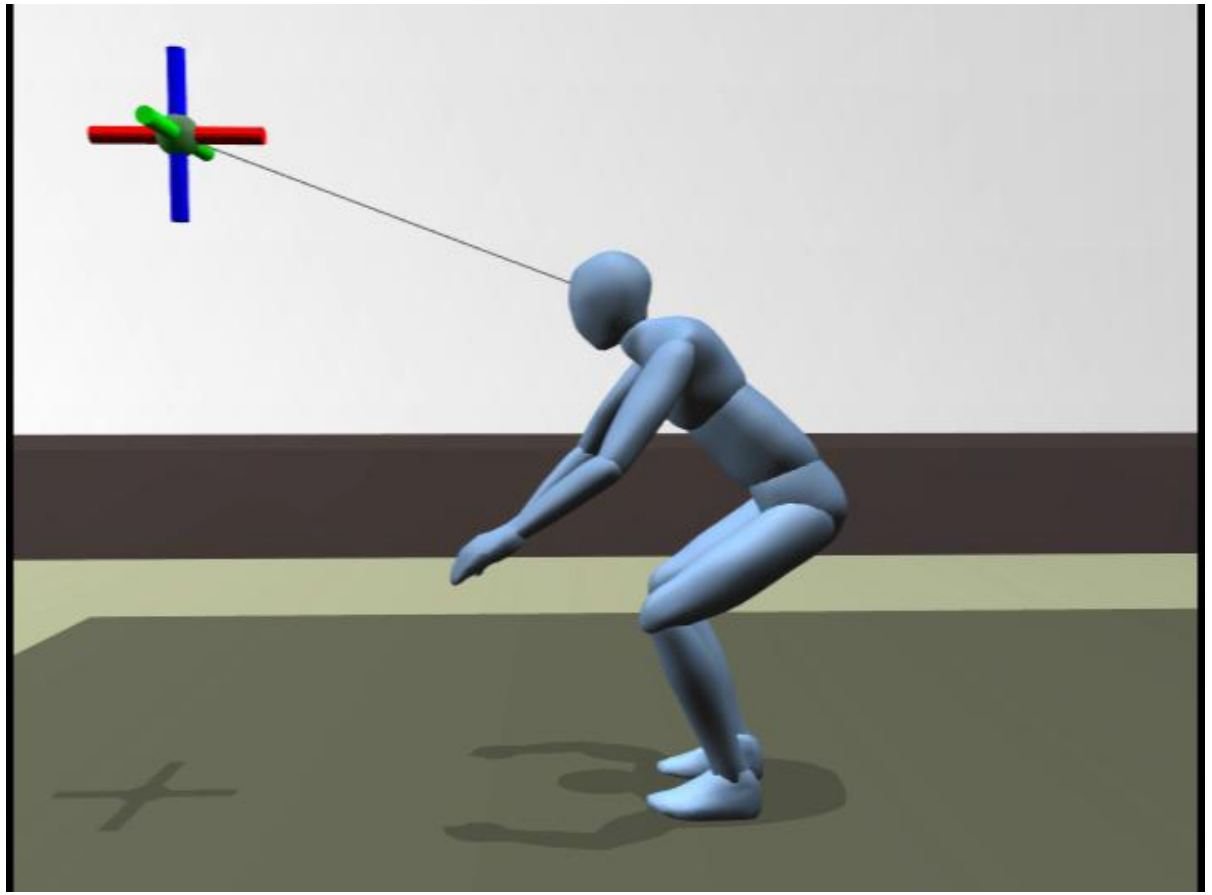




Static Balance

A fancier strategy:

- Mocap tracking as an objective function
- Controlling both the CoM position/momentum and the angular momentum
- Solve a one-step optimization problem to compute joint torques



[Macchietto et al. 2009 - Momentum Control for Balance]

Outline

- Simulating & Actuating Characters
  - Joint torques
- PD (Proportional-Derivative) control
  - Stable PD control
- Feedforward Motion Control
  - Residual force
- Feedback Motion Control
  - Static balance

Questions?